

The coppe document class

Version 4.0

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Abstract

In this work, it is described the `coppe` document class as well as other files distributed by the COPPETEX project. This class is suitable for writing academic dissertations, thesis and qualifying exams according to the formatting rules of the Alberto Luiz Coimbra Institute for Graduate Studies and Research in Engineering. The minimalist set of macro commands allows its users to concentrate most of their efforts on text composition rather than on the document layout.

1 Introduction

Writing documents in L^AT_EX may be a laborious task when the authors have to prepare their manuscripts rigorously respecting formatting rules imposed by publishers. Regardless of difficulty, a lot of thesis presented to the Coordination of Graduate Studies and Research in Engineering of the Federal University of Rio de Janeiro (COPPE/UFRJ) is typesetted in L^AT_EX. This demand motivated the creation of the COPPETEX project, which tries to facilitate and encourage the use of L^AT_EX within the COPPE/UFRJ scope.

The `coppe` document class is the main product of COPPETEX. It was designed to be clear and succinct. It enables the creation of dissertations, qualifying exams and thesis in a simple and automatic way. The main goal of the `coppe` class is to maintain authors strictly focused on text composition without worrying about margins sizes, line spacing, paper size, vertical and horizontal alignment, etc. The COPPETEX project comprehends also BibTEX and MakeIndex style files for creating lists of references, symbols and abbreviations. Although there aren't official guidelines to write qualifying exams, we provide this option just for convenience, as this exam is a requisite to obtain the DSc degree and, for some of the programs, the MSc degree.

In which follows, it is described the user interface of the `coppe` class. Some details about using the style files cited above are also given. We use the term *thesis* to generally refer to dissertation, qualifying exam, and thesis itself.

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copying, distributing or modifying the source code, among other acts covered by this license.

To see the full text of the GNU GPL license, go to the `COPYING` file attached to this package.

3 Support

We maintain a WhatsApp group where users can send questions, comments, and bugs to. More details can be found [here](#).

However, as the project maintenance was transferred to a new repository, bug reports, as well as new feature requests, should be directed to <https://github.com/COPPE-UFRJ/CoppeTeX>.

4 User interface

`\frontmatter` A thesis to be approved by the Academic Registry at COPPE/UFRJ must contain three-parts: *front*, *main* and *back* matters (COPPE/UFRJ Academic Registry guidelines). Each one of these parts is started by calling its corresponding macro `\frontmatter`, `\mainmatter` or `\backmatter`. The front matter of a thesis consists of front cover and face, cataloging page, dedication, acknowledgments, abstracts, table of contents, and lists of tables, algorithms, symbols and abbreviations. The main matter is just composed by chapters, while the back matter usually consists of bibliographic references, appendices and index.

You must invoke the `\frontmatter` macro immediately after the `\maketitle` one. The `\mainmatter` command comes right before the first chapter, and `\backmatter` must be typed before the list of references.

Cover (capa)

The ABNT cover suggested in the manual's Annex A. The `\maketitle` command builds it automatically, ahead of the title page: the university, the full COPPE name, and the graduate program (taken from `\department`), followed by the author, the title, and the city and year, all centred and in uppercase. The UFRJ and COPPE logos head the page. The capa is not counted.

Front face

The front face is unnumbered. There, it is not allowed to use hyphenation (COPPE/UFRJ Academic Registry guidelines). It is constructed by calling `\maketitle`. Next, it is described the commands used to enter the information required to create it.

`\author` The `\author` command was redefined. Here, it takes two arguments: the author's first names and surname, e.g., `\author{First Names}{Surname}`. The words should be typed with only first letters in uppercase.

`\title` The macros `\title` and `\foreigntitle` are used to enter the titles of your monograph in Brazilian Portuguese and English respectively (the two built-in languages of CoppeTeX 4.x). For documents whose main language is neither

brazilian nor english – e.g. spanish, french, italian, or a user-supplied language –, also call `\titlein{<lang>}{<text>}` to register the title in that language; the cover and folha de rosto print whichever title matches the main language declared in the class option. See the section “Multilingual support” below for the complete picture. The `babel` package is loaded automatically by `coppe.cls`, so you do not need to load it yourself.

`\advisor` Every COPPE student is coordinated by at least one advisor. M.Sc. and D.Sc. students can have at most 2 and 3 advisors, respectively. Their names must be provided by issuing the command `\advisor` as below:

```
\advisor{Title}{Advisor's Name}{Surname}{Degree}
\advisor{Title}{Second Advisor's Name}{Surname}{Degree}
\advisor{Title}{Third Advisor's Name}{Surname}{Degree}
```

The advisors are not necessarily members of the thesis examination board. Thus, it is required to enter the names of all examiners using the `\examiner` macro. The examiners' names are entered differently:

```
\examiner{Title}{First Examiner's Name Surname}{Degree}
\examiner{Title}{Second Examiner's Name Surname}{Degree}
...
\examiner{Title}{N-th Examiner's Name Surname}{Degree}
```

Remember that all names must be given before calling `\maketitle`.

`\department` The Alberto Luiz Coimbra institute is divided into 12 academic units: Biomedical Engineering (PEB), Civil Engineering (PEC), Electrical Engineering (PEE), Mechanical Engineering (PEM), Metallurgical and Materials Science Engineering (PEMM), Nuclear Engineering (PEN), Ocean Engineering (PENO), Energy Planning (PPE), Production Engineering (PEP), Chemical Engineering (PEQ), Systems Engineering and Computer Science (PESC), and Transportation Engineering (PET). You must specify your department using one of the above abbreviations, e.g., `\department{PEC}`.

`\date` This macro is used to set the month and year of defense. This information is required to create the front face, cataloging details page and abstracts. For example, October 2007 should be entered as `\date{10}{2007}`.

`\keyword` The keywords should describe the concentration areas of your work. You must provide them as follows:

```
\keyword{First Keyword}
\keyword{Second Keyword}
...
\keyword{N-th Keyword}
```

Usually, six words are enough.

Cataloging details

This page contains cataloging information useful for librarians. Fortunately, it is automatically generated from the data you entered at the time you call `\maketitle`. It is not needed in qualifying exams, though.

Dedication (optional)

`\dedication` This macro was added for convenience. The input text is placed at the right bottom of a blank page. It is emphasized and in normal size.

Abstracts

`abstract (env.)` As stated by the COPPE/UFRJ Academic Registry, abstracts must be in one page each, with at most 250 words. We recommended that they should be only one paragraph long. They must be defined inside the environments `abstract` and `foreignabstract`.

Lists of symbols and abbreviations (optional)

`\abbrev` The lists of symbols and abbreviations are optional, although highly recommended.
`\syml` It is a good practice to define a symbol/abbreviation in its first occurrence in the text. To define a symbol use `\syml[alphabetic symbol]{Symbol}{Symbol Definition}`, and for abbreviations `\abbrev[alphabetic symbol]{Abbreviation}{Abbreviation Definition}`. These commands are called *dummy*, since they don't output anything at the place they are executed, just an entry in the correspondent list.
`\makeloabbreviations` These lists are lexicographically sorted by using the MakeIndex program, which
`\makelosymbols` is part of any L^AT_EX implementation. For `\syml`, if the optional parameter is
`\printloabbreviations` provided, it will used as sort key. This was later, in 2024, implemented also for
`\printlosymbols` `\abbrev`, otherwise `Symbol`, or `Abbreviation` will be used as sort key, what can result in an undesirable order if it contains L^AT_EX commands, mathematical symbols, or mix of uppercase and lowercase. MakeIndex needs two commands to create a final sorted list: one which generates a list of entries and the other that indicates the position where the list will be printed out. To generate the lists of symbols and abbreviations, the `coppe` class provides the commands `\makeloabbreviations` and `\makelosymbols`, respectively. They must be called in the document preamble. The commands `\printlosymbols` and `\printloabbreviations` have to be invoked at the point where you want these lists appear, e.g., following the list of tables as showed in the example. Once you call `latex`, it will be created two files with extensions `abx` and `syx`, which contain MakeIndex input data. They must be processed with `makeindex` in order to get the lists correctly produced, redirecting the output to files with extension `lab` and `los` respectively:

```
makeindex -s coppe.ist -o example.lab example.abx
makeindex -s coppe.ist -o example.los example.syx
```

Note the `-s` option for specifying the style `coppe.ist`. Now, rerun `latex` twice to get the references solved and you are done.

References

COPPE_T_EX produces citations and the reference list with `biblatex` and the `biber` backend, through its own bibliography style implementing the post-2020 ABNT NBR 6023 rules. The class loads this style automatically: there is no `\bibliographystyle` to set and no Bib_T_EX `.bst` file to choose (the former `coppe-plain` and `coppe-unsrt` Bib_T_EX styles have been retired).

By default, citations follow the author–date style, e.g., “Chartier (2002)”. Passing the `numbers` class option switches to the numeric style, in which citations are bracketed numbers and the reference list is numbered in order of citation (unsorted).

Declare your bibliography database in the preamble with `\addbibresource` and print the list where it should appear with `\printbibliography`:

```
\addbibresource{example.bib}    % in the preamble
...
\printbibliography              % where the list goes
```

Entry types and fields may be written in English (`@book`, `title`, `author`) or with their unaccented Portuguese synonyms (`@livro`, `titulo`, `autor`).

Run in sequence `LATEX`, `biber`, and twice again `LATEX` to resolve the citations and references. The style is `natbib` compatible (via the `biblatex natbib=true` option), so you may freely issue `\citet` and `\citep`, as well as the other `natbib` commands.

Reference entry types

Every ABNT document category maps to a `biblatex` entry type, each with an unaccented Portuguese synonym (a worked example of every type is in the sample manuscript’s reference list):

Academic `@book` (`@livro`); `@article` (`@artigo`); `@artigodejornal` (newspaper); `@incollection` (`@partedelivro`); `@inbook` (`@emlivro`); `@inreference` (`@verbete`); `@proceedings` (`@anais`); `@inproceedings` (`@trabalhodeevento`); `@thesis` (`@tese/@dissertacao`, with the presets `@dissertacaomestrado`, `@tesedoutorado`, `@teseph`, `@tcc`); `@report` (`@relatorio`); `@periodical` (`@periodico`); `@manual`; `@misc` (`@diverso`); `@online`.

Special (NBR 6023 §4.2.5–4.2.14) `@patent` (`@patente`); `@legislation` (`@legislacao/@atonormat`); `@jurisdiction` (`@jurisprudencia`, with a `relator` field); `@legal` (`@documentojuridico`); `@standard` (`@norma`); `@music` (`@partitura`); `@map` (`@mapa`); `@image` (`@iconografico`); `@artwork` (`@obradearte`); `@video` (`@audiovisual`); `@movie` (`@filme`); `@letter` (`@correspondencia`); `@dataset` (`@conjuntodedados`).

Games and TV `@game` (`@jogo`); `@boardgame` (`@jogodetabuleiro`); `@videogame` (`@jogoeletronico`); `@tvshow` (`@programadetv`).

When an entry has no `author` or `editor`, it is entered by title, with the first significant word (and a leading article) in capitals.

Appendix and Annex (Optional)

`\appendix` Appendices and annexes are optional chapters that are part of the back matter.

`\annex` The `\appendix` command is a standard `LATEX` command used before all appendices. `COPPETEX` introduces the `\annex` command, which should be used only in the back matter (i.e., after the `\backmatter` command) and only after all existing `\appendix` chapters. This restriction is due to implementation constraints.

Therefore, the order for backmatter is:

```
\backmatter
\printbibliography
\appendix
\chapter{An Appendix}
\chapter{Another Appendix}
\annex
\chapter{First Annex}
\chapter{Second Annex}
```

Annex was introduced in v3.5

Final colophon page (`\coppetexfinalpage`)

`\coppetexfinalpage` Optional closing page modelled on the printer’s *colophon*: it records, in Portuguese, the tooling and edition used to typeset the thesis. Call it once, after every other piece of content, immediately before `\end{document}`. The macro

- forces a verso (even) page, so in a two-sided print the colophon sits opposite the rear cover;
- types the body at the foot of the page (`\vfill`);
- renders the text in Portuguese unconditionally (Brazilian `babel`), even when the thesis main language is English, Spanish, French or Italian — the colophon is institutional metadata, not body content.

Three slot macros are user-overridable. Redefine them with `\renewcommand` before `\coppetexfinalpage` if the document deviates from the defaults:

`\coppefinalengine` the T_EX engine identifier. Default: auto-detected via `iftex` (`\hologo{pdfLaTeX}`, `\hologo{LuaLaTeX}` or `\hologo{XeLaTeX}`).

`\coppefinalfont` the text body font. Default: *Computer Modern (padrão L^AT_EX)*. Redefine if you loaded `lmodern`, `mathptmx`, an OpenType math font, etc.

`\coppefinalmanual` the reference manual title (typeset in italics). Default: the current COPPE/UFRJ norm. Redefine if your work follows an earlier edition.

The macro also prints the COPPE_TE_X class version (captured into `\coppe@version` at class-load time, so it survives any later package that may redefine the global `\fileversion`) and the GitHub repository URL, <https://github.com/COPPE-UFRJ/CoppeTeX>.

5 Class options

There are some options users can specify in order to customize the appearance of the output produced by the `coppe` class. These options can be passed to `coppe` as follows: `\documentclass[option1, option2]{coppe}`. In which follows, we give a brief description of all supported options.

dsc, msc, dsccexam, msccexam The `coppe` class is able to produce thesis, dissertations, and qualifying exams, which are enabled by the `dsc`, `msc`, `msccexam`, and `dsccexam` options, respectively.

doublespacing The default line spacing is one-and-a-half. For enabling double spacing between lines, use the `doublespacing` option.

numbers The default citation style is the author–date scheme. For numbered citations, specify the `numbers` option to the `coppe` class; the reference list is then numbered in order of citation (unsorted). The matching `biblatex` style is selected automatically in either case.

brazilian, english, spanish, french, italian Pick the document’s *main* language. The class loads Babel with that language as the main one (the last in the Babel option list); inside the document `\selectlanguage{...}` and `\begin{otherlanguage}{...}` keep working as usual. The default, when no language option is given, is **brazilian**.

- **brazilian** and **english** are *built-in*: their strings ship inside `coppe.cls` and need no extra file.
- **spanish, french, italian** are shipped as *language packs*: two files per language (`coppe-lang-<lang>.def` and `<lang>-coppe.lbx`) that must sit next to `coppe.cls` (or in your project directory). They are auto-loaded by the class. Picking one of them as the main language without the matching pack files installed raises a clear class error.
- Any other Babel-known language can be plugged in by writing the same two files and either passing the language name as a class option or calling `\usecoppelanguage{<lang>}` in the preamble.

See the section “Multilingual support” below for the three-slot architecture, the `\titlein` / `\brazilianabstract` API, and how to create a new language pack.

5.1 Changing document identification

`\freeconfig` The user could *optionally* use the command `freeconfig` to modify the parameters that print the document identification. The command `freeconfig` needs all those parameters, which are degree initials, degree name, title, foreign title, local doctype, and foreign doctype as in the following example:

```
\freeconfig{Dr.}{Philosophiae Doctor}{PhD}{Doutor}{Dissertation}{Tese}
```

5.2 Multilingual support

CoppeTeX 4.x reserves three fixed language slots:

main the language of the body, captions, TOC labels and the `abstract` environment. Picked by the class option (**brazilian** by default; one of **english, spanish, french, italian** also accepted out of the box; any other Babel language possible with a user-supplied pack).

foreign the language of the `foreignabstract` environment. `english` by default for every main language; switches automatically to `brazilian` when `main = english`.

third optional. Loaded on demand with `\usecoppelanguage{<lang>}` in the preamble, useful when the body quotes or cites in a third language (e.g. a Spanish-main thesis citing a French source). Does not have a class option of its own.

`\titlein`
`brazilianabstract (env.)`
`\usecoppelanguage`

Two pieces of user-facing API come with the multilingual model:

- `\titlein{<lang>}{<text>}` records the title in any language. `\title{...}` still writes the Brazilian-Portuguese title and `\foreigntitle{...}` still writes the English one (back-compat); for a main language outside the `brazilian/english` pair, also call `\titlein{<lang>}{...}`. The cover and folha de rosto print whichever title matches `\coppe@mainlang`.
- The `brazilianabstract` environment is the optional third abstract used when the main language is neither `brazilian` nor `english` – typically a Spanish/French/Italian thesis that also wants a Portuguese resumo for UFRJ examiners. It wraps its body in `\begin{otherlanguage}{brazilian}` and uses the Portuguese template strings. A document with `main = brazilian` can simply keep using `abstract`; `brazilianabstract` is redundant then.
- `\usecoppelanguage{<lang>}` loads an extra language pack in the preamble without making it the main or foreign language.

Institutional names stay Portuguese

The cover, folha de rosto and ficha catalográfica are a Brazilian institutional template (NBR 14724 + COPPE manual). The strings `universityname`, `cityname`, `statename`, `countryname`, the departments declared with `\department` and the surrounding “apresentada/submetida ao...” wording are deliberately kept in Portuguese for every main language. Only the title (printed via `\coppe@selecttitle`) follows the main language.

Writing a new language pack

A language pack is just two files dropped next to `coppe.cls` (or anywhere on the `TEXINPUTS` path):

`coppe-lang-<lang>.def` class-level strings. Populate the dispatcher table with `\copperdefstring{<lang>}{<key>}{<value>}` for every key already populated for `brazilian/english` in `coppe.cls` (appendix, annex, frame, listframe, source, program, listprogram, references, in:, art, platform, directedby, producedby, listabbreviation, listsymbol, glossary, advisor, advisors, dept, approvedmale, approvedfemale, universityname, cityname, statename, countryname, monthname1...monthname12, algorithm2eopt, babelname). Finish with

```
\AtBeginDocument{\DeclareLanguageMapping{<lang>}{<lang>-coppe}}
```


so the `biblatex` mapping is registered after the bibliography engine is loaded by `coppe.cls` (which happens *after* the language pack is read).

`<lang>-coppe.lbx` `biblatex` localization strings. `\InheritBibliographyExtras{<lang>}` then `\DeclareBibliographyStrings` for the keys `availablefrom`, `urlseen`, `in`, `editor`, `editors`, `depositor`, `coppescale`, `mscdiss`, `dscthesi`, `phdthesis`, `tcc`. The shipped `spanish-coppe.lbx`, `french-coppe.lbx` and `italian-coppe.lbx` are the reference templates to copy from.

Tip: the `algorithm2eopt` key holds whatever `algorithm2e` calls the language. Mind the spelling: `algorithm2e` ships `portuguese`, `english`, `spanish`, `french` and `italiano` – not `italian`.

Example: a Spanish-main thesis

```
\documentclass[spanish,dsc]{coppe}
\addbibresource{example.bib}
\begin{document}
  \title{T\`i tulo em portugu\~es}      % Brazilian abstract slot
  \foreigntitle{English title}          % English abstract slot
  \titlein{spanish}{T\`i tulo en espa\~nol} % cover/folha
  \author{Juan}{P\`erez}
  \advisor{Prof.}{Mar\`a}{Garc\`a}{D.Sc.}
  \department{PESC}\date{01}{2026}
  \keyword{multilenguaje}
  \maketitle
  \frontmatter
  \begin{abstract}      Resumen principal en espa\~nol. \end{abstract}
  \begin{foreignabstract} English foreign abstract. \end{foreignabstract}
  \begin{brazilianabstract} Resumo em portugu\~es para a banca. \end{brazilianabstract}
  \tableofcontents
  \mainmatter
  \chapter{Introducci\`on} Cuerpo en espa\~nol\dots
\end{document}
```

6 Files in your project folder

In a normal install where CoppeTeX is part of your `TEXINPUTS` path (e.g. inside the local `texmf` tree of MiKTeX or TeX Live), you do not need to think about this list at all — the files travel with the class. This section is for two situations where you do:

- shipping the class together with your manuscript (a self-contained submission package); or
- installing CoppeTeX by hand, dropping the files next to `coppe.cls` in the project directory.

Always required

These files must be findable at every compilation, regardless of the main language or any class option:

`coppe.cls` the document class itself.

`coppe.dbx` custom `biblatex` datamodel (declares the ABNT-specific entry types `@game`, `@videogame`, `@boardgame`, `@tvshow` and `@map`, plus a few custom fields).

`coppe.bbx` and `coppe.cbx` `biblatex` bibliography and citation styles for the author–date scheme (the default).

`brazilian-coppe.lbx` and `english-coppe.lbx` `biblatex` localisation tables for Portuguese and English. Both are required *even when neither language is the main one*, because `coppe.bbx` registers

```
\DeclareLanguageMapping{brazilian}{brazilian-coppe}
\DeclareLanguageMapping{english}{english-coppe}
```

unconditionally, so `biblatex` tries to read both LBX files at every compilation.

Conditional on class options

`coppe-numeric.bbx` and `coppe-numeric.cbx` only when the `numbers` class option is given (`\documentclass[numbers]{coppe}`). Without it, `biblatex` never tries to read them.

`coppe.ist` only when the glossary, list-of-symbols, or list-of-abbreviations features are used. It is the `makeindex` style file invoked by the `makeindex -s coppe.ist ...` step described in subsection *Lists of symbols and abbreviations* above.

Conditional on the main language

The main language is chosen by the matching class option (`brazilian`, `english`, `spanish`, `french`, `italian`). The two built-in languages ship their strings inside `coppe.cls` and so need no extra file. The three language packs add two files each — one class-level (`coppe-lang-<lang>.def`) and one `biblatex`-level (`<lang>-coppe.lbx`):

Main language	Extra files needed
<code>brazilian</code> (default)	— (built into <code>coppe.cls</code>)
<code>english</code>	— (built into <code>coppe.cls</code>)
<code>spanish</code>	<code>coppe-lang-spanish.def</code> , <code>spanish-coppe.lbx</code>
<code>french</code>	<code>coppe-lang-french.def</code> , <code>french-coppe.lbx</code>
<code>italian</code>	<code>coppe-lang-italian.def</code> , <code>italian-coppe.lbx</code>

Loading a third language at preamble time via `\usecoppelanguage{<lang>}` brings in the same pair of files for that language. A user-supplied language pack follows the same scheme: drop `coppe-lang-<lang>.def` and `<lang>-coppe.lbx` next to `coppe.cls`.

Cover assets

The institutional title page produced by `\maketitle` needs two PDF logo files at compile time:

`ufrj-logo.pdf` and `coppe-logo.pdf` the UFRJ shield and the COPPE/UFRJ wordmark. Both ship with the CoppeTeX distribution; they must be findable from the project directory (or on `TEXINPUTS`).

7 Required L^AT_EX packages

The class is built on the standard book class (`12pt`, `a4paper`, `twoside`) and brings in the stock CTAN packages listed below with `\RequirePackage`. On a full T_EX Live or MiK_TE_X install they are all present already, so this list matters only for a minimal install — or when you simply want to know which commands are already in scope. You do *not* need to `\usepackage` any of these yourself; every command and environment they provide is available in your manuscript. Options in brackets are the ones the class passes.

Engine and encoding

`iftex` detects the running engine, so the class can branch its setup between pdfT_EX and the Unicode engines (LuaT_EX/XeT_EX).

`inputenc` (`utf8`) UTF-8 input; loaded *only under pdfT_EX* (the Unicode engines read UTF-8 natively).

`fontenc` (`T1`) T1 font encoding, for correctly composed accented glyphs and hyphenation.

Mathematics and symbols

`amsmath` the AMS mathematics environments.

`amssymb` the AMS symbol fonts; loaded *only under pdfT_EX*. Under a Unicode engine load `unicode-math` (or `amssymb`) yourself — the class warns at end of document if neither is present.

Page layout and spacing

`geometry` A4 two-side margins, binding offset and header height.

`fancyhdr` running headers and footers.

`titlesec` chapter and section title formatting.

`setspace` one-and-a-half line spacing (double with the `doublespacing` option).

Lists, tables and rules

`enumitem` customisable list spacing and labels.

`tabularx` tables with automatic-width columns.

`booktabs` professional horizontal rules (`\toprule`, `\midrule`, `\bottomrule`).

`longtable` tables that break across pages.

Floats, graphics and captions

`graphicx` image inclusion.

`float` float placement control (e.g. the `[H]` specifier).

`caption` caption typography (the ABNT caption-on-top layout).

`subcaption` subfigures and subtables.

Code, verbatim and algorithms

`listings` source-code listings.

`fancyvrb` extended verbatim.

`algorithm2e` (`linesnumbered`, `lined`, `algochapter`, `ruled`) typeset algorithms; the natural-language option tracks the main language.

Bibliography, quotations and language

`babel` multilingual typesetting; the language options are computed from the main and foreign language slots.

`csquotes` context-sensitive quotation marks (the recommended companion of `biblatex`).

`biblatex` (`backend=biber`) the bibliography engine and the ABNT `coppe` styles (`coppe` or, with the `numbers` option, `coppe-numeric`). The backend is `biber`, not `bibTeX`.

Hyperlinks

`hyperref` clickable cross-references, PDF bookmarks and document metadata.

Low-level utilities

`etoolbox` e-TeX programming toolkit.

`ltxcmds` low-level command helpers (`\ltx@ifpackageloaded`, ...).

`ifthen` the `\ifthenelse` conditional.

`hyphenat` hyphenation control (`\nohyphens`, used on the cover).

`lastpage` `\pageref{LastPage}` for the cataloging page count.

hologo T_EX-family logos (`\hologo{LaTeX}, \dots`).

xcolor colour support (used by `listings` and `hyperref`).

8 Quick, useful tips

Pictures. The default picture format of L^AT_EX is the Encapsulated PostScript (EPS). If you use pdfL^AT_EX, the default format becomes the PDF, but you can equally load PNG files. For such, you must enter the name of your image file without extension, e.g., `\includegraphics{filename}`, and pdf_lat_ex will firstly look for a file called `filename.pdf` and after for file `filename.png`. For producing high quality pictures with embedded fonts we recommend the Ipe drawing software available [here](#).

Fonts. The default font in L^AT_EX is the Computer Modern. If you would like to try its enhanced version, consider using the `lmodern` package. To use Times, it is recommended to load the package `mathptmx`, rather than the deprecated `times`. There is also an enhanced Times version available with the `tgtermes` package. You can still use the Arial font face with the package `uarial`.

Hyperref. When working with PDF's, there is the possibility to add extra information to the file as the author's name, document title, subject, keywords, etc. This is easily done with the `hyperref` package, which is loaded by the `coppe` class itself (since v4.0): the user must *not* load it again in the manuscript preamble. Author, title, subject and keyword metadata are filled in automatically by `\maketitle`.

Math (amssymb vs. unicode-math). The class always loads `amsmath`. The companion AMS symbol package `amssymb`, however, is loaded *only* under pdfL^AT_EX: it is a legacy 8-bit package built around the `msam/msbm` Type 1 fonts and clashes with `unicode-math`, the modern OpenType-math package used on Unicode engines (LuaL^AT_EX, XeL^AT_EX). On those engines the class skips `amssymb` and schedules an *end-of-document* check: if by then the author has loaded neither `unicode-math` nor `amssymb` (which they could have loaded by hand, though it is discouraged), a `ClassWarning` reminds them to add

```
\usepackage{unicode-math}
\setmathfont{Latin Modern Math} % or STIX Two Math, ...
```

to the manuscript preamble. The class deliberately does *not* auto-load `unicode-math`, because picking an OpenType math font is a typographic choice that belongs to the author. **Never load `amssymb` together with `unicode-math`:** it would re-establish the 8-bit AMS fonts on top of the OpenType math font and split symbols across two incompatible encodings.

Printing. To get your work correctly printed, you must ensure that any page scaling option (e.g., fit or shrink to printable area) isn't enabled. This kind of option often comes in print dialogs of document visualization softwares.

longquote Quotation To quote text larger than three lines, according to ABNT, you must increase the left margin to 4 cm, do not use quotation marks, and use a smaller font. The `coppe` class provides the `longquote` environment to easily make these adjustments.

9 A simple example

```

1 \*example
2 \documentclass[dsc]{coppe}
3
4 \usepackage{rotating}          % rodando coisas, como tabelas
5 \usepackage[most]{tcolorbox} % caixas de texto
6 \usepackage{tikz}             % gráficos vetoriais (o tcolorbox já o traz)
7 % Sob LuaLaTeX/XeLaTeX, descomente as duas linhas a seguir para obter os
8 % símbolos AMS (\mathbb, \hbar, \varnothing, ...) via unicode-math. Sob
9 % pdfLaTeX, a classe já carrega amssymb automaticamente; deixe comentado.
10 \usepackage{unicode-math}
11 %\setmathfont{Latin Modern Math} % ou STIX Two Math, TeX Gyre Termes Math...
12 \DeclareMathOperator{\RMSE}{RMSE} % operadores próprios (cap. de matemática)
13 \DeclareMathOperator{\diag}{diag}
14 % A classe coppe (v4.0) já carrega: amsmath, booktabs, longtable, subcaption,
15 % hyperref, biblatex, listings, fancyvrb, algorithm2e -- não recarregue esses
16 % pacotes aqui (em particular, hyperref deve ficar a cargo da classe para que
17 % a ordem com setspace e biblatex permaneça correta).
18
19 \addbibresource{example.bib}% base de referências (biblatex/biber)
20
21 \makelossymbols
22 \makeloabbreviations
23 % Algumas coisas opcionais que não cabem no coppetex.sty
24 \usepackage[indentfirst] % se quiser indentar o primeiro paragrafo das seções
25 %\renewcommand{\familydefault}{\sfdefault} % para usar fontes SEM serifa no default
26 \renewcommand{\familydefault}{\rmdefault} % para usar fontes COM serifa no default
27
28
29 \begin{document}
30   \title{Título da Tese}
31   \foreigntitle{Thesis Title}
32   \author{Nome do Autor}{Sobrenome}
33   \advisor{Prof.}{Nome do Primeiro Orientador}{Sobrenome}{D.Sc.}
34   \advisor{Prof.}{Nome do Segundo Orientador}{Sobrenome}{Ph.D.}
35   % A classe aceita de um a tres \advisor; se houver apenas um, o rotulo na
36   % capa e na folha de rosto sai como "Orientador" (singular), com dois ou
37   % mais sai como "Orientadores" (plural). Para o caso 1/3 orientadores,
38   % veja tests/test_brazilian_one_advisor.tex e tests/test_brazilian_three_advisors.tex.
39
40   \examiner{Prof.}{Nome do Primeiro Examinador Sobrenome}{D.Sc.}
41   \examiner{Prof.}{Nome do Segundo Examinador Sobrenome}{Ph.D.}
42   \examiner{Prof.}{Nome do Terceiro Examinador Sobrenome}{D.Sc.}
43   \examiner{Prof.}{Nome do Quarto Examinador Sobrenome}{Ph.D.}
44   \examiner{Prof.}{Nome do Quinto Examinador Sobrenome}{Ph.D.}
45   \department{PESC}
46   \date{06}{2026}

```

```

47
48 \keyword{Primeira palavra-chave}
49 \keyword{Segunda palavra-chave}
50 \keyword{Terceira palavra-chave}
51
52 \maketitle
53
54 \frontmatter
55 \dedication{A alguém cujo valor é digno desta dedicatória.}
56
57 \chapter*{Agradecimentos}
58
59 Gostaria de agradecer a todos os que contribu\iram para este trabalho:
60 orientadores, colegas, familiares e \as ag\~encias de fomento. Lorem ipsum
61 dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt
62 ut labore et dolore magna aliqua.
63
64 \begin{abstract}
65
66 Apresenta-se, nesta tese, um exemplo de resumo. Lorem ipsum dolor sit amet,
67 consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et
68 dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation
69 ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor
70 in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
71 pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui
72 officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis
73 iste natus error sit voluptatem accusantium doloremque laudantium, totam rem
74 aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae
75 vitae dicta sunt explicabo, nemo enim ipsam voluptatem quia voluptas sit
76 aspernatur aut odit aut fugit.
77
78 \end{abstract}
79
80 \begin{foreignabstract}
81
82 In this work, we present an example abstract. Lorem ipsum dolor sit amet,
83 consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et
84 dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation
85 ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor
86 in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
87 pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui
88 officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis
89 iste natus error sit voluptatem accusantium doloremque laudantium, totam rem
90 aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae
91 vitae dicta sunt explicabo, nemo enim ipsam voluptatem quia voluptas sit
92 aspernatur aut odit aut fugit.
93
94 \end{foreignabstract}
95
96 % Ordem pre-textual (NBR 14724/6027, R55): listas (ilustracoes, tabelas,
97 % abreviaturas e siglas, simbolos) ANTES do sumario; sumario por ultimo.
98 \listoffigures
99 \listoftables
100 \listofquadros

```

```

101 \listofprogramas
102 \listofalgorithms
103 \printloabbreviations
104 \printlosymbols
105 \tableofcontents
106
107 \mainmatter
108 \chapter{Introdução}
109
110 Este é um documento exemplo para o uso da classe CoppeTeX, destinado a ajudar os alunos da
111
112 A classe \verb|coppe| foi criada originalmente por Vicente Helano e George Ainsworth, poré
113 Em 2026, Geraldo Xexéo, com forte ajuda das ferramentas Claude 4.7 Opus e Max e 4.8 Opus M
114
115 Provavelmente Geraldo Xexéo, professor do Programa de Engenharia de Sistemas e Computação
116
117 A versão mais atual dessa classe é mantida no GitHub, no repositório \url{https://github.c
118
119 Esse documento segue a norma de formatação de teses e dissertações da COPPE. Ele também po
120
121 Há um grupo de WhatsApp para perguntas e discussões do estilo: \href{https://github.com/CO
122
123 Bugs devem ser reportados na \href{https://github.com/COPPE-UFRJ/CoppeTeX/issues}{página d
124
125 Esse documento é usado como exemplo de coisas que podem ser feitas. Ele está configurado p
126
127 É importante de notar que essa classe não foi construída sobre a classe \LaTeX \ para a A
128
129 Essa classe implementa citações ABNT modernas e vários tipos de entrada descritos na norma
130
131 Este documento não substitui, mas complementa, o documento que descreve a classe.
132
133
134
135 \chapter{Configurações Iniciais}
136
137 A primeira coisa a fazer é escolher o tipo de documento. Isso é feito como uma opção no c
138
139 Como pode ser visto nesse documento, muita coisa pode ser configurada, o que gerará o tra
140
141 Recomendo ler o documento “The \verb*|coppe| document class” para entender melhor todas
142
143 \section{Linguagem principal do texto}
144
145 Essa classe considera que o texto principal está em português e algumas partes específicas
146 \begin{itemize}
147 \item A opção \verb|english| deve ser usada no comando \verb|\documentclass|.
148 \item A bibliografia segue automaticamente o idioma do Babel (português ou inglês); não é p
149 \end{itemize}
150
151 A variação de linguagem, em inglês ou português apenas, já é suportada pela classe \CopeT
152
153
154 \section{Por que usar o \LaTeX}

```


155 Há uma grande discussão entre usuário de Word e \LaTeX, principalmente, quanto ao uso dess
156
157 Nós escolhemos o \LaTeX por alguns motivos: grande facilidade de seguir um estilo sem se p
158
159 As principais desvantagens são: idiossincrasias que podem gastar tempo, pouco controle sob
160
161
162 \section{Como e onde usar o \LaTeX}
163
164 Existem muitos tutoriais de \LaTeX, mas basicamente, em 2024, ele é usado em dois ambiente
165 \begin{enumerate}
166 \item Na sua máquina, instalando uma versão completa como o \hologo{MiKTeX}, típico do Win
167 \item Usar um ambiente na rede, como o Overleaf.
168 \end{enumerate}
169
170 Em todo caso, recomendo fortemente que, ao mesmo tempo, mantenha versões no Git e faça o b
171
172
173 \chapter{Algumas Regras da COPPE}
174
175 Todas abreviaturas e símbolos devem ser definida antes de utilizada. Isso é facilmente fei
176
177 É imprescindível definir os símbolos, tal como o
178 conjunto dos números reais $\mathbb{R}$ e o conjunto vazio $\emptyset$.
179 \syml{\mathbb{R}}{Conjunto dos números reais}
180 \syml{\emptyset}{Conjunto vazio}. Usamos esse exemplo aqui justamente para mostrar como
181
182 Para as listas de abreviaturas e símbolos funcionarem no Overleaf é necessário rodar o \ve
183
184 Como as listas de símbolos e de abreviaturas usamo o mesmo comando usado para criar índice
185
186 \section{Citações}
187 \label{sec:citacoes}
188
189 Citações curtas podem ser feitas \enquote{o comando quote} ou direto com ‘‘duas crases e do
190
191 \begin{longquote}
192 Um exemplo de citação longa nas regras da ABNT (4cm de recuo e fonte menor)
193 feita com o ambiente \verb*|longquote| The primary objective of this
194 investigation was to determine the feasibility of detecting corrosion in
195 aluminum Naval aircraft components with neutron radiographic interrogation
196 and the use of standard corrosion penetrameters. Secondary objectives
197 included the determination of the effect of object thickness on image quality,
198 the defining of minimum levels of detectability and a preliminary investigation
199 of a means whereby the degree of corrosion could be quantified with neutron
200 radiographic data. \citep{article-example}
201 \end{longquote}
202
203 Citações devem apontar as referências. Para isso, está disponível o ótimo pacote \verb*|na
204
205 Em todo caso, \textbf{deve se tomar enorme atenção com as citações, para evitar ocorrer em
206
207 Quando se cita uma obra por meio de outra (\textit{apud}), pela ABNT entra na lista de ref
208

```

209 \chapter{Floats}
210
211 Grande parte dos problemas de iniciantes, e veteranos, em \LaTeX é da localização dos \textbf
212
213 A regra geral de posicionamento é que uma figura ou quadro só pode aparecer a partir da mesma
214
215 \textbf{Pela norma atual (manual2024), a legenda} \verb|\caption| \textbf{de toda ilustração}
216
217 Quadros são ilustrações cujos dados são delimitados por linhas em \emph{todas} as bordas (ao
218
219
220 \section{Tabelas e Figuras Padrão}
221
222 Vamos ver uma tabela padrão, como a \autoref{tab:exemplo_numeros}.
223
224 \begin{table}[ht]
225 \centering % Centraliza a tabela
226 \caption{Exemplo de Tabela de Números}
227 \label{tab:exemplo_numeros}
228 \begin{tabular}{ccc} % Define a quantidade de colunas
229 \hline % Linha superior
230 \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\ \hline % Cabeçalhos
231 \hline % Linha média
232 1 & 2 & 3 \\ \hline % Primeira linha de dados
233 \hline
234 4 & 5 & 6 \\ \hline % Segunda linha de dados
235 \hline
236 7 & 8 & 9 \\ \hline % Terceira linha de dados
237 \hline
238 10 & 11 & 12 \\ \hline % Quarta linha de dados
239 \hline % Linha inferior
240 \end{tabular}
241 \fonte{Elabora\c{c}\~ao pr'opria.} % fonte obrigatoria, abaixo do float
242 \end{table}
243
244
245
246 Já a \autoref{fig:exemplo_figura} é uma figura padrão, com controle da largura.
247
248 \begin{figure}[ht]
249 \centering % Centraliza a figura
250 \caption{Exemplo de Figura com Legenda Acima} % Legenda em cima (norma atual)
251 \label{fig:exemplo_figura} % Etiqueta para referência cruzada
252 \includegraphics[width=0.5\textwidth]{coppe-logo.pdf} % Inclui a imagem com metade da largura
253 \source{COPPE/UFRJ.} % fonte obrigatoria, abaixo da figura
254 \end{figure}
255
256 Quadros têm a mesma regra (legenda acima, fonte abaixo), mas com linhas em todas as bordas.
257
258 \begin{quadro}[ht]
259 \centering
260 \caption{Exemplo de Quadro}
261 \label{qua:exemplo}
262 \begin{tabular}{|l|l|}

```

```

263 \hline
264 \textbf{Ilustração} & \textbf{Delimitação} \\
265 \hline
266 Tabela & Linhas apenas no topo e na base. \\
267 \hline
268 Quadro & Linhas em todas as bordas. \\
269 \hline
270 \end{tabular}
271 \fonte{Elabora\c{c}\~ao pr\'opria.}
272 \end{quadro}
273
274
275
276
277
278 \section{Tabelas mais elegantes}
279
280 Atualmente a tendência é usar tabelas mais leves, como \autoref{tab:exemplo_numerosbom}. Iss
281
282 \begin{table}[ht]
283 \centering % Centraliza a tabela
284 \caption{Exemplo de Tabela de Números mais elegantes}
285 \label{tab:exemplo_numerosbom}
286 \begin{tabular}{ccc} % Define a quantidade de colunas
287 \toprule % Linha superior
288 \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\ % Cabeçalhos
289 \midrule % Linha média
290 1 & 2 & 3 \\ % Primeira linha de dados
291 4 & 5 & 6 \\ % Segunda linha de dados
292 7 & 8 & 9 \\ % Terceira linha de dados
293 10 & 11 & 12 \\ % Quarta linha de dados
294 \bottomrule % Linha inferior
295 \end{tabular}
296 \fonte{Elabora\c{c}\~ao pr\'opria.}
297 \end{table}
298
299 \section{Tabelas Longas ou Largas}
300
301 Se sua tabela é muito longa ou larga, existem várias opções.
302 \begin{itemize}
303 \item alterar o tamanho da letra
304 \item Usar o longtable
305 \item rodar a tabela, fazendo ela em \textit{landscape}
306 \item fazer a tabela dentro de um minibox
307 \end{itemize}
308
309
310 \subsection{Tabelas largas demais}
311
312 É comum em teses que as tabelas sejam largas demais. Há várias formas de resolver isso.
313
314 A \autoref{tab:tabela_largafns} é larga demais, e nela isso é resolvido diminuindo a fonte p
315
316 \begin{table}[ht]

```

```

317 \centering % Centraliza a tabela
318 \caption{Exemplo de Tabela Larga com Fonte Menor}
319 \label{tab:tabela_largafns}
320 \footnotesize % Aplica uma fonte menor para a tabela
321 \begin{tabular}{ccccccc} % Aumente o número de colunas conforme necessário
322 \toprule
323 \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} & \textbf{Coluna 4} & \textbf{Coluna 5} & \textbf{Coluna 6} & \textbf{Coluna 7} \\
324 \midrule
325 Dado 1.1 & Dado 1.2 & Dado 1.3 & Dado 1.4 & Dado 1.5 & Dado 1.6 & Dado 1.7 & Dado 1.8 \\
326 Dado 2.1 & Dado 2.2 & Dado 2.3 & Dado 2.4 & Dado 2.5 & Dado 2.6 & Dado 2.7 & Dado 2.8 \\
327 Dado 3.1 & Dado 3.2 & Dado 3.3 & Dado 3.4 & Dado 3.5 & Dado 3.6 & Dado 3.7 & Dado 3.8 \\
328 \bottomrule
329 \end{tabular}
330 \fonte{Elabora\c{c}\~ao pr\'opria.}
331 \end{table}
332
333 O comando \verb|\resizebox{width}{height}{content}| permite ajustar o tamanho de qualquer conteúdo.
334
335 \begin{table}[ht]
336 \centering
337 \caption{Exemplo de Tabela Redimensionada}
338 \label{tab:examplerb}
339 \resizebox{\textwidth}{!}{%
340 \begin{tabular}{llll}
341 \toprule
342 Coluna 1 & Coluna 2 & Coluna 3 & Coluna 4 \\
343 \midrule
344 Dados 1 & Dados 2 & Dados 3 & Dados 4 \\
345 Dados 5 & Dados 6 & Dados 7 & Dados 8 \\
346 \bottomrule
347 \end{tabular}%
348 }
349 \fonte{Elabora\c{c}\~ao pr\'opria.}
350 \end{table}
351
352
353 Para rodar uma tabela muito larga em 90 graus no LaTeX, você pode usar o pacote \verb*|rotatble|.
354
355 Aqui está um exemplo de como usar o ambiente \verb*|sidewaystable| para girar uma tabela.
356
357 \begin{sidewaystable}
358 \centering
359 \caption{Sua Legenda Aqui}
360 \label{tab:sua_tabela}
361 \begin{tabular}{lll}
362 \toprule
363 Coluna 1 & Coluna 2 & Coluna 3 \\
364 \midrule
365 Item 1 & Item 2 & Item 3 \\
366 Item 4 & Item 5 & Item 6 \\
367 \bottomrule
368 \end{tabular}
369 \fonte{Elabora\c{c}\~ao pr\'opria.}
370 \end{sidewaystable}

```

```

371
372 Se a tabela for muito longa, o ambiente \verb|longtable| é o ideal. Ele fornece comandos par
373
374 % Exemplo de tabela longa que se estende por várias páginas
375 \begin{longtable}{|c|c|c|}
376 % primeiro cabeçalho (é o caption)
377 \caption{Exemplo de Tabela Longa}\label{tab:longa} \\
378 \hline \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\ \hline
379 \endfirsthead
380 % cabeçalho normal
381 \multicolumn{3}{c}%
382 {\table\thetable} -- continuação da página anterior} \\
383 \hline \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\ \hline
384 \endhead
385 % pé normal
386 \hline \multicolumn{3}{r|}{Continua na próxima página} \\ \hline
387 \endfoot
388 \hline
389 % último pé
390 \multicolumn{3}{r|}{Continua na próxima página}} \\
391 \hline \hline
392 \endlastfoot
393
394 % Conteúdo da tabela
395 1 & 2 & 3 \\
396 4 & 5 & 6 \\
397 1 & 2 & 3 \\
398 4 & 5 & 6 \\
399 1 & 2 & 3 \\
400 4 & 5 & 6 \\
401 1 & 2 & 3 \\
402 4 & 5 & 6 \\
403 1 & 2 & 3 \\
404 4 & 5 & 6 \\
405 1 & 2 & 3 \\
406 4 & 5 & 6 \\
407 1 & 2 & 3 \\
408 4 & 5 & 6 \\
409 1 & 2 & 3 \\
410 4 & 5 & 6 \\
411 1 & 2 & 3 \\
412 1 & 2 & 3 \\
413 4 & 5 & 6 \\
414 1 & 2 & 3 \\
415 4 & 5 & 6 \\
416 1 & 2 & 3 \\
417 4 & 5 & 6 \\
418 1 & 2 & 3 \\
419 4 & 5 & 6 \\
420 1 & 2 & 3 \\
421 4 & 5 & 6 \\
422 1 & 2 & 3 \\
423 4 & 5 & 6 \\
424 1 & 2 & 3

```

425 4 & 5 & 6 \\
 426 1 & 2 & 3 \\
 427 4 & 5 & 6 \\
 428 1 & 2 & 3 \\
 429 4 & 5 & 6 \\
 430 1 & 2 & 3 \\
 431 4 & 5 & 6 \\
 432 1 & 2 & 3 \\
 433 4 & 5 & 6 \\
 434 1 & 2 & 3 \\
 435 4 & 5 & 6 \\1 & 2 & 3 \\
 436 4 & 5 & 6 \\
 437 1 & 2 & 3 \\
 438 4 & 5 & 6 \\
 439 1 & 2 & 3 \\
 440 1 & 2 & 3 \\
 441 4 & 5 & 6 \\
 442 1 & 2 & 3 \\
 443 4 & 5 & 6 \\
 444 1 & 2 & 3 \\
 445 4 & 5 & 6 \\
 446 1 & 2 & 3 \\
 447 4 & 5 & 6 \\
 448 1 & 2 & 3 \\
 449 4 & 5 & 6 \\
 450 1 & 2 & 3 \\
 451 4 & 5 & 6 \\
 452 1 & 2 & 3 \\
 453 4 & 5 & 6 \\
 454 1 & 2 & 3 \\
 455 4 & 5 & 6 \\
 456 1 & 2 & 3 \\
 457 4 & 5 & 6 \\
 458 1 & 2 & 3 \\
 459 4 & 5 & 6 \\
 460 1 & 2 & 3 \\
 461 4 & 5 & 6 \\
 462 1 & 2 & 3 \\
 463 4 & 5 & 6 \\
 464 1 & 2 & 3 \\
 465 4 & 5 & 6 \\
 466 1 & 2 & 3 \\
 467 4 & 5 & 6 \\
 468 1 & 2 & 3 \\
 469 4 & 5 & 6 \\
 470 1 & 2 & 3 \\
 471 4 & 5 & 6 \\
 472 1 & 2 & 3 \\
 473 4 & 5 & 6 \\
 474 1 & 2 & 3 \\
 475 4 & 5 & 6 \\
 476 1 & 2 & 3 \\
 477 4 & 5 & 6 \\
 478 1 & 2 & 3 \\

```

479 4 & 5 & 6 \\
480 1 & 2 & 3 \\
481 4 & 5 & 6 \\
482 1 & 2 & 3 \\
483 4 & 5 & 6 \\
484 1 & 2 & 3 \\
485 4 & 5 & 6 \\
486 1 & 2 & 3 \\
487 4 & 5 & 6 \\
488 1 & 2 & 3 \\
489 4 & 5 & 6 \\
490 1 & 2 & 3 \\
491 4 & 5 & 6 \\
492 1 & 2 & 3 \\
493 4 & 5 & 6 \\
494 1 & 2 & 3 \\
495 4 & 5 & 6 \\
496 1 & 2 & 3 \\
497 4 & 5 & 6 \\
498 1 & 2 & 3 \\
499 4 & 5 & 6 \\
500 1 & 2 & 3 \\
501 4 & 5 & 6 \\
502
503 % Repetir linhas semelhantes conforme necessário para estender a tabela por 3 páginas
504 \end{longtable}
505 \fonte{Elabora\c{c}\~ao pr\'opria.}% a longtable não é float: a fonte vem após o ambiente
506
507 \section{Subfiguras (subcaption)}
508
509 Para colocar várias imagens (ou tabelas) lado a lado, cada uma com sua própria legenda, use
510
511 \begin{figure}[ht]
512   \centering
513   \begin{subfigure}[b]{0.42\textwidth}
514     \centering\includegraphics[width=\linewidth]{example-image-a}
515     \caption{Primeira vista}\label{fig:sub-a}
516   \end{subfigure}
517   \hfill
518   \begin{subfigure}[b]{0.42\textwidth}
519     \centering\includegraphics[width=\linewidth]{example-image-b}
520     \caption{Segunda vista}\label{fig:sub-b}
521   \end{subfigure}
522   \caption{Figura com duas subfiguras}
523   \label{fig:subs}
524   \fonte{Elabora\c{c}\~ao pr\'opria.}
525 \end{figure}
526
527 Referencie a figura inteira com \verb|\autoref{fig:subs}| e cada parte com \verb|\autoref{fig:sub-a}|
528
529 \section{Imagens (graphicx)}
530
531 A classe já carrega o \verb|graphicx|. Insira imagens com \verb|\includegraphics|; as opções
532 \begin{itemize}

```

```

533 \item \verb|[width=0.8\textwidth]| ou \verb|[scale=0.5]| --- tamanho;
534 \item \verb|[angle=90]| --- rotação;
535 \item \verb|[trim=l b r t, clip]| --- recorte (corta as bordas \emph{esquerda, baixo, dire
536 \end{itemize}
537 Formatos aceitos com \hologo{pdfLaTeX}/\hologo{LuaLaTeX}: \texttt{pdf}, \texttt{png}, \texttt
538
539 \section{Introdução ao tikz}
540
541 0 \verb|tikz| (disponível também via \verb|tcolorbox|) desenha gráficos vetoriais no próprio
542
543 \begin{figure}[ht]
544 \centering
545 \begin{tikzpicture}
546 \node[draw, rounded corners] (a) at (0,0) {Início};
547 \node[draw, rounded corners] (b) at (4,0) {Fim};
548 \draw[->, thick] (a) -- (b) node[midway, above] {processo};
549 \end{tikzpicture}
550 \caption{Diagrama simples em tikz}
551 \label{fig:tikz}
552 \fonte{Elabora\c{c}\~ao pr\'opria.}
553 \end{figure}
554
555 Para gráficos de dados (curvas, barras), veja o \verb|pgfplots|, construído sobre o \verb|ti
556
557 \section{Float ou direto no texto?}
558
559 Colocar a tabela ou imagem dentro de \verb|figure|/\verb|table| (um \emph{float}) deixa o \L
560
561 \chapter{Using BibLaTeX}
562 \label{cap:biblatex}
563
564 Esta classe gera as referências com BibLaTeX e o \emph{backend} \texttt{biber}, no padrão
565
566 \textbf{É obrigatório compilar com \texttt{biber}} (e não com \hologo{BibTeX}). A sequênci
567
568 Os comandos do \texttt{natbib} (\verb|\citet|, \verb|\citep|) continuam disponíveis porque
569
570 \begin{table}[h]
571 \caption{Exemplos de cita\c{c}\~oes utilizando o comando padr\~ao
572 \texttt{\textbackslash cite} do \LaTeX\ e
573 o comando \texttt{\textbackslash citep},
574 disponível pela opção \texttt{natbib} do BibLaTeX.}
575 \label{tab:citation}
576 \centering
577 {\footnotesize
578 \begin{tabular}{|c|c|c|}
579 \hline
580 Tipo da Publicação & \verb|\cite| & \verb|\citep|\\
581 \hline
582 Livro & \cite{book-example} & \citep{book-example}\\
583 Artigo & \cite{article-example} & \citep{article-example}\\
584 Relatório & \cite{techreport-example} & \citep{techreport-example}\\
585 Relatório & \cite{techreport-exampleIn} & \citep{techreport-exampleIn}\\
586 Anais de Congresso & \cite{inproceedings-example} &

```



```

587 \citep{inproceedings-example}\\
588 Séries & \cite{incollection-example} & \citep{incollection-example}\\
589 Em Livro & \cite{inbook-example} & \citep{inbook-example}\\
590 Dissertação de mestrado & \cite{mastersthesis-example} &
591 \citep{mastersthesis-example}\\
592 Tese de doutorado & \cite{phdthesis-example} & \citep{phdthesis-example}\\
593 \hline
594 \end{tabular}}
595 \fonte{Elabora\c{c}\~ao pr\'opria.}
596 \end{table}
597
598 \begin{table}[h]
599 \caption{Exemplos de cita\c{c}\~oes utilizando o comando padr\~ao}
600 \texttt{\textbackslash cite} do \LaTeX\ e
601 o comando \texttt{\textbackslash citet},
602 disponível pela opção \texttt{\textbackslash natbib} do BibLaTeX. Além disso, usando o booktabs.}
603 \label{tab:citation1}
604 \centering
605 {\footnotesize
606 \begin{tabular}{ccc}
607 \toprule
608 Tipo da Publicação & \verb|\cite| & \verb|\citet|\\
609 \midrule
610 Livro & \cite{book-example} & \citet{book-example}\\
611 Artigo & \cite{article-example} & \citet{article-example}\\
612 Relatório & \cite{techreport-example} & \citet{techreport-example}\\
613 Relatório & \cite{techreport-exampleIn} & \citet{techreport-exampleIn}\\
614 Anais de Congresso & \cite{inproceedings-example} &
615 \citet{inproceedings-example}\\
616 Séries & \cite{incollection-example} & \citet{incollection-example}\\
617 Em Livro & \cite{inbook-example} & \citet{inbook-example}\\
618 Dissertação de mestrado & \cite{mastersthesis-example} &
619 \citet{mastersthesis-example}\\
620 Tese de doutorado & \cite{phdthesis-example} & \citet{phdthesis-example}\\
621 \bottomrule
622 \end{tabular}}
623 \fonte{Elabora\c{c}\~ao pr\'opria.}
624 \end{table}
625
626 \section{Comandos de citação}
627
628 Os principais comandos são: \verb|\cite| (referência simples), \verb|\citet| (autor por ex
629
630 \subsection{Parâmetro opcional: página e “tradução nossa”}
631
632 Todos os comandos aceitam um parâmetro opcional para a localização (página, seção) e obser
633
634 \subsection{Citação de citação (\emph{apud})}
635
636 Para citar uma obra através de outra, use \verb|\citetapud| e \verb|\citeapud| (demonstra
637
638 \subsection{Múltiplas bibliografias}
639
640 Para separar as referências (por tipo, capítulo ou tema), use o ambiente \verb|refsection|

```

```

641
642 \section{Tipos de entrada bibliográfica}
643
644 Cada tipo de documento da ABNT corresponde a um tipo de entrada do BibLaTeX, com um sinônimo
645
646 \subsection{Tipos acadêmicos principais}
647 \begin{itemize}
648   \item Livro / monografia --- \verb|@book| (\verb|@livro|)
649   \item Artigo de periódico --- \verb|@article| (\verb|@artigo|); matéria de jornal --- \verb|@journal|
650   \item Capítulo de livro --- \verb|@incollection| (\verb|@partedelivro|), \verb|@inbook|
651   \item Evento no todo (anais) --- \verb|@proceedings| (\verb|@anais|); trabalho em evento
652   \item Tese/dissertação --- \verb|@thesis| (\verb|@tese|, \verb|@dissertacao|); com tipo
653   \item Relatório --- \verb|@report| (\verb|@relatorio|, \verb|@relatoriotecnico|); periódico
654   \item Manual --- \verb|@manual|; diversos --- \verb|@misc| (\verb|@diverso|); página/sítio
655 \end{itemize}
656
657 \subsection{Tipos especiais (ABNT §4.2.5--4.2.14)}
658 \begin{itemize}
659   \item Patente --- \verb|@patent| (\verb|@patente|)
660   \item Legislação / ato normativo --- \verb|@legislation| (\verb|@legislacao|, \verb|@ato|)
661   \item Norma técnica --- \verb|@standard| (\verb|@norma|); partitura --- \verb|@music| (\verb|@musica|)
662   \item Documento cartográfico (mapa) --- \verb|@map| (\verb|@mapa|, \verb|@cartografico|)
663   \item Audiovisual --- \verb|@video| (\verb|@audiovisual|); filme --- \verb|@movie| (\verb|@filme|)
664 \end{itemize}
665
666 \subsection{Jogos e TV (tipos próprios da CoppeTeX)}
667 \begin{itemize}
668   \item Jogo --- \verb|@game| (\verb|@jogo|); de tabuleiro --- \verb|@boardgame| (\verb|@jogo|)
669   \item Programa de TV / série --- \verb|@tvshow| (\verb|@programadettv|, \verb|@serietv|)
670 \end{itemize}
671
672 Os campos também aceitam sinônimos em português (sem acentos): \verb|autor|, \verb|titulo|
673
674 \chapter{Alguns outros exemplo úteis}
675
676 \begin{tcolorbox}[title=Meu Textbox]
677 Este é o conteúdo do meu textbox. Você pode adicionar qualquer texto aqui, bem como incluir
678 \end{tcolorbox}
679
680 \begin{tcolorbox}
681 Este é o conteúdo do meu textbox sem título. Você pode adicionar qualquer texto aqui, bem como
682 \end{tcolorbox}
683
684 \begin{figure}[ht]
685   \centering
686   \begin{tikzpicture}
687     \node[anchor=south west,inner sep=0] (image) at (0,0) {\includegraphics[width=0.5\textwidth]{}}
688     \begin{scope}[x={(image.south east)},y={(image.north west)}]
689       % Definindo o textbox dentro da figura
690       \node[anchor=north west, text width=0.3\textwidth, fill=white, opacity=0.7, text=
691         \begin{tcolorbox}[colback=red!5!white,colframe=red!75!black,title=Textbox de
692           Este textbox fala sobre como inserir um textbox dentro de uma figura usando
693         \end{tcolorbox}
694       ];

```

```

695         \end{scope}
696     \end{tikzpicture}
697     \caption{Figura com Textbox}
698     \label{fig:figura_com_textbox1}
699     \fonte{Elabora\c{c}\~ao pr\'opria.}
700 \end{figure}
701
702
703 \begin{figure}[ht]
704     \centering
705     \begin{tcolorbox}
706 Este é o conteúdo do meu textbox sem título. Você pode adicionar qualquer texto aqui, bem co
707 \end{tcolorbox}
708     \caption{Figura com Textbox simples}
709     \label{fig:figura_com_textbox}
710     \fonte{Elabora\c{c}\~ao pr\'opria.}
711 \end{figure}
712
713 \section{Listagens de código e algoritmos}
714
715 O CoppeTeX traz embutido o suporte a listagens de programas (pacote \verb|listings|) e a alg
716
717 \begin{python}[caption={Exemplo de programa em Python},label=prog:exemplo]
718 def saudacao(nome):
719     # imprime uma saudação acentuada
720     print(f"Olá, {nome}!")
721
722 saudacao("COPPE")
723 \end{python}
724 \source{Elaboração própria.}
725
726 Para inserir código no meio do texto use \verb|\pythoninline|, como em \pythoninline{saudaca
727
728 \begin{Verbatim}[frame=lines,numbers=left,numbersep=0.5em]
729 Olá, COPPE!
730 \end{Verbatim}
731
732 Há estilos prontos também para \verb|java|, \verb|xml|, \verb|html| e \verb|prolog|, comando
733
734 \begin{algorithm}[H]
735 \caption{Soma dos elementos de um vetor}
736 \label{alg:soma}
737 \Dados{vetor  $v$  com  $n$  elementos}
738 \Resultado{soma  $s$  dos elementos de  $v$ }
739  $s \leftarrow 0$ ;
740 \Para{ $i \leftarrow 1$  \Ate{ }  $n$ }{
741      $s \leftarrow s + v[i]$ ;
742 }
743 \Retorna  $s$ ;
744 \end{algorithm}
745
746 \chapter{Escrevendo matemática}
747
748 A classe carrega \texttt{amsmath} em qualquer motor. Sob \hologo{pdfLaTeX} ela também carr

```

```

749 \section{Alinhamento (align)}
750 O ambiente \verb|align| alinha equações pelo \verb|&| e numera cada linha (\verb|\\| quebr
751 \begin{align}
752   f(x) &= (x+1)^2 \\
753   &= x^2 + 2x + 1 .
754 \end{align}
755
756 \section{Matrizes (matrix)}
757 O \texttt{amsmath} traz \verb|pmatrix|, \verb|bmatrix|, \verb|vmatrix|, entre outros:
758 \begin{equation}
759   A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}
760   \quad
761   \det A = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}
762 \end{equation}
763
764 \section{Subequações (subequations)}
765 Para numerar um grupo de equações sob o mesmo número, com letras (a), (b):
766 \begin{subequations}
767 \begin{align}
768   a &= b + c , \\
769   d &= e - f .
770 \end{align}
771 \end{subequations}
772
773 \section{Estruturas alinhadas (array)}
774 O \verb|array| monta estruturas dentro do modo matemático, como definições por partes:
775 \begin{equation}
776   |x| = \left\{ \begin{array}{l}
777     x, \quad \text{se } x \geq 0 , \\
778     -x, \quad \text{caso contrário.}
779   \end{array} \right.
780 \end{equation}
781
782 \section{Operadores (\textbackslash DeclareMathOperator)}
783 Para operadores com a tipografia correta (romano, espaçamento adequado), declare-os no pre
784 \begin{verbatim}
785 \usepackage{amsmath}
786 \DeclareMathOperator{\RMSE}{RMSE}
787 \DeclareMathOperator{\diag}{diag}
788 \end{verbatim}
789 Assim, \verb|$ \RMSE(y, \hat{y})| produz  $\RMSE(y, \hat{y})$ , e \verb|$ \diag(\lambda_1, \dots
790
791 \section{Unicode e LuaLaTeX (unicode-math)}
792 \label{sec:unicode-math}
793 Ao migrar para \hologo{LuaLaTeX} (ou \hologo{XeLaTeX}), \textbf{não carregue} \texttt{amss
794 \begin{verbatim}
795 \usepackage{amsmath} % a classe já carrega
796 \usepackage{unicode-math} % NÃO use \usepackage{amssymb} junto
797 \setmathfont{Latin Modern Math} % ou STIX Two Math, TeX Gyre Termes Math...
798 \end{verbatim}
799 O código clássico continua valendo ( $\mathbb{R}$ ,  $\int_a^b f(x) dx$ ,  $\alpha$ ) e você ai
800
801 \chapter{Referências cruzadas, notas e índices}

```

```

803 \label{cap:refs}
804
805 Este capítulo reúne os recursos para \emph{apontar} para coisas: rótulos, notas de rodapé,
806
807 \section{Rótulos e referências cruzadas}
808
809 Marque um elemento com \verb|\label{tipo:nome}| e aponte para ele com \verb|\ref| (só o nú
810
811 \section{Notas de rodapé}
812
813 Use \verb|\footnote{texto}|\footnote{Como esta nota, que a classe imprime em espaçamento s
814
815 \section{hyperref e marcadores}
816
817 A classe carrega o \verb|hyperref|: o sumário, as citações e os \verb|\ref|/\verb|\autoref
818
819 \section{URLs}
820
821 Para endereços, use \verb|\url{https://exemplo.org}|: o \verb|hyperref| cria o link e queb
822
823 \section{Índice remissivo}
824
825 Marque os termos com \verb|\index{termo}| e imprima o índice com \verb|\printindex| (carre
826
827 \section{Traduções (NBR 10520)}
828
829 Ao citar um trecho que você mesmo traduziu, coloque a tradução no corpo do texto e escreva
830
831 \chapter{Truques de \LaTeX}
832
833 \section{Comentários mágicos (\texttt{\% !TeX})}
834 Linhas \verb|\% !TeX| no topo do arquivo dizem ao editor (\hologo{TeX}studio, Overleaf, \ho
835 \begin{verbatim}
836 % !TeX program = lualatex
837 % !TeX encoding = UTF-8
838 % !TeX spellcheck = pt_BR
839 % !BIB program = biber
840 \end{verbatim}
841 Com isso o documento compila igual em qualquer ambiente. O \texttt{latexmkrc} cumpre papel
842
843 \section{Espaço engolido}
844 Um comando sem argumento engole o espaço seguinte: \verb|\LaTeX é bom| sai como “\LaTeX é
845
846 \section{Escopo (grupos)}
847 Chaves \verb|\{...}| (ou \verb|\begingroup|\dots\verb|\endgroup|) criam um escopo: mudanças
848
849 \section{Descarregar figuras: \texttt{\textbackslash clearpage} * \texttt{\textbackslash n
850 \verb|\newpage| apenas quebra a página e leva os floats pendentes adiante, acumulando-os.
851
852 \section{\texttt{latexmkrc}, perl e Overleaf}
853 O \verb|latexmk| automatiza a compilação (roda \LaTeX, \verb|biber| e \verb|makeindex| qua
854
855 \chapter{Método Proposto}
856 \chapter{Resultados e Discussões}

```

```

857
858 \section{Algumas Demonstrações}
859
860 A Lista de Símbolos precisa usar comandos específicos. Aqui vamos usar os símbolos $\alpha$
861 \syml[beta]{Beta}{A palavra Beta mais e corrigida}
862 \syml[zzbeta]{\beta}{A letra $\beta$ corrigida}
863 \syml[beta]{A palavra beta}
864 \syml[alpha]{A palavra alpha}
865 \syml[alpha]{Alpha}{A palavra Alpha}
866 \syml[zzalpha]{\alpha}{A letra $\alpha$ corrigida}
867 \syml[marco]{Marco}{A palavra Marco corrigida}
868
869 A Lista de Abreviações segue, a partir de 2024, a mesma regra, e aqui seguem alguns exemplos
870 \abbrev{GoT}{Game of Thrones}
871 \abbrev[GOT]{GoT}{Game of Thrones ordenado como GOT}
872 \abbrev[iot]{IoT}{IoT ordenado como iot}
873 \abbrev[IoT]{IoT}{IoT ordenado como IoT}
874 \abbrev[IOT]{IoT}{IoT ordenado como IOT}
875 \abbrev{IoT}{IoT com ordenação default}
876 \abbrev[ITU]{ITU}{ITU mesmo}
877
878 \newsigla{ufrj}{UFRJ}{Universidade Federal do Rio de Janeiro}
879 \newsigla{cpgp}{CPGP}{Comissão de Pós-Graduação e Pesquisa de Engenharia}
880
881 As siglas (manual2024, seção 2.8) são expandidas na primeira ocorrência e registradas automaticamente
882
883
884
885
886 \chapter{Conclusões}
887
888 \backmatter
889
890 % Tipos exóticos da ABNT (NBR 6023): patente, legislação, norma, mapa,
891 % partitura --- declarados via sinônimos em português em example.bib.
892 \nocite{patente-example,legislacao-example,norma-example,mapa-example,partitura-example}
893 \nocite{jurisprudencia-example}
894 \nocite{videogame-example,filme-example,semautor-example,semautorart-example}
895 \nocite{jornalcaderno-example,jornal-example}
896 \printbibliography
897
898
899
900 \appendix
901
902 \chapter{Um apêndice}
903
904 Segundo a norma da ABNT (Associação Brasileira de Normas Técnicas), a definição e utilização de apêndices
905
906 Apêndice: O apêndice é um texto ou documento elaborado pelo autor do trabalho com o objetivo de complementar
907
908
909 \chapter{Outro apêndice}
910

```

```

911 \annex
912
913
914 \chapter{Um Anexo}
915 Segundo a norma da ABNT (Associação Brasileira de Normas Técnicas), a definição e utilização
916
917
918
919 Anexo: O anexo, por sua vez, consiste em um texto ou documento não elaborado pelo autor, que
920
921 No modelo \CoppeTeX os anexos devem obrigatoriamente vir depois dos apêndices e usam o comand
922
923
924 \chapter{Outro Anexo}
925
926
927 \coppetexfinalpage% colofão institucional (página par, face do contracapa)
928
929 \end{document}
930
931 \example

```

10 Implementation

10.1 The ‘coppe.cls’ file

```

932 \class
933 \def\filename{coppe.dtx}
934 \def\fileversion{v4.0}
935 \def\filedate{2026/05/28}
936 \NeedsTeXFormat{LaTeX2e}[1995/12/01]
937 \ProvidesClass{coppe}[\filedate\ \fileversion\ COPPE Dissertations and Thesis]
938 % Save the class version and date into private macros immediately, before any
939 % required package can redefine the very common \fileversion / \filedate
940 % global names (LaTeX has no convention reserving them for the host class).
941 % \coppe@version / \coppe@date are what user-facing macros (e.g.
942 % \coppetexfinalpage) should reference.
943 \let\coppe@version\fileversion
944 \let\coppe@date\filedate
945 % Document date stored in private slots so \date can set them without ever
946 % touching TeX's \month/\year primitives (which would break \today). They
947 % default to the current system month/year, so a document without \date
948 % still renders sensibly on the cover and folha de rosto.
949 \edef\coppe@docmonth{\the\month}
950 \edef\coppe@docyear{\the\year}
951 \LoadClass[12pt,a4paper,twoside]{book}
952 % setspace, hyperref and the bibliography engine (biblatex+biber) are loaded at
953 % the END of the class (after every class hook is defined), in that order:
954 % setspace -> hyperref -> biblatex. hyperref must come late so all the macros
955 % it patches (TOC, refs, lists) are already defined, and biblatex must come
956 % after hyperref so its citation links and back-references work.
957 \RequirePackage{hyphenat}
958 \RequirePackage{lastpage}
959 \RequirePackage{ifthen}

```

```

960 \RequirePackage{graphicx}
961 \RequirePackage{tabularx}
962 \RequirePackage{booktabs}          % regras de tabela profissionais (toprule/midrule/bottomrule)
963 \RequirePackage{longtable}        % tabelas que quebram entre páginas
964 \RequirePackage{etoolbox}
965 \RequirePackage{ltxcmds}
966 \RequirePackage{hologo}           % TeX-family logos: \hologo{TeX}, \hologo{LaTeX}, ...
967 % Render \TeX, \LaTeX and friends via hologo (\CoppeTeX is a custom logo, kept).
968 \def\TeX{\hologo{TeX}}
969 \def\LaTeX{\hologo{LaTeX}}
970 \def\LaTeXe{\hologo{LaTeXe}}
971 \def\BibTeX{\hologo{BibTeX}}
972 \def\pdfLaTeX{\hologo{pdfLaTeX}}
973 \def\pdfTeX{\hologo{pdfTeX}}
974 \def\LuaLaTeX{\hologo{LuaLaTeX}}
975 \def\XeLaTeX{\hologo{XeLaTeX}}
976 \def\MiKTeX{\hologo{MiKTeX}}
977 % Engine-aware UTF-8 input: pdfLaTeX needs inputenc(utf8); LuaLaTeX (and XeTeX)
978 % are UTF-8 native, so inputenc must NOT be loaded there (it would clash). Source
979 % files may use accented Latin characters (áéíóú) directly. Under pdfLaTeX we also
980 % suggest LuaLaTeX, which handles Unicode/fonts better (at the cost of speed).
981 \RequirePackage{iftex}
982 \ifPDFTeX
983   \RequirePackage[utf8]{inputenc}
984   \AtEndOfClass{\ClassWarningNoLine{coppe}{%
985     Although slower, you should try compiling with LuaLaTeX}}
986 \fi
987 \RequirePackage[T1]{fontenc}
988 % Math support. amsmath is universal and works on every engine, so the class
989 % brings it in unconditionally (it must precede hyperref, which is loaded at
990 % the end of the class). amssymb is the legacy AMS symbol package, built
991 % around the 8-bit Type 1 fonts msam/msbm; on a Unicode engine (LuaLaTeX or
992 % XeLaTeX) it clashes with unicode-math, which already provides all of its
993 % symbols through Unicode math codes. So amssymb is loaded ONLY under
994 % pdfLaTeX; Unicode-engine users get a ClassWarning pointing them at
995 % unicode-math (the class does not auto-load unicode-math because choosing
996 % an OpenType math font is a typographic decision that belongs to the author).
997 \RequirePackage{amsmath}
998 \ifPDFTeX
999   \RequirePackage{amssymb}
1000 \else
1001   % Unicode engine: defer the check to end of document. Only warn if the
1002   % user did NOT add either unicode-math (the modern OpenType-math route)
1003   % or amssymb (the legacy route, which they would have loaded manually).
1004   % This avoids nagging authors who already followed the documented
1005   % \usepackage{unicode-math} workflow.
1006   \AtEndDocument{%
1007     \@ifpackageloaded{unicode-math}{\}%
1008     \@ifpackageloaded{amssymb}{\}%
1009     \ClassWarningNoLine{coppe}{%
1010       You are compiling with a Unicode engine (LuaLaTeX or XeLaTeX) and\MessageBreak
1011       neither unicode-math nor amssymb is loaded. AMS-style symbols\MessageBreak
1012       (\string\mathbb, \string\hbar, \string\varnothing, ...) will be undefined.\MessageBreak
1013       Add to your preamble:\MessageBreak

```



```

1014         \space\space\string\usepackage{unicode-math}\MessageBreak
1015         \space\space\string\setmathfont{Latin Modern Math}\MessageBreak
1016         Do NOT load amssymb together with unicode-math -- it would clash\MessageBreak
1017         with the OpenType math font setup}}}}
1018 \fi
1019 % manual2024 margins: 3cm top, 2cm bottom, mirrored 3cm inner / 2cm outer for
1020 % two-sided printing -- realised as inner=outer=2cm plus a 1cm bindingoffset.
1021 % headheight=15pt is given here (not patched later) so geometry's own page
1022 % arithmetic stays self-consistent: fancyhdr needs >= 14.49998pt of header
1023 % height to host the running head, and the standard 12pt default triggers
1024 % its "headheight is too small" warning. Letting geometry know up front
1025 % keeps \topmargin a derived value -- we never touch it by hand.
1026 \RequirePackage[a4paper,twoside,bindingoffset=1cm,headheight=15pt,%
1027 top=3cm,bottom=2cm,inner=2cm,outer=2cm]{geometry}
1028 \RequirePackage{titlesec}
1029 % --- Section heading formats (manual2024 2.6 / proposta P22) -----
1030 % chapter (numbered): "N TITLE", ALL CAPS, bold, flush left (no "Capitulo N").
1031 % APENDICE/ANEXO chapters print "APENDICE A -- Title" (R75/R76); normal chapters
1032 % keep "N Title". \appendix and \annex raise the flag (see below).
1033 \newif\if@coppeappendix \@coppeappendixfalse
1034 \titleformat{\chapter}[hang]
1035   {\normalfont\Large\bfseries}
1036   {\if@coppeappendix\MakeUppercase{\appendixname}\ \thechapter\ \textendash\else\thechapter\
1037   {1ex}{\MakeUppercase}
1038 \titlespacing*{\chapter}{0pt}{10pt}{1.5\baselineskip}
1039 \let\coppe@orig@appendix\coppe@orig@appendix\@coppeappendixtrue}
1040 \renewcommand\appendix{\coppe@orig@appendix\@coppeappendixtrue}
1041 % Sumario (TOC) entries for APENDICE/ANEXO chapters read "APENDICE A -- Title"
1042 % / "ANEXO A -- Title" (R75/R76), not the bare "A". The label word is chosen by
1043 % which macro is written (so it is correct at .toc-render time even though the
1044 % appendixname has changed back); the chapter letter is baked in at write time.
1045 \newif\if@coppeannex \@coppeannexfalse
1046 \newcommand\coppe@tocapp[1]{%
1047   \MakeUppercase{\coppestring{appendix}}}%
1048   \nobreakspace#1\nobreakspace\textendash\nobreakspace}
1049 \newcommand\coppe@tocanx[1]{%
1050   \MakeUppercase{\coppestring{annex}}}%
1051   \nobreakspace#1\nobreakspace\textendash\nobreakspace}
1052 \def\@chapter[#1]#2{%
1053   \ifnum \c@secnumdepth >\m@ne
1054     \if@mainmatter
1055       \refstepcounter{chapter}%
1056       \typeout{\@chapapp\space\thechapter.}%
1057       \if@coppeappendix
1058         \if@coppeannex
1059           \addcontentsline{toc}{chapter}{\protect\coppe@tocanx{\thechapter}#1}%
1060         \else
1061           \addcontentsline{toc}{chapter}{\protect\coppe@tocapp{\thechapter}#1}%
1062         \fi
1063       \else
1064         \addcontentsline{toc}{chapter}{\protect\numberline{\thechapter}#1}%
1065       \fi
1066     \else
1067       \addcontentsline{toc}{chapter}{#1}%

```

```

1068     \fi
1069   \else
1070     \addcontentsline{toc}{chapter}{#1}%
1071   \fi
1072   \chaptermark{#1}%
1073   \addtocontents{lof}{\protect\addvspace{10\p@}}%
1074   \addtocontents{lot}{\protect\addvspace{10\p@}}%
1075   \if@twocolumn
1076     \@topnewpage[\@makechapterhead{#2}]%
1077   \else
1078     \@makechapterhead{#2}%
1079     \afterheading
1080   \fi}
1081 % chapter (unnumbered): centred ALL CAPS bold (errata, lists, resumo, sumario,
1082 % referencias, glossario, ... -- unnumbered headings are centred, P29-P30).
1083 \titleformat{name=chapter,numberless}[hang]
1084   {\normalfont\Large\bfseries\filcenter}{0pt}{\MakeUppercase}
1085 \titlespacing{name=chapter,numberless}{0pt}{10pt}{1.5\baselineskip}
1086 % section: ALL CAPS, normal weight.
1087 \titleformat{section}
1088   {\normalfont\large\bfseries}{\thesection}{1ex}{\MakeUppercase}
1089 % subsection: bold (author writes Title Case).
1090 \titleformat{subsection}
1091   {\normalfont\bfseries}{\thesubsection}{1ex}{}
1092 % subsubsection: bold italic.
1093 \titleformat{subsubsection}
1094   {\normalfont\bfseries\itshape}{\thesubsubsection}{1ex}{}
1095 % --- Captions (manual2024 2.10-2.11 / proposta P6): smaller font, ABNT
1096 %   "Figura 1 -- titulo" travessao separator (sources handled by \source). ---
1097 \RequirePackage{caption}
1098 \captionsetup{font=footnotesize,labelsep=endash}
1099 % subcaption (NÃO subfig/subfigure) for subfigures / subtables; load right after
1100 % caption so it picks up the ABNT captionsetup above.
1101 \RequirePackage{subcaption}
1102 % --- Floats: caption on TOP and a mandatory "Fonte:" line at the bottom.
1103 %   manual2024 2.10-2.11 (R46/R51: identification on top; R48/R52: the source
1104 %   at the bottom is mandatory). The float package's plaintop style forces the
1105 %   caption above the body for figure, table and the new quadro float. -----
1106 \RequirePackage{float}
1107 \floatstyle{plaintop}
1108 \restylefloat{figure}
1109 \restylefloat{table}
1110 % New "quadro" float -- "Quadro" (pt) / "Frame" (en) (R53) -- with its own list
1111 % file (.loq). A quadro's data is bordered on all sides (drawn by the author),
1112 % unlike a tabela (rules on top and bottom only).
1113 \providecommand{\quadroname}{\coppestring{frame}}
1114 \providecommand{\listquadroname}{\coppestring{listframe}}
1115 \newfloat{quadro}{htbp}{loq}[chapter]
1116 \floatname{quadro}{\quadroname}
1117 \providecommand{\quadroautorefname}{\quadroname}
1118 % Format the Lista de Quadros entries exactly like the Lista de Figuras ones
1119 % (float's own \l@quadro lays the number/title out incorrectly).
1120 \let\l@quadro\l@figure
1121 \newcommand\listofquadros{%

```

```

1122 \chapter*{\listquadroname}%
1123 \addcontentsline{toc}{chapter}{\listquadroname}%
1124 \mkboth{\MakeUppercase\listquadroname}{\MakeUppercase\listquadroname}%
1125 \@starttoc{loq}}
1126 % \source / \fonte: the illustration source (ABNT-mandatory). Typeset below the
1127 % float as a smaller, single-spaced, centred line; it does NOT advance any
1128 % caption counter. \cpsource/\cpfonte are clash-safe aliases (the rename rule);
1129 % the label is "Fonte" (pt) / "Source" (en).
1130 \newcommand{\cpsourcenam}{\coppestring{source}}
1131 \newcommand{\cpsource}[1]{%
1132   \par\addvspace{\smallskipamount}%
1133   {\footnotesize\singlespacing\centering\cpsourcenam: #1\par}}
1134 \let\cpfonte\cpsource
1135 \@ifundefined{source}{\let\source\cpsource}{}%
1136 \@ifundefined{fonte}{\let\fonte\cpsource}{}%
1137 % \newcoppefloat{env}{caption name}{list title}: declare an extra caption-on-top
1138 % float (e.g. "picture") that supports \source and gains a \listof<env>s command.
1139 \newcommand{\newcoppefloat}[3]{%
1140   {\floatstyle{plaintop}\newfloat{#1}{htbp}{lo#1}}%
1141   \floatname{#1}{#2}%
1142   \expandafter\newcommand\csname listof#1s\endcsname{%
1143     \chapter*{#3}\addcontentsline{toc}{chapter}{#3}%
1144     \mkboth{\MakeUppercase{#3}}{\MakeUppercase{#3}}%
1145     \@starttoc{lo#1}}}
1146 % --- Footnotes (manual2024 / R16, R8, R21): single spacing and a 5 cm rule
1147 % (filete) from the left margin; footnotesize is the book default. -----
1148 \let\coppe@orig@footnotetext\@footnotetext
1149 \long\def\@footnotetext#1{\coppe@orig@footnotetext{\singlespacing#1}}
1150 \renewcommand\footnoterule{\kern-3\p@\hrule\@width 5cm\kern 2.6\p@}
1151 % --- alineas: list keyed by lowercase letters a) b) c); subitems with "--"
1152 % (manual2024 2.6 / proposta P24). -----
1153 \RequirePackage{enumitem}
1154 \newlist{alineas}{enumerate}{2}
1155 \setlist{alineas,1}{label=\alph*},leftmargin=1.8em,itemsep=0pt}
1156 \setlist{alineas,2}{label=--,leftmargin=1.8em,itemsep=0pt}
1157 % --- Page numbering: arabic in the top OUTER corner on textual pages
1158 % (manual2024 2.7 / proposta P25-P26). Pre-textual numbering is set by
1159 % \frontmatter (unnumbered by default; roman with the numeraisromanos
1160 % class option). -----
1161 \RequirePackage{fancyhdr}
1162 % \headheight is set to 15pt by geometry up above (see the geometry call),
1163 % which is what fancyhdr needs (>= 14.49998pt) for the running head.
1164 \fancypagestyle{coppe}{\fancyhf{}\fancyhead[R0,LE]{\thepage}}%
1165 \renewcommand{\headrulewidth}{0pt}}
1166 % Blank pages inserted by \cleardoublepage (two-sided) must be empty: no page
1167 % number may leak onto them (e.g. the verso right after the cover).
1168 \renewcommand{\cleardoublepage}{%
1169   \clearpage
1170   \if@twoside
1171     \ifodd\c@page\else
1172       \hbox{}\thispagestyle{empty}\newpage
1173       \if@twocolumn\hbox{}\newpage\fi
1174     \fi
1175   \fi}

```

```

1176 \def\CoppeTeX{\rm C\kern-.05em{\sc o\kern-.025em p\kern-.025em
1177 p\kern-.025em e}}\kern-.08em
1178 T\kern-.1667em\lower.5ex\hbox{E}\kern-.125emX\spacefactor1000}

1179 \newboolean{maledoc}
1180 \setboolean{maledoc}{false}
1181 %
1182 % Class options.
1183 % Multilingual model (CoppeTeX 4.x, branch nlinguas). The class holds three
1184 % fixed language slots:
1185 % * main -- the language of the body, captions, and the main abstract.
1186 % Set by a class option: brazilian (default), english,
1187 % spanish, french, italian, or any user-supplied language
1188 % that ships a coppe-lang-<name>.def file.
1189 % * foreign -- the language used by the foreignabstract environment and
1190 % by \foreign@... registers. Defaults to english (or to
1191 % brazilian when main=english) for back-compat.
1192 % * third -- optional, loaded on demand with \usecoppelanguage{<name>}
1193 % in the preamble (for citations or quotations in a third
1194 % language). Has no class option of its own.
1195 % Note: the legacy \if@english boolean (defined and toggled here in
1196 % v3.x for backward compatibility) was removed in v4.0 -- nothing
1197 % inside the class or in the biblatex style files (.bbx/.cbx/.dbx/.ltx)
1198 % tests it. Code that needs to branch on the main language should test
1199 % \ifx\coppe@mainlang\coppe@englishstr instead.
1200 \def\coppe@mainlang{brazilian}
1201 \def\coppe@foreignlang{english}
1202 \DeclareOption{brazilian}{%
1203 \def\coppe@mainlang{brazilian}\def\coppe@foreignlang{english}}
1204 \DeclareOption{english}{%
1205 \def\coppe@mainlang{english}\def\coppe@foreignlang{brazilian}}
1206 \DeclareOption{spanish}{%
1207 \def\coppe@mainlang{spanish}\def\coppe@foreignlang{english}}
1208 \DeclareOption{french}{%
1209 \def\coppe@mainlang{french}\def\coppe@foreignlang{english}}
1210 \DeclareOption{italian}{%
1211 \def\coppe@mainlang{italian}\def\coppe@foreignlang{english}}
1212 \DeclareOption{msc}{%
1213 \newcommand{\@degree}{M.Sc.}
1214 \newcommand{\@degreename}{Mestrado}
1215 \newcommand{\local@degname}{Mestre}
1216 \newcommand{\foreign@degname}{Master}
1217 \newcommand{\local@doctype}{Disserta{\c c}{\~ a}o}
1218 \newcommand{\foreign@doctype}{Dissertation}
1219 }
1220 \DeclareOption{dsccexam}{%
1221 \newcommand{\@degree}{D.Sc.}
1222 \newcommand{\@degreename}{Doutorado}
1223 \newcommand{\local@degname}{Doutor}
1224 \newcommand{\foreign@degname}{Doctor}
1225 \setboolean{maledoc}{true}
1226 \newcommand{\local@doctype}{Exame de Qualifica{\c c}{\~ a}o}
1227 \newcommand{\foreign@doctype}{Qualifying Exam}
1228 }
1229 \DeclareOption{msccexam}{%

```

```

1230 \newcommand{\@degree}{M.Sc.}
1231 \newcommand{\@degreename}{Mestrado}
1232 \newcommand{\local@degname}{Mestre}
1233 \newcommand{\foreign@degname}{Master}
1234 \setboolean{maledoc}{true}
1235 \newcommand\local@doctype{Exame de Qualifica{\c c}{\~ a}o}
1236 \newcommand\foreign@doctype{Qualifying Exam}
1237 }
1238 \DeclareOption{dsc}{%
1239 \newcommand{\@degree}{D.Sc.}
1240 \newcommand{\@degreename}{Doutorado}
1241 \newcommand{\local@degname}{Doutor}
1242 \newcommand{\foreign@degname}{Doctor}
1243 \newcommand\local@doctype{Tese}
1244 \newcommand\foreign@doctype{Thesis}
1245 }
1246 \newif\if@coppenumeric\@coppenumericfalse
1247 \DeclareOption{numbers}{\@coppenumerictrue}
1248 % numeraisromanos: show roman numerals on the pre-textual pages. Default: the
1249 % pre-textual pages are counted but not numbered (manual2024 2.7).
1250 \newif\if@copperoman\@copperomanfalse
1251 \DeclareOption{numeraisromanos}{\@copperomantrue}

```

Default line spacing is one-and-a-half; the `doublespacing` option switches to double spacing. Since `setspace` is loaded at the end of the class (after `hyperref`), the actual `\onehalfspacing`/`\doublespacing` call is deferred: the option only flips a flag, which the late-load block reads.

```

1252 \newif\if@coppedoublespace\@coppedoublespacefalse
1253 \DeclareOption{doublespacing}{\@coppedoublespacetrue}
1254 \ProcessOptions\relax
1255 % Built-in language names (used by the loader and the dispatcher below).
1256 \def\coppe@@brazilianstr{brazilian}
1257 \def\coppe@@englishstr{english}
1258 % Load Babel with the foreign language first and the main language last
1259 % (Babel treats the LAST option in the list as the main language). With
1260 % pt-main this produces [english,brazilian]{babel} and with en-main
1261 % [brazilian,english] -- identical to the historical \if@english calls.
1262 \edef\coppe@babelopts{\coppe@foreignlang,\coppe@mainlang}
1263 \expandafter\RequirePackage\expandafter[\coppe@babelopts]{babel}
1264 % --- Multilingual string dispatcher -----
1265 % \copperdefstring{<lang>}{<key>}{<value>} adds one entry to the table for
1266 % language <lang>. The three retrievers below look strings up either at the
1267 % current babel language (\coppestring -- follows \selectlanguage), or pinned
1268 % to the main slot (\coppemainstring) or the foreign slot (\coppeforeignstring).
1269 \providecommand{\copperdefstring}[3]{%
1270 \expandafter\gdef\csname coppe@str@#1@#2\endcsname{#3}}
1271 \providecommand{\coppestring}[1]{\@nameuse{coppe@str@\language@#1}}
1272 \providecommand{\coppemainstring}[1]{\@nameuse{coppe@str@\coppe@mainlang @#1}}
1273 \providecommand{\coppeforeignstring}[1]{\@nameuse{coppe@str@\coppe@foreignlang @#1}}
1274 % Load an extra language pack at any point in the preamble (third-language
1275 % slot for citations or quotations beyond the main+foreign pair).
1276 \providecommand{\usecoppelanguage}[1]{%
1277 \def\coppe@@cur{#1}%
1278 \ifx\coppe@@cur\coppe@@brazilianstr\else

```

```

1279 \ifx\coppe@@cur\coppe@@englishstr\else
1280 \InputIfFileExists{coppe-lang-#1.def}{}
1281 {\ClassWarning{coppe}{Language pack coppe-lang-#1.def not found}}%
1282 \fi\fi}
1283 % \titlein{<lang>}{<text>}: record the title in any language. Will be
1284 % consulted by the cover/folha renderers once they are migrated in step 2.
1285 % In step 1 it is just a no-op store -- the historical \title / \foreigntitle
1286 % still drive every visible site, so behavior for pt/en is unchanged.
1287 \providecommand{\titlein}[2]{%
1288 \expandafter\gdef\csname coppe@title@#1\endcsname{#2}}
1289 % --- Built-in string tables: brazilian + english. The values mirror the
1290 % literals currently scattered through the class; step-1 only populates the
1291 % tables -- no site reads them yet, so the rendered output is unchanged. ----
1292 \copperdefstring{brazilian}{appendix}{Ap~endice}
1293 \copperdefstring{english}{appendix}{Appendix}
1294 \copperdefstring{brazilian}{annex}{Anexo}
1295 \copperdefstring{english}{annex}{Annex}
1296 \copperdefstring{brazilian}{frame}{Quadro}
1297 \copperdefstring{english}{frame}{Frame}
1298 \copperdefstring{brazilian}{listframe}{Lista de Quadros}
1299 \copperdefstring{english}{listframe}{List of Frames}
1300 \copperdefstring{brazilian}{source}{Fonte}
1301 \copperdefstring{english}{source}{Source}
1302 \copperdefstring{brazilian}{program}{Programa}
1303 \copperdefstring{english}{program}{Program}
1304 \copperdefstring{brazilian}{listprogram}{Lista de Programas}
1305 \copperdefstring{english}{listprogram}{List of Programs}
1306 \copperdefstring{brazilian}{references}{Refer~encias}
1307 \copperdefstring{english}{references}{References}
1308 \copperdefstring{brazilian}{in:}{em}
1309 \copperdefstring{english}{in:}{in}
1310 \copperdefstring{brazilian}{art}{Ilustra~c~o}
1311 \copperdefstring{english}{art}{Art}
1312 \copperdefstring{brazilian}{platform}{Plataforma}
1313 \copperdefstring{english}{platform}{Platform}
1314 \copperdefstring{brazilian}{directedby}{Dire~c~o}
1315 \copperdefstring{english}{directedby}{Directed by}
1316 \copperdefstring{brazilian}{producedby}{Produ~c~o}
1317 \copperdefstring{english}{producedby}{Produced by}
1318 \copperdefstring{brazilian}{listabbreviation}{Lista de Abreviaturas e Siglas}
1319 \copperdefstring{english}{listabbreviation}{List of Abbreviations and Acronyms}
1320 \copperdefstring{brazilian}{listsymbol}{Lista de S~i~mbolos}
1321 \copperdefstring{english}{listsymbol}{List of Symbols}
1322 \copperdefstring{brazilian}{glossary}{Gloss~a~rio}
1323 \copperdefstring{english}{glossary}{Glossary}
1324 \copperdefstring{brazilian}{advisor}{Orientador}
1325 \copperdefstring{english}{advisor}{Advisor}
1326 \copperdefstring{brazilian}{advisors}{Orientadores}
1327 \copperdefstring{english}{advisors}{Advisors}
1328 \copperdefstring{brazilian}{dept}{Programa}
1329 \copperdefstring{english}{dept}{Department}
1330 \copperdefstring{brazilian}{approvedmale}{Examinado por}
1331 \copperdefstring{brazilian}{approvedfemale}{Examinada por}
1332 \copperdefstring{english}{approvedmale}{Examined by}

```

```

1333 \copperdefstring{english} {approvedfemale}{Examined by}
1334 % Institutional names: always Portuguese on the cover/folha (UFRJ template).
1335 \copperdefstring{brazilian}{universityname}{Universidade Federal do Rio de Janeiro}
1336 \copperdefstring{english} {universityname}{Universidade Federal do Rio de Janeiro}
1337 \copperdefstring{brazilian}{cityname} {Rio de Janeiro}
1338 \copperdefstring{english} {cityname} {Rio de Janeiro}
1339 \copperdefstring{brazilian}{statename} {RJ}
1340 \copperdefstring{english} {statename} {RJ}
1341 \copperdefstring{brazilian}{countryname} {Brasil}
1342 \copperdefstring{english} {countryname} {Brasil}
1343 % Month names.
1344 \copperdefstring{brazilian}{monthname1} {Janeiro}
1345 \copperdefstring{brazilian}{monthname2} {Fevereiro}
1346 \copperdefstring{brazilian}{monthname3} {Mar{\c c}o}
1347 \copperdefstring{brazilian}{monthname4} {Abril}
1348 \copperdefstring{brazilian}{monthname5} {Maio}
1349 \copperdefstring{brazilian}{monthname6} {Junho}
1350 \copperdefstring{brazilian}{monthname7} {Julho}
1351 \copperdefstring{brazilian}{monthname8} {Agosto}
1352 \copperdefstring{brazilian}{monthname9} {Setembro}
1353 \copperdefstring{brazilian}{monthname10}{Outubro}
1354 \copperdefstring{brazilian}{monthname11}{Novembro}
1355 \copperdefstring{brazilian}{monthname12}{Dezembro}
1356 \copperdefstring{english}{monthname1} {January}
1357 \copperdefstring{english}{monthname2} {February}
1358 \copperdefstring{english}{monthname3} {March}
1359 \copperdefstring{english}{monthname4} {April}
1360 \copperdefstring{english}{monthname5} {May}
1361 \copperdefstring{english}{monthname6} {June}
1362 \copperdefstring{english}{monthname7} {July}
1363 \copperdefstring{english}{monthname8} {August}
1364 \copperdefstring{english}{monthname9} {September}
1365 \copperdefstring{english}{monthname10}{October}
1366 \copperdefstring{english}{monthname11}{November}
1367 \copperdefstring{english}{monthname12}{December}
1368 % algorithm2e package option for this language (empty -> english default).
1369 \copperdefstring{brazilian}{algorithm2eopt}{portuguese}
1370 \copperdefstring{english} {algorithm2eopt}{}
1371 % Babel package name for the foreignabstract \begin{otherlanguage}{...}.
1372 \copperdefstring{brazilian}{babelname}{brazilian}
1373 \copperdefstring{english} {babelname}{english}
1374 % Auto-load language packs for any non-built-in language listed in babelopts.
1375 % Built-ins (brazilian, english) ship their string tables inside coppe.cls
1376 % (defined just above) and need no extra file; others (spanish/french/italian
1377 % /...) ship as coppe-lang-<name>.def alongside the class. The dispatcher
1378 % must already be defined when these run, so the loader sits AFTER the tables.
1379 \@for\coppe@@lang:=\coppe@babelopts\do{%
1380 \edef\coppe@@cur{\coppe@@lang}%
1381 \ifx\coppe@@cur\coppe@@brazilianstr\else
1382 \ifx\coppe@@cur\coppe@@englishstr\else
1383 \InputIfFileExists{coppe-lang-\coppe@@cur.def}
1384 {\ClassInfo{coppe}{Loaded language pack coppe-lang-\coppe@@cur.def}}
1385 {\ClassError{coppe}
1386 {Missing language pack 'coppe-lang-\coppe@@cur.def'.}

```

```

1387         {Install it next to coppe.cls, or pick a built-in language.}}%
1388 \fi\fi}
1389 % Quotation marks: csquotes gives the language-aware \enquote; \citacao is its
1390 % Portuguese alias for short, inline quotations (proposta P3).
1391 \RequirePackage{csquotes}
1392 \newcommand{\citacao}[1]{\enquote{#1}}
1393 % --- Code listings & algorithms (TOD03 "tratar listagens"): the listings
1394 % package gives the "Programa" listing (caption on top, ABNT R46) with
1395 % \listofprogramas; algorithm2e gives algorithms. Names are babel-aware and
1396 % the language styles below are based on the user's listagens.tex. -----
1397 \RequirePackage{xcolor}
1398 \RequirePackage{listings}
1399 \RequirePackage{fancyvrb}
1400 \DeclareFixedFont{\ttb}{T1}{txtt}{bx}{n}{10}
1401 \DeclareFixedFont{\ttm}{T1}{txtt}{m}{n}{10}
1402 \lstset{captionpos=t,extendedchars=true,basicstyle=\ttfamily,inputencoding=utf8,%
1403 literate=
1404 {ã}{\~{a}}1 {á}{\~{a}}1 {â}{\~{a}}1 {ä}{\~{a}}1 {é}{\~{e}}1%
1405 {ê}{\~{e}}1 {ë}{\~{e}}1 {î}{\~{i}}1 {í}{\~{i}}1 {ô}{\~{o}}1%
1406 {ó}{\~{o}}1 {ö}{\~{o}}1 {û}{\~{u}}1 {ü}{\~{u}}1 {ç}{\~{c}}1%
1407 {Ç}{\~{C}}1 {À}{\~{A}}1 {Á}{\~{A}}1 {Â}{\~{A}}1 {Ã}{\~{A}}1 {É}{\~{E}}1%
1408 {Ê}{\~{E}}1 {Í}{\~{I}}1 {Ô}{\~{O}}1 {Õ}{\~{O}}1 {Ũ}{\~{U}}1%
1409 }
1410 }
1411 % Babel-aware listing names, set after listings' own babel setup.
1412 \AtBeginDocument{%
1413 \renewcommand\lstlistingname{\coppestring{program}}%
1414 \renewcommand\lstlistlistingname{\coppestring{listprogram}}%
1415 % \listofprogramas in the coppe list style (heading + ToC + running head).
1416 \newcommand\listofprogramas{%
1417 \chapter*{\lstlistlistingname}%
1418 \addcontentsline{toc}{chapter}{\lstlistlistingname}%
1419 \mkboth{\MakeUppercase\lstlistlistingname}{\MakeUppercase\lstlistlistingname}%
1420 \@starttoc{lol}}
1421 % Per-language styles + environments (python, java, xml, html, prolog).
1422 \newcommand\pythonstyle{\lstset{language=Python,basicstyle=\ttm,numbers=left,%
1423 numberstyle=\small\ttfamily,numbersep=0.5em,commentstyle=\color{purple},%
1424 morekeywords={self,as},keywordstyle=\ttb\color{rgb}{0,0,0.5},%
1425 emph={MyClass,__init__},emphstyle=\ttb\color{rgb}{0.6,0,0},%
1426 stringstyle=\color{rgb}{0,0.5,0},frame=tb,showstringspaces=false,%
1427 extendedchars=true,firstnumber=auto,inputencoding=utf8,breaklines=true,%
1428 escapechar=',postbreak=\mbox{\textcolor{red}{$\hookrightarrow$}\space}}%
1429 \lstnewenvironment{python}[1][\pythonstyle\lstset{#1}%
1430 \csname\@lst @SetFirstNumber\endcsname}{\csname\@lst @SaveFirstNumber\endcsname}%
1431 \newcommand\pythoninline[1]{\pythonstyle\lstinline!#1!}}
1432 \newcommand\pythonexternal[2][\pythonstyle\lstinputlisting{#1}{#2}}%
1433 \newcommand\xmlstyle{\lstset{language=XML,basicstyle=\ttm,numbers=left,%
1434 numberstyle=\small\ttfamily,numbersep=0.5em,%
1435 morekeywords={xs:schema,xs:element,xs:sequence,xs:complexType},%
1436 keywordstyle=\ttb\color{rgb}{0,0,0.5},%
1437 emph={name,type,attributeFormDefault,elementFormDefault,xmlns:xs},%
1438 emphstyle=\ttb\color{rgb}{0.6,0,0},stringstyle=\color{rgb}{0,0.5,0},frame=tb,%
1439 showstringspaces=false,extendedchars=true,inputencoding=utf8,breaklines=true,%
1440 postbreak=\mbox{\textcolor{red}{$\hookrightarrow$}\space}}}

```



```

1441 \lstnewenvironment{xml}[1][\{\xmlstyle\lstset{#1}\}}{}
1442 \newcommand\htmlstyle{\lstset{language=HTML,basicstyle=\ttm,numbers=left,%
1443   numberstyle=\small\ttfamily,numbersep=0.5em,%
1444   keywordstyle=\ttb\color{rgb}{0,0,0.5},emphstyle=\ttb\color{rgb}{0.6,0,0},%
1445   stringstyle=\color{rgb}{0,0.5,0},frame=tb,showstringspaces=false,%
1446   extendedchars=true,inputencoding=utf8,breaklines=true,%
1447   postbreak=\mbox{\textcolor{red}{\hookrightarrow}\space}}}
1448 \lstnewenvironment{html}[1][\{\htmlstyle\lstset{#1}\}}{}
1449 \newcommand\prologstyle{\lstset{language=Prolog,basicstyle=\ttm,numbers=left,%
1450   numberstyle=\small\ttfamily,numbersep=0.5em,commentstyle=\color{purple},%
1451   keywordstyle=\ttb\color{rgb}{0,0,0.5},emph={MyClass,__init__},%
1452   emphstyle=\ttb\color{rgb}{0.6,0,0},stringstyle=\color{rgb}{0,0.5,0},frame=tb,%
1453   showstringspaces=false,extendedchars=true,firstnumber=auto,inputencoding=utf8,%
1454   breaklines=true,escapechar=',%
1455   postbreak=\mbox{\textcolor{red}{\hookrightarrow}\space}}}
1456 \lstnewenvironment{prolog}[1][\{\prologstyle\lstset{#1}\}}{}
1457 \csname\@lst @SetFirstNumber\endcsname{\csname\@lst @SaveFirstNumber\endcsname}
1458 \newcommand\javastyle{\lstset{language=Java,basicstyle=\ttm,numbers=left,%
1459   numberstyle=\small\ttfamily,numbersep=0.5em,commentstyle=\color{purple},%
1460   morekeywords={self,as},keywordstyle=\ttb\color{rgb}{0,0,0.5},%
1461   emphstyle=\ttb\color{rgb}{0.6,0,0},stringstyle=\color{rgb}{0,0.5,0},frame=tb,%
1462   showstringspaces=false,extendedchars=true,firstnumber=auto,inputencoding=utf8,%
1463   breaklines=true,escapechar=',%
1464   postbreak=\mbox{\textcolor{red}{\hookrightarrow}\space}}}
1465 \lstnewenvironment{java}[1][\{\javastyle\lstset{#1}\}}{}
1466 \csname\@lst @SetFirstNumber\endcsname{\csname\@lst @SaveFirstNumber\endcsname}
1467 \newcommand\javainline[1][\{\javastyle\lstinline!#1!\}}{}
1468 \newcommand\javaexternal[2][\{\javastyle\lstinputlisting[1]{#2}\}}{}
1469 % fancyvrb's Verbatim environment is available for program output; the
1470 % \DefineVerbatimEnvironment{gxoutput}/{gxoutputs} aliases that used to live
1471 % here were author-personal (gx prefix) and were removed in v4.0. Use, for
1472 % example,
1473 % \begin{Verbatim}[frame=lines,numbers=left,numbersep=0.5em]
1474 % ...
1475 % \end{Verbatim}
1476 % (or \DefineVerbatimEnvironment{...}{Verbatim}{...} in the manuscript preamble).
1477 \renewcommand{\theFancyVerbLine}{\small\ttfamily\arabic{FancyVerbLine}}
1478 % Algorithms: algorithm2e, language driven by the main-slot \coppemainstring
1479 % lookup (key "algorithm2eopt"). Empty means use algorithm2e's default
1480 % language (English) -- needed because algorithm2e errors on an empty option.
1481 \edef\coppe@algotlang{\coppemainstring{algorithm2eopt}}
1482 \ifx\coppe@algotlang\@empty
1483 \RequirePackage[linesnumbered,lined,algochapter,ruled]{algorithm2e}
1484 \else
1485 \expandafter\RequirePackage\expandafter[\coppe@algotlang,linesnumbered,lined,algochapter,ruled]{algorithm2e}
1486 \fi
1487 \SetKwFor{Para}{para}{faça}{fim do para}
1488 \SetKwFor{Enquanto}{enquanto}{faça}{fim do enquanto}
1489 % setspace + hyperref + biblatex are loaded at the very end of the class
1490 % (see the block just before %</class>), so their hooks see every macro the
1491 % class defines, and the load order setspace -> hyperref -> biblatex is
1492 % respected.

```

\department This macro is used to set the author's affiliation. There are twelve options which correspond to all academic units at COPPE/UFRJ. It defines the current and the

foreign names of these units.

```

1493 \newcommand\department[1]{%
1494   \ifthenelse{\equal{#1}{PEB}}
1495     {\global\def\local@deptname{Engenharia Biom{\` e}dica}
1496     \global\def\foreign@deptname{Biomedical Engineering}}{}
1497   \ifthenelse{\equal{#1}{PEC}}
1498     {\global\def\local@deptname{Engenharia Civil}
1499     \global\def\foreign@deptname{Civil Engineering}}{}
1500   \ifthenelse{\equal{#1}{PEE}}
1501     {\global\def\local@deptname{Engenharia El{\` e}trica}
1502     \global\def\foreign@deptname{Electrical Engineering}}{}
1503   \ifthenelse{\equal{#1}{PEM}}
1504     {\global\def\local@deptname{Engenharia Mec{\` a}nica}
1505     \global\def\foreign@deptname{Mechanical Engineering}}{}
1506   \ifthenelse{\equal{#1}{PEMM}}
1507     {\global\def\local@deptname{Engenharia Metal{\` u}rgica e de Materiais}
1508   \global\def\foreign@deptname{Metallurgical and Materials Engineering}}{}
1509   \ifthenelse{\equal{#1}{PEN}}
1510     {\global\def\local@deptname{Engenharia Nuclear}
1511     \global\def\foreign@deptname{Nuclear Engineering}}{}
1512   \ifthenelse{\equal{#1}{PEN0}}
1513     {\global\def\local@deptname{Engenharia Oce{\` a}nica}
1514     \global\def\foreign@deptname{Ocean Engineering}}{}
1515   \ifthenelse{\equal{#1}{PPE}}
1516     {\global\def\local@deptname{Planejamento Energ{\` e}tico}
1517     \global\def\foreign@deptname{Energy Planning}}{}
1518   \ifthenelse{\equal{#1}{PEP}}
1519     {\global\def\local@deptname{Engenharia de Produ{\c c}{\` a}o}
1520     \global\def\foreign@deptname{Production Engineering}}{}
1521   \ifthenelse{\equal{#1}{PEQ}}
1522     {\global\def\local@deptname{Engenharia Qu{\` i}mica}
1523     \global\def\foreign@deptname{Chemical Engineering}}{}
1524   \ifthenelse{\equal{#1}{PESC}}
1525     {\global\def\local@deptname{Engenharia de Sistemas e Computa{\c c}{\` a}o}
1526     \global\def\foreign@deptname{Systems Engineering and Computer Science}}{}
1527   \ifthenelse{\equal{#1}{PET}}
1528     {\global\def\local@deptname{Engenharia de Transportes}
1529     \global\def\foreign@deptname{Transportation Engineering}}{}
1530   \ifthenelse{\equal{#1}{PENT}}
1531     {\global\def\local@deptname{Engenharia de Nanotecnologia}
1532     \global\def\foreign@deptname{Nanotechnology Engineering}}{}
1533 }

```

`\title` Used to enter the title in Brazilian Portuguese. Mirrors into `\coppe@title@brazilian` so that the per-language title slot used by the cover/folha renderers (`\coppe@selecttitle`) picks the right text for a pt-main thesis.

```

1534 \renewcommand\title[1]{%
1535   \global\def\local@title{#1}%
1536   \expandafter\gdef\csname coppe@title@brazilian\endcsname{#1}%
1537 }

```

`\foreigntitle` Used to enter the foreign title. Also mirrors into `\coppe@title@english` so that the cover renders the right text for an english-main thesis.

```

1538 \newcommand\foreigntitle[1]{%
1539   \global\def\foreign@title{#1}%
1540   \expandafter\gdef\csname coppe@title@english\endcsname{#1}%
1541 }

```

\coppe@selecttitle Expands to the title stored for the current main language. For pt-main this is `\local@title`; for en-main `\foreign@title`; for any other main language the user must call `\titlein{<lang>}{...}` to populate it.

```

1542 \newcommand\coppe@selecttitle{\@nameuse{coppe@title@\coppe@mainlang}}

```

\advisor Defines globally the title, name and academic degree of the advisors.

```

1543 \newcount\@advisor\@advisor0
1544 \newcommand\advisor[4]{%
1545   \global\@namedef{CoppeAdvisorTitle:\expandafter\the\@advisor}{#1}
1546   \global\@namedef{CoppeAdvisorName:\expandafter\the\@advisor}{#2}
1547   \global\@namedef{CoppeAdvisorSurname:\expandafter\the\@advisor}{#3}
1548   \global\@namedef{CoppeAdvisorDegree:\expandafter\the\@advisor}{#4}
1549   \global\advance\@advisor by 1
1550   \ifnum\@advisor>1
1551     \renewcommand\local@advisorstring{Orientadores}
1552     \renewcommand\foreign@advisorstring{Advisors}
1553   \fi
1554 }

```

\examiner

```

1555 \newcount\@examiner\@examiner0
1556 \newcommand\examiner[3]{%
1557   \global\@namedef{CoppeExaminer:\expandafter\the\@examiner}{#1\ #2}
1558   \global\advance\@examiner by 1
1559 }

```

\author It was redefined to allow the identification of the author's first names and surname.

```

1560 \renewcommand\author[2]{%
1561   \global\def\@authname{#1}
1562   \global\def\@authsurn{#2}
1563 }

```

\date Sets the *document's* nominal month and year (used by the cover, the folha de rosto, and the catalog). Stores them in private slots (`\coppe@docmonth` / `\coppe@docyear`) rather than mutating TeX's `\month` / `\year` primitives, which would clobber `\today`.

```

1564 \renewcommand\date[2]{%
1565   \def\coppe@docmonth{#1}%
1566   \def\coppe@docyear{#2}}

```

\local@monthname

```

1567 \newcommand\local@monthname{\ifcase\coppe@docmonth\or
1568   Janeiro\or Fevereiro\or Mar{c}o\or Abril\or Maio\or Junho\or
1569   Julho\or Agosto\or Setembro\or Outubro\or Novembro\or Dezembro\fi}

```

\foreign@monthname

```

1570 \newcommand\foreign@monthname{\ifcase\coppe@docmonth\or
1571   January\or February\or March\or April\or May\or June\or
1572   July\or August\or September\or October\or November\or December\fi}

```

`\keyword`

```
1573 \newcounter{keywords}
1574 \newcommand\keyword[1]{%
1575   \global\@namedef{CoppeKeyword:\expandafter\the\c@keywords}{#1}
1576   \global\addtocounter{keywords}{1}
1577 }
```

`\freeconfig` This command allows easy changing of core class parameters.

```
1578 \newcommand\freeconfig[6]{%
1579   \providecommand\@degree{}
1580   \renewcommand\@degree{#1}
1581   \providecommand\@degreename{}
1582   \renewcommand\@degreename{#2}
1583   \providecommand\local@degname{}
1584   \renewcommand\local@degname{#3}
1585   \providecommand\foreign@degname{}
1586   \renewcommand\foreign@degname{#4}
1587   \providecommand\local@doctype{}
1588   \renewcommand\local@doctype{#5}
1589   \providecommand\foreign@doctype{}
1590   \renewcommand\foreign@doctype{#6}%
1591 }
1592 % \end{macrocode}
1593 % \end{macro}
1594 %
1595 % \begin{macro}{\frontmatter}
1596 % The number of pages for both frontmatter and mainmatter printed
1597 % in the cataloging details page is computed by means of simple
1598 % LaTeX\ labels.
1599 % \begin{macrocode}
1600 \renewcommand\frontmatter{%
1601   \cleardoublepage
1602   \@mainmatterfalse
1603   \pagenumbering{roman}
1604   \thispagestyle{empty}
1605   % folha de rosto = counted page 1 (ABNT: the capa is not counted). An odd
1606   % start keeps the page-counter parity aligned with the physical (recto) side,
1607   % so \cleardoublepage lands every pre-textual element on an odd (front) page.
1608   \setcounter{page}{1}
1609   \makefrontpage
1610   \clearpage
1611   \if@copperoman
1612     \fancypagestyle{plain}{\fancyhf{}\fancyhead[R0,LE]{\thepage}}%
1613     \renewcommand{\headrulewidth}{0pt}}%
1614     \pagestyle{coppe}%
1615   \else
1616     \fancypagestyle{plain}{\fancyhf{}\renewcommand{\headrulewidth}{0pt}}%
1617     \pagestyle{empty}%
1618   \fi
1619 }
```

`\mainmatter`

```
1620 \renewcommand\mainmatter{%
1621   \coppe@mainBegin
```

```

1622 \cleardoublepage
1623 \@mainmattertrue
1624 \fancypagestyle{plain}{\fancyhf{}\fancyhead[R0,LE]{\thepage}%
1625     \renewcommand{\headrulewidth}{0pt}}%
1626 \pagestyle{coppe}%
1627 \pagenumbering{arabic}}

```

\backmatter

```

1628 \renewcommand\backmatter{%
1629     \if@openright
1630         \cleardoublepage
1631     \else
1632         \clearpage
1633     \fi}
1634 %

```

\coppetexfinalpage Optional colophon-style closing page intended to face the rear cover. The macro forces a verso (even) page so that, in a two-sided print, the reader sees the colophon on the left and the inside of the rear cover on the right; pushes the text to the bottom of the page with `\vfill`; and types it in Portuguese unconditionally (the colophon is institutional metadata, not body content, so it is rendered in pt-BR regardless of the thesis main language). Three slot macros are user-overrideable: `\coppefinalengine` (engine name; auto-detected via `iftex`), `\coppefinalfont` (text body font; defaults to Computer Modern, the L^AT_EX default), and `\coppefinalmanual` (the reference manual title; defaults to the current COPPE/UFRJ norm). Override them before `\coppetexfinalpage` with `\renewcommand` if the document uses a non-default font or follows an earlier edition of the norm.

```

1635 \providecommand{\coppefinalengine}{%
1636     \ifPDFTeX\hologo{pdfLaTeX}\fi
1637     \ifLuaTeX\hologo{LuaLaTeX}\fi
1638     \ifXeTeX\hologo{XeLaTeX}\fi}
1639 \providecommand{\coppefinalfont}{Computer Modern (pdr\~ao LaTeX{})}
1640 \providecommand{\coppefinalmanual}{%
1641     \emph{Norma para a Elabora\c c\~ao Gr\'afica de Teses e Disserta\c c\~oes
1642     da COPPE/UFRJ}}
1643 \newcommand{\coppetexfinalpage}{%
1644     \clearpage
1645     % Force the colophon onto a verso (even) page so it faces the rear cover.
1646     % If the current page is odd (recto), fill it with an empty verso first.
1647     \if@twoside
1648         \ifodd\c@page
1649             \hbox{}\thispagestyle{empty}\newpage
1650         \fi
1651     \fi
1652     \thispagestyle{empty}%
1653     \null\vfill
1654     \begin{otherlanguage}{brazilian}%
1655         \begin{center}
1656             {\footnotesize\setstretch{1.0}%
1657             \textsc{Colof\~ao}\par
1658             \vspace{0.7em}
1659             \begin{minipage}{0.75\textwidth}\centering

```

```

1660     Este documento foi composto em \LaTeX{} com \coppefinalengine{}
1661     em \today, na fonte \coppefinalfont. A bibliografia foi processada
1662     por Bib\LaTeX{} e Biber. Foi utilizada a classe \CoppeTeX,
1663     vers\~ao\coppe@version, dispon\’i vel no reposit\’orio GitHub
1664     \mbox{\url{https://github.com/COPPE-UFRJ/CoppeTeX}}, em
1665     conformidade com a \coppefinalmanual.
1666     \end{minipage}\par}
1667   \end{center}
1668 \end{otherlanguage}%
1669 \vspace*{2cm}%
1670 \clearpage}

```

`\makecover` The ABNT cover (capa, NBR 14724) suggested in the manual’s Anexo A. It is the institution block — the university, the full COPPE name, and the graduate program taken from `\department` — followed by the author, the title, and the city and year, all centred and uppercased. The optional UFRJ and COPPE logos head the page. The capa is not counted: it precedes the (roman) front matter and is produced before the `titlepage` (folha de rosto).

```

1671 \newcommand\makecover{%
1672   \begin{titlepage}%
1673   \centering
1674   \makebox[\linewidth]{%
1675     \includegraphics[height=2cm]{ufrj-logo}\hfill
1676     \includegraphics[height=2cm]{coppe-logo}}\par
1677   \vspace{1.5cm}
1678   {\singlespacing
1679   \MakeUppercase{\local@universityname}\par
1680     \nohyphens{\MakeUppercase{Instituto Alberto Luiz Coimbra de \
1681       P{\’ o}s-Gradua{\c c}{\~ a}o e Pesquisa de Engenharia}}\par
1682     \MakeUppercase{Programa de \local@deptname}\par}
1683   \vspace{\stretch{2}}
1684   \textbf{\MakeUppercase{\textbf{\@authname\ \@authsur}}}\par
1685   \vspace{\stretch{2}}
1686   \nohyphens{\MakeUppercase{\coppe@selecttitle}}\par
1687   \vspace{\stretch{3}}
1688   \MakeUppercase{\local@cityname}\par
1689   \coppe@docyear
1690   \end{titlepage}%
1691 }

```

`\maketitle`

```

1692 \renewcommand\maketitle{%
1693   \pagenumbering{alph}
1694   \ltx@ifpackageloaded{hyperref}{\coppe@hypersetup}{}%
1695   \makecover
1696   \begin{titlepage}
1697   \begin{flushleft}
1698     \vspace*{1.5mm}
1699     \setlength\baselineskip{0pt}
1700     \setlength\parskip{1mm}
1701     \makebox[\linewidth]{%
1702       \includegraphics[height=2cm]{ufrj-logo}\hfill
1703       \includegraphics[height=2cm]{coppe-logo}}

```

```

1704 \end{flushleft}
1705 \vspace{1.05cm}
1706 \begin{center}
1707 \nohyphens{\@authname\ \@authsurname}\par
1708 \vspace*{3cm}
1709 \nohyphens{\MakeUppercase{\coppe@selecttitle}}\par
1710 \end{center}
1711 \vspace*{2.1cm}
1712 \begin{flushright}
1713 \begin{minipage}{8.45cm}
1714 \frontcover@maintext
1715 \end{minipage}\par
1716 \vspace*{7.5mm}
1717 \nohyphens{%
1718 \begin{tabularx}{8.45cm}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
1719 \local@advisorstring: &
1720 \count1=0
1721 \toks@={}
1722 \@whilenum \count1<\@advisor \do{%
1723 \ifcase\count1 % same as \ifnum0=\count1
1724 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1725 \expandafter\endcsname\expandafter\space%
1726 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1727 \else
1728 \toks@=\expandafter\expandafter\expandafter{%
1729 \expandafter\the\expandafter\toks@%
1730 \expandafter&\expandafter\space%
1731 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1732 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1733 }%
1734 \fi
1735 \advance\count1 by 1}
1736 \the\toks@
1737 \end{tabularx}}\par
1738 \end{flushright}
1739 \vspace*{\fill}
1740 \begin{center}
1741 \local@cityname\par
1742 \local@monthname\ de \coppe@docyear
1743 \end{center}
1744 \end{titlepage}
1745 % Ficha catalográfica on the verso of the folha de rosto (NBR 14724, R57):
1746 % the title page is a recto, so this lands on the immediately following verso,
1747 % ahead of the approval sheet produced by \frontmatter.
1748 \ifthenelse{\boolean{maledoc}}{\thispagestyle{empty}\makecatalog}%
1749 \global\let\maketitle\relax%
1750 \global\let\and\relax}

1751 \newcommand\makefrontpage{%
1752 \begin{center}
1753 \sloppy\nohyphens{\MakeUppercase{\coppe@selecttitle}}\par
1754 \vspace*{7mm}
1755 {\@authname\ \@authsurname}\par
1756 \end{center}\par

```

```

1757 \vspace*{4mm}
1758 \frontpage@maintext
1759 \vspace*{16mm}
1760 \nohyphens{%
1761 \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }}>{\raggedright\arraybackslash}X@{}}
1762 \local@advisorstring: &
1763 \count1=0
1764 \toks@={}
1765 \@whilenum \count1<\@advisor \do{%
1766 \ifcase\count1 % same as \ifnum0=\count1
1767 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1768 \expandafter\endcsname\expandafter\space%
1769 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1770 \else
1771 \toks@=\expandafter\expandafter\expandafter{%
1772 \expandafter\the\expandafter\toks@%
1773 \expandafter&\expandafter\space%
1774 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1775 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1776 }%
1777 \fi
1778 \advance\count1 by 1}
1779 \the\toks@
1780 \end{tabularx}}\par
1781 \vspace*{20mm}
1782 \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }}>{\raggedright\arraybackslash}X@{}}
1783 Aprovada por: &
1784 \count1=0
1785 \toks@={}
1786 \@whilenum \count1<\@examiner \do{%
1787 \ifcase\count1 % same as \ifnum0=\count1
1788 \toks@=\expandafter{\csname CoppeExaminer:\the\count1%
1789 \expandafter\endcsname\expandafter\\}
1790 \else
1791 \toks@=\expandafter\expandafter\expandafter{%
1792 \expandafter\the\expandafter\toks@%
1793 \expandafter&\expandafter\space%
1794 \csname CoppeExaminer:\the\count1\expandafter\endcsname%
1795 \expandafter\space\\}
1796 }%
1797 \fi
1798 \advance\count1 by 1}
1799 \the\toks@
1800 \end{tabularx}}\par
1801 \vspace*{\fill}
1802 \frontpage@bottomtext}

1803 \newcommand\coppe@hypersetup{%
1804 \beginingroup
1805 % changes to \toks@ and \count@ are kept local;
1806 % it's not necessary for them, but it is usually the case
1807 % for \count1, because the first ten counters are written
1808 % to the DVI file, thus you got lucky because of PDF output
1809 \toks@={} % in this special case not necessary

```



```

1810 \count@=0 %
1811 \@whilenum\count@<\value{keywords}\do{%
1812   % * a keyword separator is not necessary,
1813   %   if there is just one keyword
1814   % * \csname CoppeKeyword:\the\count@\endcsname must be expanded
1815   %   at least once, to get rid of the loop depended \count@
1816   \ifcase\count@ % same as \ifnum0=\count@
1817     \toks@=\expandafter{\csname CoppeKeyword:\the\count@\endcsname}%
1818   \else
1819     \toks@=\expandafter\expandafter\expandafter{%
1820       \expandafter\the\expandafter\toks@
1821       \expandafter;\expandafter\space
1822       \csname CoppeKeyword:\the\count@\endcsname
1823     }%
1824   \fi
1825   \advance\count@ by 1 %
1826 }%
1827 \edef\x{\endgroup
1828   \noexpand\hypersetup{%
1829     pdfkeywords={\the\toks@}%
1830   }%
1831 }%
1832 \x
1833 \hypersetup{%
1834   pdfauthor={\@authname\ \@authsur},
1835   pdftitle={\local@title},
1836   pdfsubject={\local@doctype\ de \@degree\ em \local@deptname\ da COPPE/UFRJ},
1837   pdfcreator={LaTeX with CoppeTeX toolkit},
1838   breaklinks={true},
1839   raiselinks={true},
1840   pageanchor={true},
1841 }}

```

`\makecatalog` When the document has illustrations, it is required to insert “: il,” between the number of pages of the textual part and the page dimension. We have created a label to flag the existence of lists of figures. It is checked to be undefined using the plain T_EX command `\isundefined` (cf. the T_EX FAQ).

```

1842 \newcommand\makecatalog{%
1843   \vspace*{\fill}
1844   \begin{center}
1845     \setlength{\fboxsep}{5mm}
1846     \framebox[120mm][c]{\makebox[5mm][c]{}%
1847       \begin{minipage}[c]{105mm}
1848         \setlength{\parindent}{5mm}
1849         \noindent\sloppy\nohyphens\@authsur,
1850         \nohyphens\@authname\par
1851         \nohyphens{%
1852           \coppe@selecttitle/\@authname\ \@authsur. -- \local@cityname:
1853           UFRJ/COPPE, \coppe@docyear.}\par
1854         \pageref{front:pageno},
1855         \pageref{LastPage}
1856         p.\@ifundefined{r@cat:lofflag}{\pageref{cat:lofflag}}{$29,7$cm.\par
1857         % There is an issue here. When the last entry must be split between lines,
1858         % the spacing between it and the next paragraph becomes smaller.

```

```

1859 % Should we manually introduce a fixed space? But how could we know that
1860 % a name was split? Is this happening yet?
1861 \nohyphens{%
1862 \begin{tabularx}{100mm}[b]{@{}l@{ }}>{\raggedright\arraybackslash}X@{}}
1863 \local@advisorstring: &
1864 \count1=0
1865 \toks@={}
1866 \@whilenum \count1<\@advisor \do{%
1867 \ifcase\count1 % same as \ifnum0=\count1
1868 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1869 \expandafter\endcsname\expandafter\space%
1870 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1871 \else
1872 \toks@=\expandafter\expandafter\expandafter{%
1873 \expandafter\the\expandafter\toks@
1874 \expandafter&\expandafter\space
1875 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1876 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1877 }%
1878 \fi
1879 \advance\count1 by 1}
1880 \the\toks@
1881 \end{tabularx}}\par
1882 \nohyphens{\local@doctype\ ({\MakeLowercase\@degreename}) --
1883 UFRJ/COPPE/Programa de \local@deptname, \coppe@docyear.}\par
1884 Refer{\^ e}ncias Bibliogr{\` a}ficas: p. \pageref{bib:begin} -- \pageref{bib:end}.\par
1885 \count1=0
1886 \count2=1
1887 \nohyphens{\@whilenum \count1<\value{keywords} \do {%
1888 \number\count2. \csname CoppeKeyword:\the\count1 \endcsname.
1889 \advance\count1 by 1
1890 \advance\count2 by 1}
1891 I. \csname CoppeAdvisorSurname:0\endcsname,%
1892 \ \csname CoppeAdvisorName:0\endcsname%
1893 \ifthenelse{\@advisor>1}{\ \emph{et~al.}}{}}.
1894 II. \local@universityname, COPPE, Programa de \local@deptname.
1895 III. T{\` i}tulo.}
1896 \end{minipage}}
1897 \end{center}
1898 \vspace*{\fill}}

```

\dedication

```

1899 \newcommand\dedication[1]{
1900 \gdef\@dedic{#1}
1901 \cleardoublepage
1902 \vspace*{\fill}
1903 \begin{flushright}
1904 \begin{minipage}{60mm}
1905 \raggedleft \it \normalsize \@dedic
1906 \end{minipage}
1907 \end{flushright}}

```

abstract (*env.*) This is a specialization of the abstract in the article standard class.

```

1908 \newenvironment{abstract}{%

```

```

1909 \cleardoublepage
1910 \thispagestyle{plain}
1911 \abstract@toptext\par
1912 \vspace*{8.6mm}
1913 \begin{center}
1914   \sloppy\nohyphens{\MakeUppercase\local@title}\par
1915   \vspace*{13.2mm}
1916   \@authname\ \@authsurn \par
1917   \vspace*{7mm}
1918   \local@monthname/\coppe@docyear
1919 \end{center}\par
1920 \vspace*{1.5cm}
1921 \noindent%
1922 \begin{tabularx}{\textwidth}[b]{@{}l@{ }>\raggedright\arraybackslash}X@{}}
1923   \local@advisorstring: &
1924   \count1=0
1925   \toks@={}
1926   \@whilenum \count1<\@advisor \do{%
1927     \ifcase\count1 % same as \ifnum0=\count1
1928       \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1929         \expandafter\endcsname\expandafter\space%
1930         \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1931     \else
1932       \toks@=\expandafter\expandafter\expandafter{%
1933         \expandafter\the\expandafter\toks@
1934         \expandafter&\expandafter\space
1935         \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1936         \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1937     }%
1938   \fi
1939   \advance\count1 by 1}
1940   \the\toks@
1941 \end{tabularx}\par
1942 \vspace*{2mm}
1943 \noindent\local@deptstring: \local@deptname\par
1944 \vspace*{7mm}}{\vspace*{\fill}}

```

foreignabstract (*env.*)

```

1945 \newenvironment{foreignabstract}{%
1946   \cleardoublepage
1947   \thispagestyle{plain}
1948   \begin{otherlanguage}{\coppe@foreignlang}
1949   \foreignabstract@toptext\par
1950   \vspace*{8.6mm}
1951   \begin{center}
1952     \sloppy\nohyphens{\MakeUppercase\foreign@title}\par
1953     \vspace*{13.2mm}
1954     \@authname\ \@authsurn \par
1955     \vspace*{7mm}
1956     \foreign@monthname/\coppe@docyear
1957   \end{center}\par
1958   \vspace*{1.5cm}
1959   \noindent%
1960   \begin{tabularx}{\textwidth}[b]{@{}l@{ }>\raggedright\arraybackslash}X@{}}

```

```

1961 \foreign@advisorstring: &
1962 \count1=0
1963 \toks@={}
1964 \@whilenum \count1<\@advisor \do{%
1965 \ifcase\count1 % same as \ifnum0=\count1
1966 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1967 \expandafter\endcsname\expandafter\space%
1968 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1969 \else
1970 \toks@=\expandafter\expandafter\expandafter{%
1971 \expandafter\the\expandafter\toks@
1972 \expandafter&\expandafter\space
1973 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1974 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
1975 }%
1976 \fi
1977 \advance\count1 by 1}
1978 \the\toks@
1979 \end{tabularx}\par
1980 \vspace*{2mm}
1981 \noindent\foreign@deptstring: \foreign@deptname\par
1982 \vspace*{7mm}}{%
1983 \end{otherlanguage}
1984 \vspace*{\fill}
1985 \global\let\@author\@empty
1986 \global\let\@date\@empty
1987 \global\let\foreign@title\@empty
1988 \global\let\foreign@title\relax
1989 \global\let\local@title\@empty
1990 \global\let\local@title\relax
1991 \global\let\author\relax
1992 \global\let\author\relax
1993 \global\let\date\relax}

```

brazilianabstract (*env.*) Optional third abstract for non-pt main languages: a Brazilian Portuguese abstract printed in addition to **abstract** (main language) and **foreignabstract** (English by default). Wraps its body in `\begin{otherlanguage}{brazilian}` so the babel typography (hyphenation, quotation marks) is correct regardless of the current main language. Uses the same `\local@...` Portuguese strings the **abstract** env uses today. Meaningful only when main is neither **brazilian** nor **english**; harmless otherwise (the document just won't include it).

```

1994 \newenvironment{brazilianabstract}{%
1995 \cleardoublepage
1996 \thispagestyle{plain}
1997 \begin{otherlanguage}{brazilian}
1998 \abstract@toptext\par
1999 \vspace*{8.6mm}
2000 \begin{center}
2001 \sloppy\nohyphens{\MakeUppercase\local@title}\par
2002 \vspace*{13.2mm}
2003 \@authname\ \@authsurn \par
2004 \vspace*{7mm}
2005 \local@monthname/\coppe@docyear

```

```

2006 \end{center}\par
2007 \vspace*{1.5cm}
2008 \noindent%
2009 \begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X{}}
2010 \local@advisorstring: &
2011 \count1=0
2012 \toks@={}
2013 \@whilenum \count1<\@advisor \do{%
2014 \ifcase\count1
2015 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
2016 \expandafter\endcsname\expandafter\space%
2017 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2018 \else
2019 \toks@=\expandafter\expandafter\expandafter{%
2020 \expandafter\the\expandafter\toks@
2021 \expandafter&\expandafter\space
2022 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
2023 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
2024 }%
2025 \fi
2026 \advance\count1 by 1}
2027 \the\toks@
2028 \end{tabularx}\par
2029 \vspace*{2mm}
2030 \noindent\local@deptstring: \local@deptname\par
2031 \vspace*{7mm}}{%
2032 \end{otherlanguage}
2033 \vspace*{\fill}}

```

\listoffigures

```

2034 \renewcommand\listoffigures{%
2035 \coppe@hasLof
2036 \if@twocolumn
2037 \@restonecoltrue\onecolumn
2038 \else
2039 \@restonecolfalse
2040 \fi
2041 \chapter*{\listfigurename}%
2042 \addcontentsline{toc}{chapter}{\listfigurename}%
2043 \@mkboth{\MakeUppercase\listfigurename}%
2044 {\MakeUppercase\listfigurename}%
2045 \@starttoc{lof}%
2046 \if@restonecol\twocolumn\fi
2047 }

```

\listoftables

```

2048 \renewcommand\listoftables{%
2049 \if@twocolumn
2050 \@restonecoltrue\onecolumn
2051 \else
2052 \@restonecolfalse
2053 \fi
2054 \chapter*{\listtablename}%
2055 \addcontentsline{toc}{chapter}{\listtablename}%

```

```

2056      \@mkboth{%
2057          \MakeUppercase\listtablename}%
2058          {\MakeUppercase\listtablename}%
2059      \@starttoc{lot}%
2060      \if@restonecol\twocolumn\fi
2061      }

\printlosymbols

2062 \newcommand\printlosymbols{%
2063 \renewcommand\glossaryname{\listsymbolname}%
2064 \@input@{\jobname.los}}

\makelosymbols

2065 \def\makelosymbols{%
2066   \newwrite\@losfile
2067   \immediate\openout\@losfile=\jobname.syx
2068   \newcommand\syml[3][\@bsphack\beginingroup
2069   \ifstrepty{##1}{\def\@tempsymb1{##2=}}{\def\@tempsymb1{##1=}}%
2070       \@sanitize%
2071       \@wrls{\@tempsymb1}{##2}{##3}}\typeout%
2072   {Writing index of symbols file \jobname.syx}%
2073   \let\makelosymbols\@empty%
2074 }%
2075 \@onlypreamble\makelosymbols

2076 \AtBeginDocument{%
2077 \@ifpackageloaded{hyperref}{%
2078   \newcommand\@wrls[3]{%
2079     \protected@write\@losfile{%
2080       {\string\indexentry{#1[#2] #3|hyperpage}{\thepage}}%
2081     \endgroup%
2082     \@esphack}}{%
2083   \newcommand\@wrls[3]{%
2084     \protected@write\@losfile{%
2085       {\string\indexentry{#1[#2] #3}{\thepage}}%
2086     \endgroup%
2087     \@esphack}}}%

\printloabbreviations

2088 \newcommand\printloabbreviations{%
2089 \renewcommand\glossaryname{\listabbreviationname}%
2090 \@input@{\jobname.lab}}

\makeloabbreviations

2091 \def\makeloabbreviations{%
2092   \newwrite\@labfile
2093   \immediate\openout\@labfile=\jobname.abx
2094   \newcommand\abbrev[3][\@bsphack\beginingroup
2095   \ifstrepty{##1}{\def\@tempsymb1{##2=}}{\def\@tempsymb1{##1=}}
2096       \@sanitize
2097       \@wrlab{\@tempsymb1}{##2}{##3}}\typeout
2098   {Writing index of abbreviations file \jobname.abx}%
2099   \let\makeloabbreviations\@empty
2100 }
2101 \@onlypreamble\makeloabbreviations

```

```

2102 \AtBeginDocument{%
2103 \ifpackageloaded{hyperref}{%
2104   \newcommand\wrlab[3]{%
2105     \protected@write\@labfile{%
2106       {\string\indexentry{#1[#2] #3|hyperpage}{\thepage}}%
2107     \endgroup%
2108     \@esphack}}{%
2109   \newcommand\wrlab[3]{%
2110     \protected@write\@labfile{%
2111       {\string\indexentry{#1[#2] #3}{\thepage}}%
2112     \endgroup%
2113     \@esphack}}}%

\newsigla Acronyms (siglas), manual2024 2.8 / R43. \newsigla{key}{SIGLA}{full form}
\sigla declares one; \sigla{key} prints “full form (SIGLA)” on first use and registers it in
the list of abbreviations and acronyms (through \abbrev, so \makeloabbreviations
must be active), then just “SIGLA” on later uses. \sigla*{key} forces the full
form. The first-use flag is set globally so it survives groups (floats, footnotes).

2114 \newcommand\newsigla[3]{%
2115   \@namedef{coppe@sigla@s#1}{#2}%
2116   \@namedef{coppe@sigla@l#1}{#3}%
2117   \newcommand\coppe@sigla@reg[1]{%
2118     \ifundefined{abbrev}{}%
2119     {\abbrev{\@nameuse{coppe@sigla@s#1}}{\@nameuse{coppe@sigla@l#1}}}%
2120   \newcommand\coppe@sigla@full[1]{%
2121     \@nameuse{coppe@sigla@l#1}\space(\@nameuse{coppe@sigla@s#1})}
2122   \newcommand\coppe@sigla@undef[1]{%
2123     \PackageWarning{coppe}{Sigla '#1' nao declarada com \string\newsigla}%
2124     \textbf{???#1???}}
2125   \newcommand\coppe@sigla@first[1]{%
2126     \global\@namedef{coppe@sigla@done#1}{\coppe@sigla@reg{#1}\coppe@sigla@full{#1}}
2127   \newcommand\coppe@sigla@norm[1]{%
2128     \ifundefined{coppe@sigla@s#1}{\coppe@sigla@undef{#1}}%
2129     {\@ifundefined{coppe@sigla@done#1}{\coppe@sigla@first{#1}}%
2130      {\@nameuse{coppe@sigla@s#1}}}%
2131   \newcommand\coppe@sigla@star[1]{%
2132     \ifundefined{coppe@sigla@s#1}{\coppe@sigla@undef{#1}}%
2133     {\global\@namedef{coppe@sigla@done#1}{\coppe@sigla@full{#1}}}%
2134   \newcommand\sigla{\@ifstar\coppe@sigla@star\coppe@sigla@norm}

2135 %%% \AtBeginDocument{%
2136 %%% \ifpackageloaded{hyperref}{%
2137 %%%   \def\wrlab#1#2{%
2138 %%%     \protected@write\@labfile{%
2139 %%%       {\string\indexentry{[#1] #2|hyperpage}{\thepage}}%
2140 %%%     \endgroup
2141 %%%     \@esphack}}{%
2142 %%%   \def\wrlab#1#2{%
2143 %%%     \protected@write\@labfile{%
2144 %%%       {\string\indexentry{[#1] #2}{\arabic{page}}}%
2145 %%%     \endgroup
2146 %%%     \@esphack}}

2147 % Catalog-label writers. Two language-related traps must be avoided here:
2148 % 1. The label NAMES contain ':' and '.', which babel makes active in

```

```

2149 %      some languages (e.g. Brazilian's "shorthand" set). If the write
2150 %      argument is tokenised while one of those is active, pdfTeX aborts
2151 %      with "<inserted text> }\endwrite" because the shorthand expansion
2152 %      injects unbalanced braces. \detokenize emits its argument as
2153 %      catcode-12 characters regardless of the current activation, so the
2154 %      label NAMES survive whatever language is current at writer time.
2155 % 2. Spanish babel redefines \@roman (and therefore \roman{...}) to
2156 %      produce small-caps roman numerals via \es@scroman, whose expansion
2157 %      contains an \uppercase group that is unbalanced inside \write. The
2158 %      same crash signature appears. The fix is to bypass LaTeX's
2159 %      \roman/\arabic counter wrappers entirely and use the raw e-TeX
2160 %      primitives \romannumeral and \number directly on the \c@page count
2161 %      register --- both are fully expandable, language-agnostic, and
2162 %      always produce plain ASCII digits or lowercase roman letters.
2163 %      \Roman{page} is left in place because babel-spanish only redefines
2164 %      \@Roman under the (non-default) ucroman option.
2165 \AtBeginDocument{%
2166   \ltx@ifpackageloaded{hyperref}{%
2167     \def\coppe@bibEnd{%
2168       \immediate\write\@auxout{%
2169         \string\newlabel{\detokenize{bib:end}}{%
2170           {}{\number\c@page}}{\detokenize{page.}\number\c@page}}}}}%
2171   \def\coppe@bibBegin{%
2172     \immediate\write\@auxout{%
2173       \string\newlabel{\detokenize{bib:begin}}{%
2174         {}{\number\c@page}}{\detokenize{page.}\number\c@page}}}}}%
2175   \def\coppe@mainBegin{%
2176     \immediate\write\@auxout{%
2177       \string\newlabel{\detokenize{front:pageno}}{%
2178         {}{\Roman{page}}{\detokenize{page.}\romannumeral\c@page}}}}}%
2179   \def\coppe@hasLof{%
2180     \immediate\write\@auxout{%
2181       \string\newlabel{\detokenize{cat:lofflag}}{%
2182         {}{\detokenize{:~il.}}{\detokenize{page.}\romannumeral\c@page}}}}}%
2183 }{%
2184   \def\coppe@bibEnd{%
2185     \immediate\write\@auxout{%
2186       \string\newlabel{\detokenize{bib:end}}{}}{\number\c@page}}}}}%
2187   \def\coppe@bibBegin{%
2188     \immediate\write\@auxout{%
2189       \string\newlabel{\detokenize{bib:begin}}{}}{\number\c@page}}}}}%
2190   \def\coppe@mainBegin{%
2191     \immediate\write\@auxout{%
2192       \string\newlabel{\detokenize{front:pageno}}{}}{\Roman{page}}}}}}}%
2193   \def\coppe@hasLof{%
2194     \immediate\write\@auxout{%
2195       \string\newlabel{\detokenize{cat:lofflag}}{}}{\detokenize{:~il.}}}}}}}%
2196 }%
2197 }
2198 % The reference list is now produced by biblatex's \printbibliography (see the
2199 % bibliography setup after babel, above). The former thebibliography
2200 % redefinition and its \bibindent are no longer used and have been removed; the
2201 % bib:begin/bib:end catalog labels are now written by \AtBeginBibliography /
2202 % \AtEndBibliography via \coppe@bibBegin and \coppe@bibEnd, defined just above.

```


longquote (*env.*)

```
2203 \newlength{\recuolongquote}
2204 \setlength{\recuolongquote}{4cm}
2205
2206 \newenvironment*{longquote}[1][default]{%
2207 \begin{list}{}{%
2208 \setlength{\leftmargin}{\recuolongquote}%
2209 \setlength{\rightmargin}{0pt}%
2210 \setlength{\itemindent}{0pt}%
2211 \setlength{\listparindent}{0pt}%
2212 \setlength{\parsep}{0pt}%
2213 \setlength{\topsep}{\baselineskip}%
2214 }%
2215 \item\relax
2216 \footnotesize
2217 \singlespacing
2218 \ifthenelse{not\equal{#1}{default}}{%
2219 {\itshape\selectlanguage{#1}}%
2220 }%
2221 \ignorespaces
2222 }{%
2223 \end{list}%
2224 \ignorespacesafterend
2225 }

2226 \newenvironment{theglossary}{%
2227 \if@twocolumn%
2228 \@restonecoltrue\onecolumn%
2229 \else%
2230 \@restonecolfalse%
2231 \fi%
2232 \@mkboth{\MakeUppercase\glossaryname}%
2233 {\MakeUppercase\glossaryname}%
2234 \chapter*{\glossaryname}%
2235 \addcontentsline{toc}{chapter}{\glossaryname}
2236 \list{}
2237 {\setlength{\listparindent}{0in}%
2238 \setlength{\labelwidth}{1.0in}%
2239 \setlength{\leftmargin}{1.5in}%
2240 \setlength{\labelsep}{0.5in}%
2241 \setlength{\itemindent}{0in}}%
2242 \sloppy}%
2243 {\if@restonecol\twocolumn\fi%
2244 \endlist}
2245 %
2246 \renewenvironment{theindex}{%
2247 \if@twocolumn
2248 \@restonecolfalse
2249 \else
2250 \@restonecoltrue
2251 \fi
2252 \twocolumn[\@makeschapterhead{\indexname}]%
2253 \@mkboth{\MakeUppercase\indexname}%
2254 {\MakeUppercase\indexname}%
```

```

2255 \thispagestyle{plain}\parindent\z@
2256 \addcontentsline{toc}{chapter}{\indexname}
2257 \parskip\z@ \@plus .3\p@\relax
2258 \columnseprule \z@
2259 \columnsep 35\p@
2260 \let\item\@idxitem}
2261 {\if@restonecol\onecolumn\else\clearpage\fi}
2262 \newcommand\listabbreviationname{\coppemainstring{listabbreviation}}
2263 \newcommand\listsymbolname{\coppemainstring{listsymbol}}
2264 \newcommand\glossaryname{\coppemainstring{glossary}}
2265 %
2266 \newcommand\local@advisorstring{Orientador}
2267 \newcommand\foreign@advisorstring{Advisor}
2268 \ifthenelse{\boolean{maledoc}}{%
2269 \newcommand\local@approvedname{Examinado por}%
2270 }{%
2271 \newcommand\local@approvedname{Examinada por}%
2272 }
2273 \newcommand\local@universityname{Universidade Federal do Rio de Janeiro}
2274 \newcommand\local@deptstring{Programa}
2275 \newcommand\foreign@deptstring{Department}
2276 \newcommand\local@cityname{Rio de Janeiro}
2277 \newcommand\local@statename{RJ}
2278 \newcommand\local@countryname{Brasil}
2279 %
2280 \newcommand\frontcover@maintext{
2281 \sloppy\nohyphens{\local@doctype\ de \@degreename\
2282 \ifthenelse{\boolean{maledoc}}{apresentado}{apresentada}
2283 ao Programa de P{\ ' o}s-gradua{\c c}{\~ a}o em \local@deptname,
2284 COPPE, da \local@universityname, como parte dos requisitos
2285 necess{\ ' a}rios {\ ' a} obten{\c c}{\~ a}o do t{\ ' i}tulo de
2286 \local@degname\ em \local@deptname.}
2287 }
2288 %
2289 \newcommand\frontpage@maintext{
2290 \noindent {\MakeUppercase\local@doctype}
2291 \ifthenelse{\boolean{maledoc}}{SUBMETIDO}{SUBMETIDA}
2292 \sloppy\nohyphens{AO CORPO DOCENTE DO INSTITUTO ALBERTO LUIZ COIMBRA
2293 DE P{\ ' O}S-GRADUA{\c C}{\~ A}O E PESQUISA DE ENGENHARIA DA
2294 UNIVERSIDADE FEDERAL DO RIO DE JANEIRO COMO PARTE DOS REQUISITOS
2295 NECESS{\ ' A}RIOS PARA A OBTEN{\c C}{\~ A}O DO GRAU DE
2296 {\MakeUppercase\local@degname} EM CI{\^E}NCIAS EM
2297 {\MakeUppercase\local@deptname.\par}}%
2298 }
2299 %
2300 \newcommand\frontpage@bottomtext{%
2301 \begin{center}
2302 {\MakeUppercase{\local@cityname, \local@statename\ -- \local@countryname}}\par
2303 {\MakeUppercase\local@monthname\ DE \coppe@docyear}
2304 \end{center}}%
2305 }
2306 %
2307 \newcommand\abstract@toptext{%
2308 \noindent Resumo \ifthenelse{\boolean{maledoc}}{do}{da}

```

```

2309 \local@doctype\ \ifthenelse{\boolean{maledoc}}{apresentado}{apresentada}
2310 \sloppy\nohyphens{{\' a} COPPE/UFRJ como parte dos requisitos
2311 necess{{\' a}rios para a obten{{c c}}{\^ a}o do grau de
2312 \local@degname\ em Cif{\^ e}ncias (\@degree)}
2313 }
2314 \newcommand\foreignabstract@toptext{%
2315 \noindent \sloppy\nohyphens{Abstract of \foreign@doctype\ presented to
2316 COPPE/UFRJ as a partial fulfillment of the requirements for the
2317 degree of \foreign@degname\ of Science (\@degree)}
2318 }
2319 %

```

Late-load packages: setspace, hyperref, biblatex

These three packages *must* be the last things the class loads, and in this exact order:

- **setspace** – supplies `\singlespacing`, `\onehalfspacing` and friends. It is loaded here so the line-spacing state it carries around is settled before **hyperref** patches all the cross-reference and TOC plumbing.
- **hyperref** – patches an enormous number of LaTeX internals (`\section`, `\label`, `\ref`, `\bibcite`, `list-of-*` commands, ...). Loading it as late as possible is the standard rule of thumb so it sees the final version of those macros, including the ones the **coppe** class redefines (e.g. `\@chapter`, `\listofquadros`).
- **biblatex** – must come after **hyperref** so its citation links and back-references hook into **hyperref** correctly; otherwise the `\cite` commands print as plain (non-clickable) text.

```

2320 \RequirePackage{setspace}
2321 % Apply the default line spacing now that setspace is available. The flag was
2322 % set above by \DeclareOption{doublespacing}; absent the option, one-and-a-half.
2323 \if@coppedoublespace\doublespacing\else\onehalfspacing\fi
2324 \RequirePackage{hyperref}
2325 % Bibliography engine: the full coppe BibLaTeX/biber ABNT style, replacing the
2326 % former natbib+bibtex setup. The numbers class option selects the numeric
2327 % citation style; otherwise the author-date default is used. natbib=true keeps
2328 % \citet/\citep available for users migrating from natbib.
2329 \if@coppenumeric
2330 \RequirePackage[backend=biber,datamodel=coppe,%
2331 bibstyle=coppe-numeric,citestyle=coppe-numeric,natbib=true]{biblatex}
2332 \else
2333 \RequirePackage[backend=biber,datamodel=coppe,%
2334 bibstyle=coppe,citestyle=coppe,natbib=true]{biblatex}
2335 \fi
2336 % Reference-list heading "Referencias" (pt) / "References" (en) -- manual2024
2337 % 3.1.4.1 requires the post-textual list to be titled "Referencias", not
2338 % "Bibliografia" (babel's default \bibname). The title is rendered through
2339 % \coppestring so it follows the current main language in BOTH the chapter
2340 % heading and the .toc entry (biblatex's \bibstring proved unreliable across
2341 % the .toc write, and babel's \bibname caption override was equally fragile).
2342 \defbibheading{bibliography}{%

```

```

2343 \chapter*{\coppestring{references}}%
2344 \addcontentsline{toc}{chapter}{\coppestring{references}}
2345 % The catalog page range (bib:begin/bib:end) was written by the old
2346 % thebibliography redefinition, which biblatex's \printbibliography does not
2347 % use. biblatex offers \AtBeginBibliography but no end counterpart, so the
2348 % begin label uses that hook and the end label is emitted by wrapping
2349 % \printbibliography to write it just after the reference list.
2350 \AtBeginBibliography{\coppe@bibBegin}
2351 \let\coppe@orig@printbibliography\printbibliography
2352 \renewcommand\printbibliography[1][]{\%
2353 \coppe@orig@printbibliography[#1]\%
2354 \coppe@bibEnd}
2355 \</class>

2356 \<*glossary>
2357 actual '='
2358 quote '!'
2359 level '>'
2360 %%% delim_0 " , p. "
2361 delim_0 "\dotfill "
2362 lethead_flag 0
2363 headings_flag 0
2364 preamble
2365 "\n\begin{theglossary}\n \makeatletter"
2366 postamble
2367 "\n \end{theglossary}\n"
2368 \</glossary>

2369 \<*class>
2370 \newcommand{\annex}{
2371 \renewcommand{\appendixname}{\coppestring{annex}}%
2372 \appendix
2373 \@coppeannextrue
2374 }
2375 \</class>

```

BibLaTeX style files

Generated from this .dtx by coppe.ins (one docstrip module each).

```

2376 \<*dbx>
2377 \ProvidesFile{coppe.dbx}[2026/05/28 v4.0 CoppeTeX ABNT datamodel]
2378 % Custom entry types beyond biblatex's standard datamodel (games, TV, maps).
2379 % map = cartographic document (NBR 6023 4.2.12); biblatex has no native @map.
2380 % The remaining ABNT "exotic" types (legislation, jurisdiction, legal, patent,
2381 % standard, music, audio, image, artwork, video, dataset, letter, periodical,
2382 % performance) are already in biblatex's standard datamodel.
2383 \DeclareDatamodelEntrytypes{game,videogame,boardgame,tvshow,map}
2384 % Custom fields.
2385 \DeclareDatamodelFields[type=field,datatype=literal]{course} % "(... em <course>)"
2386 \DeclareDatamodelFields[type=field,datatype=literal]{coppedegree} % degree word
2387 \DeclareDatamodelFields[type=field,datatype=literal]{platform} % videogame platform
2388 \DeclareDatamodelFields[type=field,datatype=literal]{scale} % map scale (escala)
2389 \DeclareDatamodelFields[type=list,datatype=literal]{artist} % game/illustration art
2390 \DeclareDatamodelFields[type=list,datatype=literal]{director} % movie/tvshow/video

```

```

2391 \DeclareDatamodelFields[type=list,datatype=literal]{producer}
2392 \DeclareDatamodelFields[type=field,datatype=literal]{caderno}      % newspaper section
2393 \DeclareDatamodelFields[type=field,datatype=literal]{relator}      % jurisprudence rapporteur
2394 \DeclareDatamodelEntryfields{course,coppedegree}
2395 \DeclareDatamodelEntryfields[game,videogame,boardgame,software,movie,tvshow,video]%
2396   {artist,platform,director,producer,version}
2397 \DeclareDatamodelEntryfields[map]{scale}
2398 \DeclareDatamodelEntryfields[article]{caderno}
2399 \DeclareDatamodelEntryfields[jurisdiction]{relator}
2400 </dbx>
2401 <*bbx>
2402 \ProvidesFile{coppe.bbx}[2026/05/28 v4.0 CoppeTeX ABNT bibliography style]
2403 % The reference list is single-spaced (ABNT 2.4) via \singlespacing in
2404 % \AtBeginBibliography. coppe.cls already loads setspace, but the style must work
2405 % standalone too (e.g. \usepackage[bibstyle=coppe]{biblatex} in a plain article),
2406 % so load setspace here as well (idempotent).
2407 \RequirePackage{setspace}
2408 % ABNT reference list keeps the date at the end (imprenta), same for author-date
2409 % and numeric modes, so we derive from the standard style, not authoryear.
2410 \RequireBibliographyStyle{standard}
2411
2412 % Note: standard style never dashes repeated authors, which already matches the
2413 % post-2020 ABNT rule (repeat the entry in full), so no 'dashed' option is needed.
2414 \ExecuteBibliographyOptions{%
2415   maxbibnames=99,      % ABNT 4.3.1.3: list all authors by default
2416   giveninits=false,
2417   uniquename=false,
2418   uniquelist=false,
2419   labeldateparts=true, % needed for author-date extradate (1999a/1999b)
2420   sorting=nyt,
2421 }
2422
2423 % Language strings (Disponível em / Acesso em / In, etc.)
2424 \DeclareLanguageMapping{brazilian}{brazilian-coppe}
2425 \DeclareLanguageMapping{english}{english-coppe}
2426
2427 % --- Names: all reversed "FAMILY, Given", separated by ";" in the list ---
2428 \DeclareNameAlias{author}{family-given}
2429 \DeclareNameAlias{editor}{family-given}
2430 \DeclareNameAlias{translator}{family-given}
2431 \DeclareNameAlias{default}{family-given}
2432 \DeclareDelimFormat[bib,biblist]{multinamedelim}{\addsemicolon\space}
2433 \DeclareDelimFormat[bib,biblist]{finalnamedelim}{\addsemicolon\space}
2434
2435 % ABNT (NBR 6023 8.1.1.4): a collection entered under the name of its
2436 % organizer prints the kind of responsibility in parentheses after the
2437 % name, abbreviated, lowercase and in the singular even for several
2438 % organizers -- e.g. "SOUSA, A. M. de; OLIVEIRA, L. da C. (org.)". The
2439 % standard style prints ", Org." (comma, no parentheses) when the editor
2440 % takes the author slot, so override editor+others to wrap the role string
2441 % in "(...)". The host position ("In: NAME (org.)") already parenthesises
2442 % the role via coppe:bookeditor below. The role word itself is the
2443 % lowercase, singular "org." (and "ed.", "cur.", ...) set in the language
2444 % packs (brazilian-coppe.lbx et al.).

```

```

2445 \renewbibmacro*{editor+others}{%
2446   \ifboolexpr{ test \ifuseeditor and not test {\ifnameundef{editor}} }
2447     {\printnames{editor}%
2448      \setunit*{\addspace}%
2449      \printtext{\mkbibparens{\usebibmacro{editor+othersstrg}}}%
2450      \clearname{editor}}
2451   {}}
2452 % Same parenthesised role for drivers that take the editor through the plain
2453 % editor macro instead of editor+others (e.g. @manual).
2454 \renewbibmacro*{editor}{%
2455   \ifboolexpr{ test \ifuseeditor and not test {\ifnameundef{editor}} }
2456     {\printnames{editor}%
2457      \setunit*{\addspace}%
2458      \printtext{\mkbibparens{\usebibmacro{editorstrg}}}%
2459      \clearname{editor}}
2460   {}}
2461
2462 % Uppercase the surname (family + prefix) only inside the bibliography list;
2463 % citations keep normal caps (post-2020 rule).
2464 % ABNT date (NBR 6023 §4.3.5.5.2): "dia mês. ano" -- without the brazilian "de"
2465 % between the parts (e.g. "4 fev. 1992", "abr. 1996"). #1=year, #2=month, #3=day
2466 % field names; the month stays abbreviated via \mkbibmonth.
2467 \protected\def\coppe@abntdate#1#2#3{%
2468   \iffieldundef{#3}{\stripzeros{\thefield{#3}}\nobreakspace}%
2469   \iffieldundef{#2}{\mkbibmonth{\thefield{#2}}\iffieldundef{#1}{\space}}%
2470   \iffieldundef{#1}{\stripzeros{\thefield{#1}}}}
2471 \AtBeginBibliography{%
2472   \singlespacing % entries single-spaced internally (2.4)
2473   \setlength{\bibitemsep}{\baselineskip}% one blank line between entries (2.4)
2474   \renewcommand*{\mkbibnamefamily}[1]{\MakeUppercase{#1}}%
2475   \renewcommand*{\mkbibnameprefix}[1]{\MakeUppercase{#1}}%
2476   \let\mkbibdatelong\coppe@abntdate % ABNT date format (drop the "de")
2477   \let\mkbibdateshort\coppe@abntdate
2478 }
2479
2480 % --- Titles: bold for whole-document types; plain for parts; entry-by-title
2481 % (no author/editor) -> not bold, first word in CAPS (ABNT 4.2). ---
2482 % \coppeUCtitle uppercases the first SIGNIFICANT word of the title (ABNT 4.2).
2483 % If the title starts with an article (o/a/os/as/um/uma/uns/umas, the/an), the
2484 % article AND the next word are uppercased; otherwise just the first word. The
2485 % delimited macros split off words; \coppeUCkeep strips the trailing-space
2486 % sentinel so no spurious space is left before the following punctuation.
2487 \protected\def\coppeUCtitle#1{\coppeUCtitleA#1 \coppeUCstop}
2488 \protected\def\coppeUCtitleA#1 #2\coppeUCstop{%
2489   \coppeUCifart{#1}
2490     {\MakeUppercase{#1}\ifblank{#2}{\space\coppeUCtitleB#2\coppeUCstop}}%
2491     {\MakeUppercase{#1}\ifblank{#2}{\space\coppeUCkeep#2\coppeUCstop}}%
2492   \protected\def\coppeUCtitleB#1 #2\coppeUCstop{%
2493     \MakeUppercase{#1}\ifblank{#2}{\space\coppeUCkeep#2\coppeUCstop}}
2494   \protected\def\coppeUCkeep#1 \coppeUCstop{#1}
2495   % \coppeUCifart{word}{if-article}{if-not}: case-insensitive membership test.
2496   \protected\def\coppeUCifart#1{%
2497     \begingroup
2498     \lowercase{\def\coppeUC@w{#1}}%

```

```

2499 \edef\coppeUC@w{\coppeUC@w,}%
2500 \expandafter\in@\expandafter{\coppeUC@w}{,a,o,as,os,um,uma,uns,umas,the,an,}%
2501 \ifin@\aftergroup\@firstoftwo\else\aftergroup\@secondoftwo\fi
2502 \endgroup}
2503 \DeclareFieldFormat{title}{%
2504 \ifboolexpr{ test {\ifnameundef{labelname}} }
2505     {#1}%
2506     {\mkbibbold{#1}}}
2507 % First word of the title in CAPS for entry-by-title (ABNT 4.2). The titlecase
2508 % format receives the RAW title text (the title format above only sees the
2509 % field-printing commands), so \coppeUCtitle can split off and uppercase the
2510 % first word here.
2511 \DeclareFieldFormat{titlecase}{%
2512 \ifboolexpr{ test {\ifnameundef{labelname}} }
2513     {\coppeUCtitle{#1}}%
2514     {#1}}
2515 \DeclareFieldFormat[article,incollection,inbook,inproceedings,supperiodical,review]{title}{%
2516 \DeclareFieldFormat[thesis]{title}{\mkbibbold{#1}}
2517 % biblatex's standard style quotes patent (and unpublished) titles; ABNT bolds
2518 % the title of a whole document, so override those back to bold.
2519 \DeclareFieldFormat[patent]{title}{\mkbibbold{#1}}
2520 % Legal documents (NBR 6023 4.2.6): the law's identification/ementa is NOT
2521 % highlighted; the entry leads with the jurisdiction (entity author, CAPS).
2522 \DeclareFieldFormat[legislation,jurisdiction,legal]{title}{#1}
2523 \DeclareFieldFormat[journaltitle]{\mkbibbold{#1}}
2524 \DeclareFieldFormat{booktitle}{\mkbibbold{#1}}
2525 \DeclareFieldFormat{maintitle}{\mkbibbold{#1}}
2526
2527 % --- Pages: ABNT uses "p." (also for ranges); theses use "f." for pagetotal ---
2528 \DeclareFieldFormat{pagetotal}{#1\addnbspace p\adddot}
2529 \DeclareFieldFormat[thesis]{pagetotal}{#1\addnbspace f\adddot}
2530 \DeclareFieldFormat{pages}{p\adddot\addnbspace#1}
2531
2532 % --- Periodical volume / number prefixes ---
2533 \DeclareFieldFormat[article]{volume}{v\adddot\addnbspace#1}
2534 \DeclareFieldFormat[article]{number}{n\adddot\addnbspace#1}
2535
2536 % --- "In:" before host title (incollection/inproceedings/inbook) ---
2537 \renewbibmacro*{in:}{\printtext{\bibstring{in}\addcolon\space}}
2538
2539 % --- Online: "Disponível em: <url>." + "Acesso em: <date>." via lbx strings ---
2540 \DeclareFieldFormat{url}{\bibstring{availablefrom}\addcolon\space\url{#1}}
2541
2542 % --- ABNT article driver: no "In:"; Journal, Local, v., n., p., date, DOI/URL.
2543 % Newspapers (entrysubtype=newspaper, set by the @artigodejornal source map)
2544 % differ: the newspaper name (journaltitle) and city, then either
2545 % "<date>. <caderno>, p. <pages>" (when a caderno/section is given), or
2546 % "p. <pages>, <date>" (page before date, when none),
2547 % per NBR 6023 §4.2.3.6. ---
2548 \DeclareBibliographyDriver{article}{%
2549 \usebibmacro{bibindex}%
2550 \usebibmacro{begentry}%
2551 \usebibmacro{author/translator+others}%
2552 \setunit{\labelnamepunct}\newblock

```

```

2553 \usebibmacro{title}%
2554 \newunit\newblock
2555 \printfield{journaltitle}%
2556 \setunit{\addcomma\space}\printlist{location}%
2557 \iffieldequalstr{entrysubtype}{newspaper}
2558   {\iffieldundef{caderno}
2559     {\setunit{\addcomma\space}\printfield{pages}%
2560       \setunit{\addcomma\space}\printdate}%
2561     {\setunit{\addcomma\space}\printdate%
2562       \setunit{\adddot\space}\printfield{caderno}%
2563       \setunit{\addcomma\space}\printfield{pages}}}%
2564   {\setunit{\addcomma\space}\printfield{volume}%
2565     \setunit{\addcomma\space}\printfield{number}%
2566     \setunit{\addcomma\space}\printfield{pages}%
2567     \setunit{\addcomma\space}\printdate}%
2568 \newunit\newblock
2569 \printfield{doi}%
2570 \newunit\newblock
2571 \usebibmacro{url+urldate}%
2572 \usebibmacro{finentry}}
2573
2574 % --- incollection: ABNT puts the host organizer before the booktitle, as
2575 %   "AUTOR. Parte. In: ORGANIZADOR (Org.). Booktitle. Local: Editora, ano,
2576 %   p. X-Y." The organizer role string is "Org./Orgs." (see the .lbx). ---
2577 \newbibmacro*{coppe:bookeditor}{%
2578   \ifnameundef{editor}
2579     {}
2580     {\printnames{editor}%
2581       \setunit*{\addspace}\printtext{\mkbibparens{\usebibmacro{editorstrg}}}%
2582       \clearname{editor}%
2583       \setunit{\adddot\space}}}%
2584 \newbibmacro*{coppe:partdriver}{%
2585   \usebibmacro{bibindex}\usebibmacro{begentry}%
2586   \usebibmacro{author/translator+others}%
2587   \setunit{\labelnamepunct}\newblock
2588   \usebibmacro{title}%
2589   \newunit\newblock
2590   \usebibmacro{in:}%
2591   \usebibmacro{coppe:bookeditor}%
2592   \printfield{booktitle}%
2593   \newunit\newblock
2594   \printfield{edition}%
2595   \newunit\newblock
2596   \printlist{location}%
2597   \setunit*{\addcolon\space}\printlist{publisher}%
2598   \setunit*{\addcomma\space}\printdate%
2599   \newunit\newblock
2600   \printfield{pages}%
2601   \setunit{\addcomma\space}\printfield{chapter}%
2602   \newunit\newblock
2603   \usebibmacro{url+urldate}%
2604   \usebibmacro{finentry}}
2605 \DeclareBibliographyDriver{incollection}{\usebibmacro{coppe:partdriver}}
2606 \DeclareBibliographyDriver{inbook}{\usebibmacro{coppe:partdriver}}

```



```

2607
2608 % --- type field: resolve a known bibliography string (babel-aware), else literal ---
2609 \DeclareFieldFormat{type}{\ifbibstring{#1}{\bibstring{#1}}{#1}}
2610
2611 % --- Thesis: "<type> (em <course>)" using the custom course field (ABNT). ---
2612 \DeclareFieldFormat{course}{\mkbibparens{\coppestring{in:}\addspace#1}}
2613 \DeclareBibliographyDriver{thesis}{%
2614   \usebibmacro{bibindex}\usebibmacro{begentry}%
2615   \usebibmacro{author}%
2616   \setunit{\labelnamepunct}\newblock
2617   \usebibmacro{title}%
2618   \newunit\newblock
2619   \printfield{type}\setunit*{\addspace}\printfield{course}%
2620   \newunit\newblock
2621   \printlist{institution}%
2622   \setunit*{\addcomma\space}\printlist{location}%
2623   \setunit*{\addcomma\space}\printdate%
2624   \newunit\newblock
2625   \printfield{pagetotal}%
2626   \newunit\newblock
2627   \printfield{note}%
2628   \newunit\newblock
2629   \usebibmacro{url+urldate}%
2630   \usebibmacro{finentry}}
2631 % --- Games / audiovisual: render the custom fields (artist, platform,
2632 %       director, producer, version) the generic misc driver ignores. ---
2633 \newbibmacro*{coppe:media}{%
2634   \iflistundef{artist}{\setunit{\addperiod\space}%
2635     \printtext{\coppestring{art}\addcolon\space}\printlist{artist}}%
2636   \iffieldundef{platform}{\setunit{\addperiod\space}%
2637     \printtext{\coppestring{platform}\addcolon\space}\printfield{platform}}%
2638   \iflistundef{director}{\setunit{\addperiod\space}%
2639     \printtext{\coppestring{directedby}\addcolon\space}\printlist{director}}%
2640   \iflistundef{producer}{\setunit{\addperiod\space}%
2641     \printtext{\coppestring{producedby}\addcolon\space}\printlist{producer}}%
2642 \newbibmacro*{coppe:mediadriv}{%
2643   \usebibmacro{bibindex}\usebibmacro{begentry}%
2644   \usebibmacro{author/translator+others}%
2645   \setunit{\labelnamepunct}\newblock
2646   \usebibmacro{title}%
2647   \newunit\newblock
2648   \usebibmacro{coppe:media}%
2649   \newunit\newblock
2650   \printlist{publisher}%
2651   \setunit*{\addcomma\space}\printlist{location}%
2652   \setunit*{\addcomma\space}\printdate%
2653   \newunit\newblock
2654   \printfield{version}%
2655   \newunit\newblock
2656   \printfield{note}%
2657   \newunit\newblock
2658   \usebibmacro{url+urldate}%
2659   \usebibmacro{finentry}}
2660 \DeclareBibliographyDriver{game}{\usebibmacro{coppe:mediadriv}}

```

```

2661 \DeclareBibliographyDriver{videogame}{\usebibmacro{coppe:mediadrivere}}
2662 \DeclareBibliographyDriver{boardgame}{\usebibmacro{coppe:mediadrivere}}
2663 \DeclareBibliographyDriver{movie}{\usebibmacro{coppe:mediadrivere}}
2664 \DeclareBibliographyDriver{tvshow}{\usebibmacro{coppe:mediadrivere}}
2665 % Cartographic document (map, NBR 6023 4.2.12): like a book (author/title/
2666 % imprenta) plus the mandatory scale ("Escala 1:N").
2667 \DeclareFieldFormat{scale}{\bibstring{coppescale}\adnbspace#1}
2668 \DeclareBibliographyDriver{map}{%
2669   \usebibmacro{bibindex}\usebibmacro{begentry}%
2670   \usebibmacro{author/translator+others}%
2671   \setunit{\labelnamepunct}\newblock
2672   \usebibmacro{title}%
2673   \newunit\newblock
2674   \printlist{location}%
2675   \setunit*{\addcolon\space}\printlist{publisher}%
2676   \setunit*{\addcomma\space}\printdate%
2677   \newunit\newblock
2678   \printfield{scale}%
2679   \newunit\newblock
2680   \usebibmacro{url+urldate}%
2681   \usebibmacro{finentry}}
2682 % Patent (NBR 6023 4.2.5): inventor. \textbf{title}. Depositante: holder. type
2683 % number. date. The holder (depositante) is printed in normal case, not the
2684 % ALL-CAPS family treatment the bibliography gives the leading author.
2685 \newbibmacro*{coppe:depositor}{%
2686   \ifnameundef{holder}
2687     {}
2688     {\printtext{\bibstring{depositor}\addcolon\space}%
2689       \begingroup
2690         \def\mkbibnamefamily##1{##1}%
2691         \def\mkbibnameprefix##1{##1}%
2692         \printnames{holder}%
2693       \endgroup}}
2694 \DeclareBibliographyDriver{patent}{%
2695   \usebibmacro{bibindex}%
2696   \usebibmacro{begentry}%
2697   \usebibmacro{author/translator+others}%
2698   \setunit{\labelnamepunct}\newblock
2699   \usebibmacro{title}%
2700   \newunit\newblock
2701   \usebibmacro{coppe:depositor}%
2702   \newunit\newblock
2703   \printfield{type}%
2704   \setunit*{\addspace}\printfield{number}%
2705   \newunit\newblock
2706   \printdate%
2707   \newunit\newblock
2708   \usebibmacro{url+urldate}%
2709   \usebibmacro{finentry}}
2710 % Exotic ABNT types that biblatex's standard style leaves without a driver:
2711 % give each a baseline (book-like for published docs: norma/partitura; misc for
2712 % the rest) so they print. Per-type ABNT refinements can follow later.
2713 \DeclareBibliographyDriver{standard}{\usedriver{}{book}}
2714 \DeclareBibliographyDriver{music}{\usedriver{}{book}}

```

```

2715 % Legal documents (NBR 6023 4.2.6-4.2.8): jurisdiction/entity (CAPS author),
2716 % the law id + ementa (plain, not bold), then location and date.
2717 \newbibmacro*{coppe:legaldriver}{%
2718   \usebibmacro{bibindex}\usebibmacro{begentry}%
2719   \usebibmacro{author/translator+others}%
2720   \setunit{\labelnamepunct}\newblock
2721   \usebibmacro{title}%
2722   \newunit\newblock
2723   % Jurisprudência: "Relator: <nome>." after the decision title (NBR 6023).
2724   \iffielddundef{relator}{\}\printtext{Relator\addcolon\space}\printfield{relator}\newunit\ne
2725   \printlist{location}%
2726   \setunit{\addcomma\space}\printdate%
2727   \newunit\newblock
2728   \usebibmacro{url+urldate}%
2729   \usebibmacro{finentry}}
2730 \DeclareBibliographyDriver{legislation}{\usebibmacro{coppe:legaldriver}}
2731 \DeclareBibliographyDriver{jurisdiction}{\usebibmacro{coppe:legaldriver}}
2732 \DeclareBibliographyDriver{legal}{\usebibmacro{coppe:legaldriver}}
2733 \DeclareBibliographyDriver{audio}{\usedriver{}{misc}}
2734 \DeclareBibliographyDriver{video}{\usedriver{}{misc}}
2735 \DeclareBibliographyDriver{software}{\usedriver{}{misc}}
2736 \DeclareBibliographyDriver{image}{\usedriver{}{misc}}
2737 \DeclareBibliographyDriver{artwork}{\usedriver{}{misc}}
2738 \DeclareBibliographyDriver{performance}{\usedriver{}{misc}}
2739 \DeclareBibliographyDriver{dataset}{\usedriver{}{misc}}
2740
2741 % --- biber source map: Portuguese synonyms (entry types + fields) + thesis family ---
2742 \DeclareSourcemap{%
2743   \maps[datatype=bibtex]{%
2744     % entry-type PT synonyms -> canonical English
2745     \map{step[typesource=livro,typetarget=book]}
2746     \map{step[typesource=artigo,typetarget=article]}
2747     \map{step[typesource=artigodejornal,typetarget=article,final]\step[fieldset=entrysubtyp
2748     \map{step[typesource=partedelivro,typetarget=incollection]}
2749     \map{step[typesource=emlivro,typetarget=inbook]}
2750     \map{step[typesource=verbete,typetarget=inreference]}
2751     \map{step[typesource=anais,typetarget=proceedings]}
2752     \map{step[typesource=trabalhodeevento,typetarget=inproceedings]}
2753     \map{step[typesource=periodico,typetarget=periodical]}
2754     \map{step[typesource=relatorio,typetarget=report]}
2755     \map{step[typesource=relatoriotecnico,typetarget=report]}
2756     \map{step[typesource>manual,typetarget>manual]}
2757     \map{step[typesource=correspondencia,typetarget=letter]}
2758     \map{step[typesource=patente,typetarget=patent]}
2759     \map{step[typesource=diverso,typetarget=misc]}
2760     \map{step[typesource=jogo,typetarget=game]}
2761     \map{step[typesource=jogoeletronico,typetarget=videogame]}
2762     \map{step[typesource=jogodetabuleiro,typetarget=boardgame]}
2763     \map{step[typesource=filme,typetarget=movie]}
2764     \map{step[typesource=serietv,typetarget=tvshow]}
2765     \map{step[typesource=programadetelevisao,typetarget=tvshow]}
2766     % Exotic ABNT types (NBR 6023 4.2.5-4.2.14); targets are biblatex-native
2767     % except map (declared in coppe.dbx).
2768     \map{step[typesource=legislacao,typetarget=legislation]}

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2769 \map{\step[typesource=atonormativo,typetarget=legislation]}
2770 \map{\step[typesource=jurisprudencia,typetarget=jurisdiction]}
2771 \map{\step[typesource=documentojuridico,typetarget=legal]}
2772 \map{\step[typesource=documentocivil,typetarget=legal]}
2773 \map{\step[typesource=norma,typetarget=standard]}
2774 \map{\step[typesource=partitura,typetarget=music]}
2775 \map{\step[typesource=iconografico,typetarget=image]}
2776 \map{\step[typesource=obradearte,typetarget=artwork]}
2777 \map{\step[typesource=tridimensional,typetarget=artwork]}
2778 \map{\step[typesource=cartografico,typetarget=map]}
2779 \map{\step[typesource=mapa,typetarget=map]}
2780 \map{\step[typesource=audiovisual,typetarget=video]}
2781 \map{\step[typesource=conjuntodedados,typetarget=dataset]}
2782 \map{\step[typesource=fasciculo,typetarget=supperperiodical]}
2783 \map{\step[typesource=trabalhoapresentado,typetarget=unpublished]}
2784 % thesis family: retype to thesis and preset the (babel-aware) type word
2785 \map{\step[typesource=tese,typetarget=thesis]}
2786 \map{\step[typesource=dissertacao,typetarget=thesis]}
2787 \map{\step[typesource=masterdissertation,typetarget=thesis,final]\step[fieldset=type,fiel
2788 \map{\step[typesource=mscdissertation,typetarget=thesis,final]\step[fieldset=type,fiel
2789 \map{\step[typesource=dissertacaomestrado,typetarget=thesis,final]\step[fieldset=type,fi
2790 \map{\step[typesource=dscthesis,typetarget=thesis,final]\step[fieldset=type,fieldvalue=d
2791 \map{\step[typesource=tesedoutorado,typetarget=thesis,final]\step[fieldset=type,fieldval
2792 \map{\step[typesource=phdthesis,typetarget=thesis,final]\step[fieldset=type,fieldvalue=p
2793 \map{\step[typesource=teseph,typetarget=thesis,final]\step[fieldset=type,fieldvalue=phd
2794 \map{\step[typesource=tcc,typetarget=thesis,final]\step[fieldset=type,fieldvalue=tcc]}
2795 % field PT synonyms (unaccented) -> canonical English
2796 \map{\step[fieldsource=periodico,fieldtarget=journaltitle]}
2797 \map{\step[fieldsource=revista,fieldtarget=journaltitle]}
2798 \map{\step[fieldsource=jornal,fieldtarget=journaltitle]}
2799 \map{\step[fieldsource=titulo,fieldtarget=title]}
2800 \map{\step[fieldsource=subtitulo,fieldtarget=subtitle]}
2801 \map{\step[fieldsource=autor,fieldtarget=author]}
2802 \map{\step[fieldsource=editora,fieldtarget=publisher]}
2803 \map{\step[fieldsource=local,fieldtarget=location]}
2804 \map{\step[fieldsource=ano,fieldtarget=year]}
2805 \map{\step[fieldsource=data,fieldtarget=date]}
2806 \map{\step[fieldsource=edicao,fieldtarget=edition]}
2807 \map{\step[fieldsource=paginas,fieldtarget=pages]}
2808 \map{\step[fieldsource=numero,fieldtarget=number]}
2809 \map{\step[fieldsource=tradutor,fieldtarget=translator]}
2810 \map{\step[fieldsource=organizador,fieldtarget=editor]}
2811 \map{\step[fieldsource=instituicao,fieldtarget=institution]}
2812 \map{\step[fieldsource=tipo,fieldtarget=type]}
2813 \map{\step[fieldsource=nota,fieldtarget=note]}
2814 \map{\step[fieldsource=dataacesso,fieldtarget=urldate]}
2815 \map{\step[fieldsource=curso,fieldtarget=course]}
2816 \map{\step[fieldsource=grau,fieldtarget=coppedegree]}
2817 \map{\step[fieldsource=artista,fieldtarget=artist]}
2818 \map{\step[fieldsource=plataforma,fieldtarget=platform]}
2819 \map{\step[fieldsource=diretor,fieldtarget=director]}
2820 \map{\step[fieldsource=produtor,fieldtarget=producer]}
2821 \map{\step[fieldsource=versao,fieldtarget=version]}
2822 \map{\step[fieldsource=escala,fieldtarget=scale]}

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2823 }%
2824 }
2825
2826 % --- ABNT list layout: flush left, single spacing, blank line between entries ---
2827 \setlength{\bibhang}{0pt}
2828 \setlength{\bibitemsep}{\baselineskip}
2829
2830 </bbx>
2831 <*cbx>
2832 \ProvidesFile{coppe.cbx}[2026/05/28 v4.0 CoppeTeX ABNT author-date citations]
2833 % Author-date is the CoppeTeX 4.0 default. authoryear-comp gives compressed
2834 % author-year cites with normal-caps names (post-2020 rule) out of the box.
2835 \RequireCitationStyle{authoryear-comp}
2836
2837 % --- NBR 10520:2023 author separator -----
2838 % Parenthetical citations separate authors with ";" -> (Silva; Souza, 2020).
2839 % Narrative citations (\textcite, natbib \citet) keep comma + "e/and" ->
2840 % Silva, Souza e Oliveira (2020). The reference list keeps its own ";" via
2841 % coppe.bbx ([bib,biblist]); these contextless formats cover the cite side.
2842 \DeclareDelimFormat{multinamedelim}{\addsemicolon\space}
2843 \DeclareDelimFormat{finalnamedelim}{\addsemicolon\space}
2844 \DeclareDelimFormat[textcite]{multinamedelim}{\addcomma\space}
2845 \DeclareDelimFormat[textcite]{finalnamedelim}{\addspace\bibstring{and}\space}
2846
2847 % --- apud: citation of a citation (ABNT NBR 6023 §4.1.2.2 / NBR 10520) -----
2848 % ABNT lists only the work actually CONSULTED. So only the consulted work (#4, a
2849 % .bib key) is cited and enters the reference list; the ORIGINAL work (#2 author,
2850 % #3 year) was not consulted and is given as plain text -- it is never a cite key,
2851 % so it is never added to the bibliography.
2852 % \citetapud[<page>]{<orig author>}{<orig year>}{<consulted key>}
2853 % -> Orig Author (year) apud Consulted (yr[, page]) [narrative]
2854 % \citepapud[<page>]{<orig author>}{<orig year>}{<consulted key>}
2855 % -> (Orig Author, year apud CONSULTED, yr[, page]) [parenthetical]
2856 % #1 = optional postnote (page/loc) on the consulted work; "apud" is latin -> italic.
2857 \DeclareRobustCommand{\citetapud}[4][]{\%
2858 % #2\space(#3)\space\mkbibemph{apud}\space\textcite[#1]{#4}}
2859 \DeclareRobustCommand{\citepapud}[4][]{\%
2860 % \mkbibparens{#2\addcomma\space#3\space\mkbibemph{apud}\space
2861 % \citeauthor{#4}\addcomma\space\citeyear{#4}}
2862 % \ifblank{#1}{\}\addcomma\space\mkpageprefix[pagination]{#1}}}}
2863 </cbx>
2864 <*lbr>
2865 \ProvidesFile{brazilian-coppe.lbx}[2026/05/28 v4.0 CoppeTeX brazilian strings]
2866 \InheritBibliographyExtras{brazilian}
2867 \NewBibliographyString{availablefrom,mscdiss,dscthesi,tcc,depositor,coppescale}
2868 \DeclareBibliographyStrings{%
2869 inherit = {brazilian},
2870 availablefrom = {{Dispon\'}{\i}vel em}{Dispon\'}{\i}vel em}},
2871 urlseen = {{Acesso em}{Acesso em}},
2872 in = {{In}{In}},
2873 editor = {{organizador}{org\adddot}},
2874 editors = {{organizadores}{org\adddot}},
2875 depositor = {{Depositante}{Depositante}},
2876 coppescale = {{Escala}{Escala}},

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2877 mscdiss      = {{Disserta\c{c}\~{a}o de Mestrado}{Disserta\c{c}\~{a}o de Mestrado}},
2878 dscthesi      = {{Tese de Doutorado}{Tese de Doutorado}},
2879 phdthesis     = {{Tese de Doutorado}{Tese de Doutorado}},
2880 tcc           = {{Trabalho de Conclus\~{a}o de Curso}{Trabalho de Conclus\~{a}o de Curso}},
2881 }
2882 </lbxbbr>
2883 <*lbxen>
2884 \ProvidesFile{english-coppe.lbx}[2026/05/28 v4.0 CoppeTeX english strings]
2885 \InheritBibliographyExtras{english}
2886 \NewBibliographyString{availablefrom,mscdiss,dscthesis,tcc,depositor,coppescale}
2887 \DeclareBibliographyStrings{%
2888   inherit      = {english},
2889   availablefrom = {{Available from}{Available from}},
2890   urlseen      = {{Access on}{Access on}},
2891   in           = {{In}{In}},
2892   editor       = {{editor}{ed\adddot}},
2893   editors      = {{editors}{ed\adddot}},
2894   depositor    = {{Depositor}{Depositor}},
2895   coppescale   = {{Scale}{Scale}},
2896   mscdiss      = {{M\adddot Sc\adddot\ Dissertation}{M\adddot Sc\adddot\ Diss\adddot}},
2897   dscthesi     = {{D\adddot Sc\adddot\ Thesis}{D\adddot Sc\adddot\ Thesis}},
2898   phdthesis    = {{Ph\adddot D\adddot\ Thesis}{Ph\adddot D\adddot\ Thesis}},
2899   tcc          = {{Undergraduate Thesis}{Undergraduate Thesis}},
2900 }
2901 </lbxen>
2902 <*lbxes>
2903 \ProvidesFile{spanish-coppe.lbx}[2026/05/28 v4.0 CoppeTeX spanish strings]
2904 \InheritBibliographyExtras{spanish}
2905 \NewBibliographyString{availablefrom,mscdiss,dscthesis,tcc,depositor,coppescale}
2906 \DeclareBibliographyStrings{%
2907   inherit      = {spanish},
2908   availablefrom = {{Disponible en}{Disponible en}},
2909   urlseen      = {{Consultado el}{Consultado el}},
2910   in           = {{En}{En}},
2911   editor       = {{editor}{ed\adddot}},
2912   editors      = {{editores}{ed\adddot}},
2913   depositor    = {{Solicitante}{Solicitante}},
2914   coppescale   = {{Escala}{Escala}},
2915   mscdiss      = {{Tesis de Maestr\'{i}a}{Tesis de Maestr\'{i}a}},
2916   dscthesi     = {{Tesis Doctoral}{Tesis Doctoral}},
2917   phdthesis    = {{Tesis Doctoral}{Tesis Doctoral}},
2918   tcc          = {{Trabajo de Fin de Carrera}{Trabajo de Fin de Carrera}},
2919 }
2920 </lbxes>
2921 <*lbxfr>
2922 \ProvidesFile{french-coppe.lbx}[2026/05/28 v4.0 CoppeTeX french strings]
2923 \InheritBibliographyExtras{french}
2924 \NewBibliographyString{availablefrom,mscdiss,dscthesis,tcc,depositor,coppescale}
2925 \DeclareBibliographyStrings{%
2926   inherit      = {french},
2927   availablefrom = {{Disponible \'a}{Disponible \'a}},
2928   urlseen      = {{Consult\'e le}{Consult\'e le}},
2929   in           = {{Dans}{Dans}},
2930   editor       = {{\'editeur}{\ed\adddot}},

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2931 editors      = {{\'}editeurs}{\'ed\adddot}},
2932 depositor     = {{D\'eposant}{D\'eposant}},
2933 coppescale    = {{\'Echelle}{\'Echelle}},
2934 msdiss        = {{M\'emoire de Master}{M\'emoire de Master}},
2935 dscthesi      = {{Th\'ese de Doctorat}{Th\'ese de Doctorat}},
2936 phdthesis     = {{Th\'ese de Doctorat}{Th\'ese de Doctorat}},
2937 tcc           = {{M\'emoire de fin d\'\'etudes}{M\'emoire de fin d\'\'etudes}},
2938 }
2939 </lbxfr>
2940 <*lbxit>
2941 \ProvidesFile{italian-coppe.lbx}[2026/05/28 v4.0 CoppeTeX italian strings]
2942 \InheritBibliographyExtras{italian}
2943 \NewBibliographyString{availablefrom,msdiss,dscthesi,tcc,depositor,coppescale}
2944 \DeclareBibliographyStrings{%
2945   inherit      = {italian},
2946   availablefrom = {{Disponibile da}{Disponibile da}},
2947   urlseen      = {{Consultato il}{Consultato il}},
2948   in           = {{In}{In}},
2949   editor       = {{curatore}{cur\adddot}},
2950   editors      = {{curatori}{cur\adddot}},
2951   depositor    = {{Depositante}{Depositante}},
2952   coppescale   = {{Scala}{Scala}},
2953   msdiss       = {{Tesi di Laurea Magistrale}{Tesi di Laurea Magistrale}},
2954   dscthesi     = {{Tesi di Dottorato}{Tesi di Dottorato}},
2955   phdthesis    = {{Tesi di Dottorato}{Tesi di Dottorato}},
2956   tcc          = {{Tesi di Laurea Triennale}{Tesi di Laurea Triennale}},
2957 }
2958 </lbxit>
2959 <*langes>
2960 %% coppe-lang-spanish.def -- CoppeTeX 4.x Spanish class-string pack.
2961 %% Loaded by coppe.cls when \documentclass[spanish]{coppe} (or any other
2962 %% spanish slot) is active. Populates the dispatcher table for the
2963 %% spanish language and registers the biblalex language mapping.
2964 %% Institutional names (universityname/cityname/statename/countryname)
2965 %% are deliberately kept Portuguese: the COPPE/UFRJ cover and folha de
2966 %% rosto are a Brazilian institutional template that does not translate.
2967 \ProvidesFile{coppe-lang-spanish.def}[2026/05/28 v4.0 CoppeTeX spanish class strings]
2968 \copperdefstring{spanish}{appendix}      {Ap\'endice}
2969 \copperdefstring{spanish}{annex}         {Anexo}
2970 \copperdefstring{spanish}{frame}         {Cuadro}
2971 \copperdefstring{spanish}{listframe}     {Lista de Cuadros}
2972 \copperdefstring{spanish}{source}        {Fuente}
2973 \copperdefstring{spanish}{program}       {Programa}
2974 \copperdefstring{spanish}{listprogram}   {Lista de Programas}
2975 \copperdefstring{spanish}{references}    {Referencias}
2976 \copperdefstring{spanish}{in:}          {en}
2977 \copperdefstring{spanish}{art}           {Arte}
2978 \copperdefstring{spanish}{platform}      {Plataforma}
2979 \copperdefstring{spanish}{directedby}    {Direcci\'on}
2980 \copperdefstring{spanish}{producedby}    {Producci\'on}
2981 \copperdefstring{spanish}{listabbreviation}{Lista de Abreviaturas y Siglas}
2982 \copperdefstring{spanish}{listsymbol}    {Lista de S\'imbolos}
2983 \copperdefstring{spanish}{glossary}      {Glosario}
2984 \copperdefstring{spanish}{advisor}       {Director}

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2985 \copperdefstring{spanish}{advisors}      {Directores}
2986 \copperdefstring{spanish}{dept}          {Programa}
2987 \copperdefstring{spanish}{approvedmale}   {Examinado por}
2988 \copperdefstring{spanish}{approvedfemale} {Examinada por}
2989 \copperdefstring{spanish}{universityname} {Universidade Federal do Rio de Janeiro}
2990 \copperdefstring{spanish}{cityname}       {Rio de Janeiro}
2991 \copperdefstring{spanish}{statename}      {RJ}
2992 \copperdefstring{spanish}{countryname}    {Brasil}
2993 \copperdefstring{spanish}{monthname1}     {enero}
2994 \copperdefstring{spanish}{monthname2}     {febrero}
2995 \copperdefstring{spanish}{monthname3}     {marzo}
2996 \copperdefstring{spanish}{monthname4}     {abril}
2997 \copperdefstring{spanish}{monthname5}     {mayo}
2998 \copperdefstring{spanish}{monthname6}     {junio}
2999 \copperdefstring{spanish}{monthname7}     {julio}
3000 \copperdefstring{spanish}{monthname8}     {agosto}
3001 \copperdefstring{spanish}{monthname9}     {septiembre}
3002 \copperdefstring{spanish}{monthname10}    {octubre}
3003 \copperdefstring{spanish}{monthname11}    {noviembre}
3004 \copperdefstring{spanish}{monthname12}    {diciembre}
3005 \copperdefstring{spanish}{algorithm2eopt}{spanish}
3006 \copperdefstring{spanish}{babelname}      {spanish}
3007 %% biblatex is loaded by coppe.cls AFTER this .def file is read; defer the
3008 %% language-mapping registration until \begin{document} so biblatex's
3009 %% \DeclareLanguageMapping is available.
3010 \AtBeginDocument{\DeclareLanguageMapping{spanish}{spanish-coppe}}
3011 \end{langes}
3012 \end{langfr}
3013 %% coppe-lang-french.def -- CoppeTeX 4.x French class-string pack.
3014 %% See coppe-lang-spanish.def for the structure. Institutional names
3015 %% remain Portuguese; only document-content strings translate.
3016 \ProvidesFile{coppe-lang-french.def}[2026/05/28 v4.0 CoppeTeX french class strings]
3017 \copperdefstring{french}{appendix}          {Annexe}
3018 \copperdefstring{french}{annex}             {Annexe}
3019 \copperdefstring{french}{frame}              {Encadr\'e}
3020 \copperdefstring{french}{listframe}          {Liste des encadr\'es}
3021 \copperdefstring{french}{source}             {Source}
3022 \copperdefstring{french}{program}            {Programme}
3023 \copperdefstring{french}{listprogram}        {Liste des programmes}
3024 \copperdefstring{french}{references}         {R\'ef\'erences}
3025 \copperdefstring{french}{in:}               {dans}
3026 \copperdefstring{french}{art}               {Illustration}
3027 \copperdefstring{french}{platform}          {Plateforme}
3028 \copperdefstring{french}{directedby}        {R\'ealis\'e par}
3029 \copperdefstring{french}{producedby}        {Produit par}
3030 \copperdefstring{french}{listabbreviation}  {Liste des abr\'eviations et acronymes}
3031 \copperdefstring{french}{listsymbol}        {Liste des symboles}
3032 \copperdefstring{french}{glossary}          {Glossaire}
3033 \copperdefstring{french}{advisor}           {Directeur}
3034 \copperdefstring{french}{advisors}          {Directeurs}
3035 \copperdefstring{french}{dept}              {Programme}
3036 \copperdefstring{french}{approvedmale}      {Examin\'e par}
3037 \copperdefstring{french}{approvedfemale}    {Examin\'ee par}
3038 \copperdefstring{french}{universityname}    {Universidade Federal do Rio de Janeiro}

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3039 \copperdefstring{french}{cityname} {Rio de Janeiro}
3040 \copperdefstring{french}{statename} {RJ}
3041 \copperdefstring{french}{countryname} {Brasil}
3042 \copperdefstring{french}{monthname1} {janvier}
3043 \copperdefstring{french}{monthname2} {f\`evrier}
3044 \copperdefstring{french}{monthname3} {mars}
3045 \copperdefstring{french}{monthname4} {avril}
3046 \copperdefstring{french}{monthname5} {mai}
3047 \copperdefstring{french}{monthname6} {juin}
3048 \copperdefstring{french}{monthname7} {juillet}
3049 \copperdefstring{french}{monthname8} {ao\`ut}
3050 \copperdefstring{french}{monthname9} {septembre}
3051 \copperdefstring{french}{monthname10}{octobre}
3052 \copperdefstring{french}{monthname11}{novembre}
3053 \copperdefstring{french}{monthname12}{d\`ecembre}
3054 \copperdefstring{french}{algorithm2eopt}{french}
3055 \copperdefstring{french}{babelname} {french}
3056 \AtBeginDocument{\DeclareLanguageMapping{french}{french-coppe}}
3057 \langfr
3058 \*langit
3059 %% coppe-lang-italian.def -- CoppeTeX 4.x Italian class-string pack.
3060 %% See coppe-lang-spanish.def for the structure. The algorithm2e
3061 %% language option for Italian is spelled "italiano" -- that's the
3062 %% spelling algorithm2e ships, even though babel calls the language
3063 %% "italian".
3064 \ProvidesFile{coppe-lang-italian.def}[2026/05/28 v4.0 CoppeTeX italian class strings]
3065 \copperdefstring{italian}{appendix} {Appendice}
3066 \copperdefstring{italian}{annex} {Allegato}
3067 \copperdefstring{italian}{frame} {Riquadro}
3068 \copperdefstring{italian}{listframe} {Elenco dei riquadri}
3069 \copperdefstring{italian}{source} {Fonte}
3070 \copperdefstring{italian}{program} {Programma}
3071 \copperdefstring{italian}{listprogram} {Elenco dei programmi}
3072 \copperdefstring{italian}{references} {Riferimenti}
3073 \copperdefstring{italian}{in:} {in}
3074 \copperdefstring{italian}{art} {Illustrazione}
3075 \copperdefstring{italian}{platform} {Piattaforma}
3076 \copperdefstring{italian}{directedby} {Regia di}
3077 \copperdefstring{italian}{producedby} {Prodotto da}
3078 \copperdefstring{italian}{listabbreviation}{Elenco di abbreviazioni e sigle}
3079 \copperdefstring{italian}{listsymbol} {Elenco dei simboli}
3080 \copperdefstring{italian}{glossary} {Glossario}
3081 \copperdefstring{italian}{advisor} {Relatore}
3082 \copperdefstring{italian}{advisors} {Relatori}
3083 \copperdefstring{italian}{dept} {Programma}
3084 \copperdefstring{italian}{approvedmale} {Esaminato da}
3085 \copperdefstring{italian}{approvedfemale} {Esaminata da}
3086 \copperdefstring{italian}{universityname}{Universidade Federal do Rio de Janeiro}
3087 \copperdefstring{italian}{cityname} {Rio de Janeiro}
3088 \copperdefstring{italian}{statename} {RJ}
3089 \copperdefstring{italian}{countryname} {Brasil}
3090 \copperdefstring{italian}{monthname1} {gennaio}
3091 \copperdefstring{italian}{monthname2} {febbraio}
3092 \copperdefstring{italian}{monthname3} {marzo}

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3093 \copperdefstring{italian}{monthname4} {aprile}
3094 \copperdefstring{italian}{monthname5} {maggio}
3095 \copperdefstring{italian}{monthname6} {giugno}
3096 \copperdefstring{italian}{monthname7} {luglio}
3097 \copperdefstring{italian}{monthname8} {agosto}
3098 \copperdefstring{italian}{monthname9} {settembre}
3099 \copperdefstring{italian}{monthname10}{ottobre}
3100 \copperdefstring{italian}{monthname11}{novembre}
3101 \copperdefstring{italian}{monthname12}{dicembre}
3102 \copperdefstring{italian}{algorithm2eopt}{italiano}
3103 \copperdefstring{italian}{babelname} {italian}
3104 \AtBeginDocument{\DeclareLanguageMapping{italian}{italian-coppe}}
3105 </langit>
3106 <*numbbx>
3107 \ProvidesFile{coppe-numeric.bbx}[2026/05/28 v4.0 CoppeTeX ABNT numeric bibliography]
3108 % Numeric reference list for the ABNT numeric system (NBR 10520). It reuses the
3109 % full coppe ABNT style (drivers, CAPS surnames, bold titles, "In:", p./f.,
3110 % the PT->EN source map and the language mappings) and only adds the [n] list
3111 % label and the numeric sorting on top.
3112 \RequireBibliographyStyle{coppe}
3113
3114 % ABNT numeric system lists references in the order they are first cited
3115 % (NBR 10520), i.e. unsorted. Override with sorting=nty for an alphabetical
3116 % numbered list. 'labelnumber' generates the [n] field.
3117 \ExecuteBibliographyOptions{labelnumber,sorting=none}
3118
3119 % Bracketed labels, both in the list and for any shorthand entries.
3120 \DeclareFieldFormat{labelnumberwidth}{\mkbibbrackets{#1}}
3121 \DeclareFieldFormat{shorthandwidth}{\mkbibbrackets{#1}}
3122
3123 % Reference list driven by the [n] label (copied from biblatex's numeric.bbx),
3124 % keeping the ABNT blank line between entries via \bibitemsep set in coppe.bbx.
3125 \defbibenvironment{bibliography}
3126 {
3127   \list
3128     {\printtext[labelnumberwidth]{%
3129       \printfield{labelprefix}%
3130       \printfield{labelnumber}}}
3131     {\setlength{\labelwidth}{\labelnumberwidth}%
3132      \setlength{\leftmargin}{\labelwidth}%
3133      \setlength{\labelsep}{\biblabelsep}%
3134      \addtolength{\leftmargin}{\labelsep}%
3135      \setlength{\itemsep}{\bibitemsep}%
3136      \setlength{\parsep}{\bibparsep}}%
3137     {\renewcommand*{\makelabel}[1]{\hss#1}}
3138   {\endlist}
3139   {\item}
3140 }
3141 <*numcbx>
3142 \ProvidesFile{coppe-numeric.cbx}[2026/05/28 v4.0 CoppeTeX ABNT numeric citations]
3143 % Numeric citation system (ABNT NBR 10520): in-text references are bracketed
3144 % numbers [1], compressed and sorted within a single cite group. This is the
3145 % opt-in alternative to the author-date default (coppe.cbx); it pairs with
3146 % bibstyle=coppe-numeric. With the biblatex natbib option, \citep -> [1] and

```

```

3147 % \citet -> Author [1].
3148 \RequireCitationStyle{numeric-comp}
3149 </numcbx>

```

10.2 The bibliographic databases

The two BibLaTeX/biber databases shipped with the class — `example.bib` (the sample database cited by the example monograph) and `coppe.bib` (the references of this manual) — are generated from this source too, so the whole distribution still has a single editable origin. They live inside an `\iffalse` block below: the documentation driver never typesets them, and only `coppe.ins` extracts them, through the `examplebib` and `coppebib` docstrip targets. Edit the entries here, never the generated `.bib` files.

Acknowledgments

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References

- [1] *Manual para elaboração e normalização de trabalhos acadêmicos*, 9 edition.
- [2] CPGP/COPPE/UFRJ. *Norma para a Elaboração Gráfica de Teses*. Rio de Janeiro, RJ, Brasil, Julho 2008. Revisada em 10/09/2010.
- [3] Donald E. Knuth. *The TeXbook*. Addison-Wesley, Reading, MA, 1984.
- [4] Leslie Lamport. *LaTeX: A Document Preparation System*. Addison-Wesley, Reading, MA, 1986.
- [5] Oren Patashnik. Bibtexing. Documentation for general BibTeX users, February 1988.
- [6] Oren Patashnik. Designing bibtex styles. The part of BibTeX’s documentation that’s not meant for general users, February 1988.
- [7] William Strunk, Jr. and E. B. White. *The Elements of Style*. Macmillan, 3 edition, 1979.
- [8] TeX Frequently Asked Questions.

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Change History

v0.0	General: Creation Date.	1	v0.4	Generalization.	47
v0.1	General: Documentation: bibliography fixed, title translation.	1	v0.5	General: Beta documentation. . . .	1
	Sourceforge submission.	1		abstract : Changed from macro to environment.	51
v0.2	General: Unification of the code for the list of symbols and abbreviations.	1		\backmatter : Added mainmatter pages counter.	45
				foreignabstract : Changed from macro to environment.	52
v0.3	General: Added ‘draft’ option. . . .	1	v1.0	General: First COPPE _{TEX} release. .	1
	\maketitle : Added number of examiners test.	47	v2.0	General: COPPE _{TEX} release 2.0. . .	1

v2.1	General: COPPE _{TeX} release 2.1: Matching the new rules.	1	v4.0	General: Catalog-label writers (coppe@mainBegin, coppe@bibBegin, coppe@bibEnd, coppe@hasLof) made language-agnostic: detokenize wrap on label names protects ':' and ',' from babel shorthand activation, and the raw etex primitives romannumeral/number replace LaTeX's roman/arabic to bypass spanish-babel's small-caps @roman redefinition that would otherwise crash pdfTeX with "Unbalanced write command" on every es-main document.	1
v2.2	General: Matching new guidelines, including new logo.	1			
v2.2.2	General: Fixed some text constants in .bib and documented it here.	1			
v3.0	General: Added support for monographs in English.	41			
	New approval page layout.	48			
	\department: Added new course on Nanotechnology.	42			
	\examiner: Examiners expansion without degree.	43		Class-wide string dispatcher (copperdefstring, coppestring, coppemainstring, coppeforeignstring) replaces scattered iflanguage and if english branches; pt/en outputs remain byte-identical.	1
v3.1	\department: Included a sort key in symb	42			
v3.2	General: Fixed version problem between cls and dtx.	1			
v3.3	General: Extend abbrev to work like symb and accept a sorting key	1		Multilingual model: three fixed slots (main, foreign, optional third); five built-in language options (brazilian, english, spanish, french, italian); plug-in mechanism for any other Babel language via coppe-lang-lang.def plus lang-coppe.lbx.	1
v3.4	General: Some examples for figures, tables, longtables, etc.	1			
v3.4.1	General: Fix english option to consider brazilian a second language, describe it in document	41		New user API: titlein, usecoppelanguage, brazilianabstract environment.	1
v3.5	General: New command annex	75			
v3.5.1	General: New command annex now supports brazilian or english	75		\coppetexfinalpage: Initial coppetexfinalpage colophon page.	46